
Dropout Universality: Scaling Laws and Optimal Scheduling at the Edge-of-Chaos

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Abstract

We develop a mean-field theory of dropout as a perturbation of critical signal propagation at the edge of chaos, and use it to motivate a simple, no-cost change to standard practice: *front-loaded* dropout schedules cut final test loss by 18–35% in MLPs and 4–6% in Vision Transformers over constant dropout at fixed budget. The theoretical mechanism is that dropout shifts the perfect-alignment fixed point, making the depth scale for information propagation finite even at critical initialization. We derive critical and crossover scaling laws for correlation decay and establish that smooth activations and kinked, ReLU-like activations constitute distinct universality classes, with different critical exponents and a universal two-parameter scaling collapse in detuning and dropout strength. The distinction traces to the analytic structure of the correlation map: smooth activations admit a Taylor expansion near perfect alignment, while kinked activations develop a branch point with universal non-analyticity. A local-stationary propagation approximation favors saturated dropout profiles under fixed budget; an additive regularization-reach heuristic then selects front-loaded schedules, with accuracy gains as a secondary effect in the reported CIFAR comparisons. We also discuss how the same Gaussian-kernel structure extends the theory beyond MLPs toward CNNs and residual architectures.

1. Introduction

Mean-field analyses of randomly initialized deep networks reveal an order-to-chaos phase diagram that controls both signal propagation and gradient penetration with depth

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(Poole et al., 2016; Schoenholz et al., 2017; Bahri et al., 2020; 2024b). In this work, we study how this picture is modified by dropout, and use the resulting theory to motivate a no-cost change to standard practice: we find that front-loaded dropout schedules cut final test loss by 18–35% over constant dropout at fixed budget in MLPs, with smaller improvements in Vision Transformers (Table 3). Using the representation-group (coarse-graining) language of (Roberts et al., 2022; Bahri et al., 2024b), we show that dropout behaves as a relevant perturbation: it displaces the critical fixed point and grows under depth coarse-graining, pushing the dynamics away from criticality and determining the macroscopic phase. Concretely, dropout deforms the correlation map (Schoenholz et al., 2017) by adding a correlation-independent shift at perfect alignment, so that perfect correlation between inputs is no longer a fixed point for any nonzero dropout. Thus, even at the edge-of-chaos (Sompolinsky et al., 1988; Packard, 1988; Bertschinger & Natschläger, 2004), the depth correlation length is finite. We interpret this shift using Landau theory, as an equation of state for the decorrelation order parameter $m \equiv 1 - c^*$ (with c^* the asymptotic inter-signal correlation), derive the resulting scaling laws, and show that smooth and kinked, ReLU-like activations exhibit the same qualitative phase diagram but different critical scaling in the presence of dropout.

Additionally, we show that the two control parameters (dropout strength and distance to criticality) admit a scaling collapse into a single universal form,¹ and we illustrate how these predictions compare with mean-field recursions.

The main conceptual contributions are threefold: first, while dropout was previously shown to destroy the order-to-chaos critical point (Schoenholz et al., 2017), we show that the deformed mean-field map still retains a nontrivial $c^* < 1$ fixed point. This allows us to have a well-defined correlation length recursion even in the presence of dropout, and yields a Landau equation of state for $m = 1 - c^*$ together with a two-parameter scaling collapse. Second, we show

¹This is analogous to the homogeneous scaling of connected correlation functions near a critical point; related scaling forms hold for one-particle-irreducible vertex functions after the usual Legendre transform (Zinn-Justin, 2002).

that universality classes and scaling laws are set by the analytic structure of activations: smooth and kinked activations have distinct critical exponents (Table 2), a distinction reflected in their Hermite spectral structure (App. C). Since the same Gaussian activation kernels arise beyond MLPs, including CNNs and residual branches (App. A.4), we expect the qualitative smooth/kinked split to be a well-motivated heuristic. Third, we treat dropout as a depth-dependent dynamical field; treating classically fixed hyperparameters as fields that can change through the network opens a new dimension for hyperparameter optimization across depth, and potentially over training time.

This work complements prior studies of depth-dependent regularization: stochastic depth drops whole residual blocks with increasing probability (Huang et al., 2016), curriculum dropout anneals dropout over training time (Morero et al., 2017), and LayerDrop removes transformer layers for efficiency (Fan et al., 2020). Our schedules are spatial depth profiles, so they are complementary to temporal curricula and layer-dropping mechanisms. Qualitatively different edge-of-chaos behavior for smooth and ReLU-like activations was previously observed in (Hayou et al., 2019); here we extract the scaling exponents and identify the corresponding universality classes.²

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2. Mean-Field Theory Background

We first state the mean-field assumptions for MLPs, which provide the baseline for the dropout analysis (Poole et al., 2016; Schoenholz et al., 2017; Bahri et al., 2020; 2024b). We consider fully-connected MLPs at random initialization, where preactivations are well-approximated as Gaussian random variables with self-consistently determined statistics (means, variances, covariances). In the strict infinite width limit this Gaussian description becomes exact (Lee et al., 2018), while at large but finite width deviations from Gaussianity can be organized perturbatively in the depth-to-width aspect ratio (schematically L/N), as treated comprehensively in (Roberts et al., 2022; Bahri et al., 2024b). The intuition behind this treatment is that as we add more activations, through the CLT, the statistics will become more and more Gaussian, approaching this distribution asymptotically in the large N limit.

We consider an MLP of depth L , N_l neurons in layer l , and an activation $\phi : \mathbb{R} \rightarrow \mathbb{R}$ (e.g. tanh, ReLU, etc.). We use

²The original notebooks, saved results, and seed-level confidence-interval calculations are available in the `icml` repository. See App. D for the full reproducibility statement.

the standard i.i.d. Gaussian initialization

$$W_{ij}^l \sim \mathcal{N}\left(0, \frac{\sigma_w^2}{N_{l-1}}\right), \quad b_i^l \sim \mathcal{N}(0, \sigma_b^2), \quad (1)$$

take the input as $y_{i;a}^0 = x_{i;a}$, and propagate forward via

$$z_{i;a}^l = \sum_{j=1}^{N_{l-1}} W_{ij}^l y_{j;a}^{l-1} + b_i^l, \quad y_{i;a}^l = \phi(z_{i;a}^l), \quad (2)$$

where $W^l \in \mathbb{R}^{N_l \times N_{l-1}}$ and a labels the input. Thus the weight variance is scaled by the number of incoming units. Throughout, expectations $\mathbb{E}[\cdot]$ are over the random weights and biases at initialization, with inputs held fixed. We also use the standard Gaussian measure

$$\int Dz(\cdots) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dz e^{-z^2/2}(\cdots), \quad (3)$$

and similarly for $\int Dz_1 Dz_2$.

In the infinite-width limit this becomes exact layer-by-layer, while at finite width the first layer is exactly Gaussian (being a linear map of fixed inputs) and subsequent layers are only approximately Gaussian, with controlled non-Gaussian corrections parametrized by the depth-to-width ratio (Roberts et al., 2022). A fuller treatment of the MFT background is given in App. A.2.

We track the single-input preactivation variance q_{aa}^l , the two-input covariance q_{ab}^l , and the induced correlation c_{ab}^l . For $l \geq 2$, their recursions are

$$q_{aa}^l = \sigma_w^2 \int Dz \phi^2\left(\sqrt{q_{aa}^{l-1}} z\right) + \sigma_b^2 \quad (4)$$

$$q_{ab}^l = \sigma_w^2 \int Dz_1 Dz_2 \phi(u_1) \phi(u_2) + \sigma_b^2 \quad (5)$$

$$u_1 = \sqrt{q_{aa}^{l-1}} z_1, \quad c_{ab}^{l-1} = \frac{q_{ab}^{l-1}}{\sqrt{q_{aa}^{l-1} q_{bb}^{l-1}}}, \quad (6)$$

$$u_2 = \sqrt{q_{bb}^{l-1}} \left(c_{ab}^{l-1} z_1 + \sqrt{1 - (c_{ab}^{l-1})^2} z_2 \right). \quad (7)$$

The first layer is initialized directly from the inputs, $q_{ab}^1 = \sigma_w^2 N_0^{-1} \sum_j x_{j;a} x_{j;b} + \sigma_b^2$. In the absence of dropout, for a wide array of activations q_{ab}^l, c_{ab}^l settle to fixed points q^*, c^* ; in particular, there exists a $c^* = 1$ fixed point as a straight-forward calculation shows. The variance fixed point sets the typical activation scale, while the correlation fixed point describes whether distinct inputs become indistinguishable under depth iteration. We denote the recurrence relation for c_{ab}^l by $c_{ab}^l = F(c_{ab}^{l-1})$.

To probe its stability, one linearizes the map at $c = 1$, defining the (angular) susceptibility

$$\chi_{\perp} \equiv \left. \frac{\partial c_{ab}^l}{\partial c_{ab}^{l-1}} \right|_{c_{ab}=1} = \sigma_w^2 \int Dz [\phi'(\sqrt{q^*} z)]^2. \quad (8)$$

This creates a phase diagram with three regimes: $\chi_\perp < 1$ is the ordered regime, $\chi_\perp > 1$ is chaotic, and $\chi_\perp = 1$ defines the critical regime (the so-called "edge-of-chaos"). The depth scale controlling cross-input correlations can become arbitrarily large in the latter, allowing information to penetrate deep into the network. For ReLU, the criticality condition gives $\sigma_w^2 = 2$, coinciding with the variance used in He initialization (He et al., 2015)³ (see App. A.2), a more fundamental viewpoint than variance preservation.

This leads to an emergent characteristic depth scale ξ_c , which parametrizes both signal propagation and gradient flow. Iterating random affine maps induces an effective coarse-graining in depth akin to renormalization-group flow, called representation group flow in the ML context (Roberts et al., 2022); information about norms and angles relaxes exponentially fast,

$$|c^* - c_{ab}^l| \propto e^{-l/\xi_c}, \quad \xi_c^{-1} \equiv -\log |r_c|. \quad (9)$$

An analogous correlation length ξ_q controls how quickly single-input norms settle to q^* . Heuristically, ξ_c controls how far distinctions between different inputs survive with depth; in the ordered phase $c^* = 1$, so $u_1^* = u_2^*$ and $\xi_c^{-1} = -\log \chi_\perp$, hence $\chi_\perp \rightarrow 1$ implies $\xi_c \rightarrow \infty$. Consequently, the edge-of-chaos is primarily about ξ_c rather than ξ_q , except in special cases. For ReLU, the curvature vanishes everywhere away from the origin, leading to a coincidence between ξ_c and ξ_q , both diverging as $\chi_\perp \rightarrow 1$, provided a finite q^* exists. This is one concrete reason why ReLU-style near-critical initializations are comparatively forgiving in practice.

As previously discussed, mean-field theory is an expansion in non-Gaussian corrections suppressed by powers of L/N . Therefore, in our experiments we work in a controlled regime with $L/N \ll 1$ (i.e., $N \gg L$), where the large-width expansion is perturbatively sound (Roberts et al., 2022). The correlation length diverges as we approach criticality, amplifying finite-width and other non-Gaussian corrections. Thus, when probing critical power laws with dropout, we keep the dropout rate small but non-negligible: infinitesimal dropout would require prohibitively deep networks to estimate the correlation length and related quantities.

Although we derive these results for MLPs, analogous large-width limits exist elsewhere. CNNs are the canonical convolutional setting (LeCun et al., 1998), and at infinite channel count their covariance recursions again close through Gaussian activation kernels (Xiao et al., 2018). ResNets are especially suggestive: Yang and Schoenholz (Yang & Schoenholz, 2017) find qualitatively different tanh/ReLU large-depth behavior. Skip connections alter global depth dynamics, easing the information propagation by injecting

³This is not true for general activations, but rather an idiosyncrasy of ReLU.

the original signal after every layer, changing exponential convergence to subexponential or polynomial laws and allowing norm drift, but each residual branch still inherits the local nonlinear Gaussian kernel. Thus our smooth/kinked universality split offers a possible theoretical explanation for their observed dichotomy; a dropout-deformed ResNet theory is left for future work. For transformers (Vaswani et al., 2017), related infinite-width attention analyses use large hidden dimension and/or many heads (Hron et al., 2020). Because the exponents are controlled by the local analytic structure of this channel, we expect the mechanism to persist when it controls the wide-limit recursion. In Sec. 3.3, we test this extrapolation in transformers.

In App. D we probe realistic scenarios across architectures, datasets, and schedules. We focus on overparameterized regimes where dropout improves generalization and accuracy, and thus where it is used in practice, and we explicitly study the trade-off between dropout-driven regularization and stable signal propagation. The local-stationary propagation objective introduced below favors sparse, step-like allocation at fixed budget and is permutation invariant in depth. To choose an ordering within this approximation, we use a regularization-reach heuristic: at a common background decay scale, early masks expose more downstream layers to their perturbation. This motivates testing whether the first layers are the most useful place to inject noise.

3. Universality Classes and Critical Scaling under Dropout

Dropout (Srivastava et al., 2014; Wager et al., 2013) is often used as a regularizer that reduces overfitting by randomly masking activations during training, thereby reducing the network's reliance on any single subset of pathways and discouraging co-adaptation. In the previous literature, dropout was explored in MFT (Huang et al., 2020; Schoenholz et al., 2017), but not in the universality class sense. We formalize dropout as multiplicative Bernoulli noise applied post-activation,

$$p_j^{(\ell)} \sim \text{Bernoulli}(\rho), \quad y_j^{(\ell)} \mapsto \tilde{y}_j^{(\ell)} \equiv \frac{p_j^{(\ell)}}{\rho} y_j^{(\ell)}, \quad (10)$$

i.e. the standard inverted-dropout convention in which $\mathbb{E}_p[\tilde{y}_j^{(\ell)}] = y_j^{(\ell)}$, and hence $\mathbb{E}_p[\tilde{z}_i^{\ell+1} | W, b, y^\ell] = z_i^{\ell+1}$, so the mean preactivation is invariant to the dropout rate (Schoenholz et al., 2017). Throughout, ρ is the keep probability. When comparing two inputs, we draw the dropout masks independently for each input⁴; this independence is the source of the decorrelation effect analyzed below.

⁴Correlated dropout masks can alleviate this detuning: if the same mask is shared across the batch, the $c = 1$ fixed point is preserved and near-critical scaling can be recovered by retuning, but the noise becomes batch coupled and the regularization is

This distinction originally emerged from the following variational question: among activations of fixed scale, which functions make the correlation length of the near-critical Gaussian channel as large as possible? As a tractable part of that question, we extremize the angular susceptibility at fixed variance scale; the resulting eigenfunctions are Hermite polynomials. When standard nonlinearities are decomposed in that basis, two qualitatively different spectra emerge. Smooth activations such as $\tanh$ have rapidly decaying Hermite coefficients, so much of their spectral weight is concentrated in the first few modes. Kinked activations such as ReLU have power-law tails instead: the kink keeps exciting higher Hermite modes, leaving a broader spectrum in the correlation map. This spectral split was the original diagnostic for the smooth/kinked universality classes, which later reappears in the dropout scaling laws. See App. C for detailed calculations.

Our classification is orthogonal to the scale-invariant vs. non-scale-invariant distinction of (Roberts et al., 2022): the multi-scale Hermite spectrum highlighted above is controlled primarily by analytic structure, not by scale invariance per se. The criterion is based on two facts: (a) near-critical scaling laws differ between smooth and kinked activations, and (b) ordinary kinks produce power-law Hermite tails, whereas the smooth examples considered here have much faster decay (Apps. B and C).

Extending the aforementioned mean-field framework to include dropout, one finds that for a single input dropout largely acts as a renormalization of the effective weight variance, $\sigma_w^2 \mapsto \sigma_w^2/\rho$, but for two inputs the perfect-alignment fixed point at $c^* = 1$ is lost for any $\rho < 1$ when the two dropout masks are chosen independently (Schoenholz et al., 2017). In other words, dropout acts as an explicit symmetry-breaking “field” for alignment: even if two inputs are perfectly aligned at some depth, the next layer produces $\bar{c}_{ab} < 1$, cutting off the divergence of the depth correlation length at the edge.

Introducing multiplicative Bernoulli noise with keep probability ρ , the perturbed preactivations read

$$z_i^\ell = \frac{1}{\rho} \sum_{j=1}^{N_{\ell-1}} W_{ij}^\ell p_j^{\ell-1} y_j^{\ell-1} + b_i^\ell. \quad (11)$$

The single-input variance recursion becomes

$$\bar{q}_{aa}^\ell = \frac{\sigma_w^2}{\rho} \int Dz \phi^2 \left(\sqrt{\bar{q}_{aa}^{\ell-1}} z \right) + \sigma_b^2, \quad (12)$$

where the bar reminds us that the variance fixed point is shifted by dropout. We denote the (putative) dropout vari-
correspondingly weaker and more structured. We leave detailed analysis of correlated masks to future work.

ance fixed point by

$$\bar{q}^* = \frac{\sigma_w^2}{\rho} \int Dz \phi^2 (\sqrt{\bar{q}^*} z) + \sigma_b^2. \quad (13)$$

Assuming this variance fixed point exists and is positive, two distinct inputs $a \neq b$ acquire a dropout shift at perfect alignment. If at some depth $c_{ab}^\ell = 1$, then at the next layer one finds (Schoenholz et al., 2017)

$$\bar{F}_\rho(1) = \bar{c}_{ab}^{\ell+1} = 1 - \frac{1-\rho}{\rho \bar{q}^*} \sigma_w^2 \int Dz \phi^2 (\sqrt{\bar{q}^*} z). \quad (14)$$

It is therefore natural to define the dropout field

$$\begin{aligned} h &\equiv 1 - \bar{F}_\rho(1) \\ &= \frac{1-\rho}{\rho \bar{q}^*} \sigma_w^2 \int Dz \phi^2 (\sqrt{\bar{q}^*} z). \end{aligned} \quad (15)$$

For a nondegenerate channel with $\sigma_w^2 \int Dz \phi(\sqrt{\bar{q}^*} z)^2 > 0$, construction gives $\rho < 1 \Rightarrow h > 0$, so the fixed point at $c^* = 1$ is immediately spoiled: even if two inputs are perfectly aligned at some depth, the next layer produces $\bar{c}_{ab} = 1 - h < 1$. Since we focus on weak dropout, it is useful to perform the small- $\delta\rho$ expansion. Writing $\delta\rho \equiv 1 - \rho$ with $\delta\rho \ll 1$, suppose the no-dropout variance fixed point is positive, isolated and stable, with variance-map derivative $|\chi_q| < 1$. The implicit-function theorem then gives $\bar{q}^* = q^* + \mathcal{O}(\delta\rho)$, so to leading order

$$h = \delta\rho \frac{\sigma_w^2}{\bar{q}^*} \int Dz \phi^2 (\sqrt{\bar{q}^*} z) + \mathcal{O}(\delta\rho^2). \quad (16)$$

This variance expansion excludes a marginal variance fixed point. For bias-free ReLU at unchanged He initialization, dropout gives $q_{\ell+1} = q_\ell/\rho$; its normalized correlation map can still be analyzed without a converged variance, as in App. B.2. The backward pass has the same source of asymmetry. Let $\delta_{j,a}^\ell$ be the backpropagated preactivation gradient for input a . With inverted dropout,

$$\delta_{j,a}^\ell = \frac{p_{j,a}^\ell}{\rho} \phi'(z_{j,a}^\ell) \sum_i W_{ij}^{\ell+1} \delta_{i,a}^{\ell+1}. \quad (17)$$

For equal hidden-layer widths, let $G_{ab}^\ell \equiv \mathbb{E}[\delta_{j,a}^\ell \delta_{j,b}^\ell]$ denote the covariance per coordinate. Under the usual wide-limit gradient-independence approximation, the recursion contains

$$G_{ab}^\ell = \sigma_w^2 \mathbb{E} \left[\frac{p_a^\ell p_b^\ell}{\rho^2} \phi'(u_a) \phi'(u_b) \right] G_{ab}^{\ell+1}. \quad (18)$$

Thus independent masks give a diagonal factor $\mathbb{E}[p_a^2]/\rho^2 = 1/\rho$ but an off-diagonal factor $\mathbb{E}[p_a p_b]/\rho^2 = 1$. Dropout inflates single-sample gradient variance without similarly inflating cross-sample covariance, mirroring the forward

asymmetry that produced h . After normalization this suggests a backward field $h_{\nabla}(\rho)$ that moves perfect gradient alignment below one. We do not identify it with the forward field: the backward kernel is built from ϕ' , not ϕ , and for ReLU ϕ' is discontinuous. A full dropout-deformed backward critical theory, including finite-width gradient susceptibilities and training-time mask correlations, is left for future work.

3.1. Landau Theory and Critical Exponents

For smooth activations, the dropout-deformed correlation edge admits a simple Landau-style description in terms of three mean-field variables:

$$m \equiv 1 - c^*, \quad t \equiv \chi_\rho - 1, \quad h \equiv 1 - \bar{F}_\rho(1). \quad (19)$$

The fields (m, t, h) are chosen to mirror statistical-mechanics notation: t plays the role of a reduced temperature, m is the order parameter (magnetization), and h is an external field conjugate to m that explicitly biases the system away from perfect alignment.

We assume inverted dropout with keep probability $\rho \in (0, 1]$, with masks drawn i.i.d. across units and independently across inputs. Let $\bar{q}^*$ denote the dropout variance fixed point (13). For an activation for which Price's theorem applies and $\phi', \phi'' \in L^2(\mathcal{N}(0, \bar{q}^*))$ (for instance, $\phi \in C^2$ with sufficient Gaussian integrability) repeated application of Price's theorem gives the first two derivatives of the correlation map at perfect alignment,⁵

$$\chi_\rho \equiv \bar{F}'_\rho(1) = \sigma_w^2 \int Dz [\phi'(\sqrt{\bar{q}^*}z)]^2, \quad (20)$$

$$g_\rho \equiv \bar{F}''_\rho(1) = \sigma_w^2 \bar{q}^* \int Dz [\phi''(\sqrt{\bar{q}^*}z)]^2. \quad (21)$$

For the $\mathcal{O}(m^3)$ remainder used below, we additionally assume that the third application of Price's theorem is valid and that $\phi''' \in L^2(\mathcal{N}(0, \bar{q}^*))$. We take $\bar{q}^*$ and g_ρ to have finite positive limits near the reference critical point. If $g_\rho = 0$, as for a linear activation, the quadratic Landau term is absent and the leading nonzero term determines a degenerate case rather than the generic smooth universality class. For kinked, ReLU-like activations, $\bar{F}''_\rho(1)$ is not finite and the analytic expansion below is not the correct organizing principle; we treat that universality class separately.

The Landau equation of state follows by expanding $\bar{F}_\rho(c)$ around $c = 1$. Writing $c^* = 1 - m$ with $0 < m \ll 1$ acting as the order parameter,

$$\bar{F}_\rho(1 - m) = \bar{F}_\rho(1) - \chi_\rho m + \frac{g_\rho}{2} m^2 + \mathcal{O}(m^3) \quad (22)$$

$$= (1 - h) - \chi_\rho m + \frac{g_\rho}{2} m^2 + \mathcal{O}(m^3), \quad (23)$$

⁵The ρ -dependence is implicit through $\bar{q}^*$. The full computation is given in App. A.3.

imposing the fixed-point condition $1 - m = \bar{F}_\rho(1 - m)$ yields

$$h + tm - \frac{g_\rho}{2} m^2 = 0, \quad \Longleftrightarrow \quad h = \frac{g_\rho}{2} m^2 - tm, \quad (24)$$

with $t \equiv \chi_\rho - 1$. The physical (nonnegative) solution is

$$m(t, h) = \frac{t + \sqrt{t^2 + 2g_\rho h}}{g_\rho}. \quad (25)$$

At the correlation edge $t = 0$, dropout displaces the fixed point by

$$m(0, h) = \sqrt{\frac{2h}{g_\rho}}. \quad (26)$$

Table 1. Definitions of critical exponents in the thermodynamic limit ($t \rightarrow 0, h \rightarrow 0$).

Exponent	Mathematical Definition
β	$m \sim t ^\beta \quad (h = 0)$
δ	$m \sim h ^{1/\delta} \quad (t = 0)$
γ	$\chi_M \equiv \frac{\partial m}{\partial h} \Big _{h \rightarrow 0} \sim t ^{-\gamma}$
α	$C \equiv -\frac{\partial^2 f}{\partial t^2} \sim t ^{-\alpha}$
ν_t	$\xi \sim t ^{-\nu_t} \quad (h = 0)$
ν_ρ	$\xi \sim h ^{-\nu_\rho} \quad (t = 0)$
θ_{rel}	$m \sim \ell^{-\theta_{\text{rel}}} \quad (t = 0, h = 0)$

For the α column we define the singular Landau potential f_{sing} through the equation of state. In the smooth case,

$$\frac{\partial f_{\text{sing}}}{\partial m} = -h - tm + \frac{g_\rho}{2} m^2, \quad (27)$$

and in the kinked case,

$$\frac{\partial f_{\text{sing}}}{\partial m} = -h - tm + \kappa m^{3/2}. \quad (28)$$

Evaluating f_{sing} on the stable branch at $h = 0$ gives $f_{\text{sing}} \sim |t|^3$ for the smooth class and $f_{\text{sing}} \sim |t|^5$ for the kinked class, hence $\alpha = -1$ and $\alpha = -3$, respectively.

This square-root power law is our first encounter with a larger family of critical exponents familiar in the physics literature, defined in Table 1. The key exponents for our analysis are those governing the depth correlation length $\xi_{c,\rho}$, as it sets the characteristic depth scale over which distinct inputs remain distinguishable and therefore provides an effective upper bound on the depth over which

the network can reliably propagate information (and learn). Their derivations are straightforward but cumbersome, so we collect them in App. B and simply quote the resulting exponent values in Table 2. For reference, we also list the standard mean-field (Landau) exponents of the Ising universality class (Ising, 1925; Landau & Lifshitz, 1980) and the corresponding mean-field exponents of spin-glass theory (Edwards & Anderson, 1975; Parisi, 1979; 1980; 1983) for comparison.

Table 2. Critical exponents for neural network initialization, with statistical-mechanics comparisons. Neural ν exponents describe depth; the comparison rows use spatial correlation length. Their relaxation entries are omitted because a dynamics must be specified separately. The kinked ν_ρ is a normal-form field exponent; ordinary bias-free ReLU dropout follows the constrained path $t = -h$ but has the same leading $\xi \sim h^{-1/3}$ law (App. B).

Universality Class	ν_t	β	θ_{rel}	γ	δ	ν_ρ	α
Smooth activation	1	1	1	1	2	$\frac{1}{2}$	-1
Kinked activation	1	2	2	1	$\frac{3}{2}$	$\frac{1}{3}$	-3
Spin glass (SK MF)	$\frac{1}{2}$	1	—	1	2	$\frac{1}{4}$	-1
Ising-like	$\frac{1}{2}$	$\frac{1}{2}$	—	1	3	$\frac{1}{3}$	0

Figure 1 summarizes these power laws from direct mean-field recursion and shows where higher-order corrections become visible away from the asymptotic regime.

3.2. Scaling Collapse and Crossover Region

At leading normal-form order, dropout and detuning enter the same local equation of state. The collapse below is exact for the truncated Landau normal form, while the full mean-field recursion receives higher-order corrections such as $\mathcal{O}(m^3)$ in the smooth case, $\mathcal{O}(m^2)$ in the kinked case, and corrections from the nonlinear microscopic map $h = \Delta(\rho)$. These corrections become visible away from the asymptotic near-critical regime. From the field-theory point of view, the resulting collapse is the standard signature of criticality: in the massless limit, connected n -point functions become homogeneous scaling functions of their separations, so the data collapse after rescaling by the appropriate critical dimensions (Zinn-Justin, 2002). In practice, experiments are rarely tuned exactly to the correlation edge $t = 0$ while simultaneously taking the dropout field to zero, $h \rightarrow 0$. The useful near-critical description is therefore a two-parameter scaling function.

For smooth activations, the order parameter $m \equiv 1 - c^*$ obeys (24),

$$h = \frac{g_\rho}{2} m^2 - tm, \quad t \equiv \chi_\rho - 1. \quad (29)$$

The explicit crossover solution is the physical branch in (25).

Define rescaled variables

$$\tilde{m} \equiv m \sqrt{\frac{g_\rho}{2h}}, \quad \tilde{t} \equiv -\frac{t}{\sqrt{2g_\rho h}}. \quad (30)$$

Then all curves collapse onto the universal scaling function

$$\tilde{m} = \sqrt{1 + \tilde{t}^2} - \tilde{t}. \quad (31)$$

The crossover scale is

$$|t| \sim \sqrt{g_\rho h}. \quad (32)$$

For $|t| \ll \sqrt{g_\rho h}$, one is in the field-dominated regime

$$m \sim \sqrt{\frac{2h}{g_\rho}}. \quad (33)$$

For $|t| \gg \sqrt{g_\rho h}$, the asymptotics depend on the sign of t :

$$t < 0 : \quad m \simeq \frac{h}{|t|}, \quad (34)$$

$$t > 0 : \quad m \simeq \frac{2t}{g_\rho} + \frac{h}{t} \dots \quad (35)$$

Thus, at either side of the crossover window the system is either in the linear-response regime on the subcritical side ($t < 0$), or it sits on the chaotic fixed-point branch on the supercritical side ($t > 0$).

For kinked activations, the leading non-linearity is non-analytic and the equation of state takes the form (see (101)),

$$h = \kappa m^{3/2} - tm + \mathcal{O}(m^2). \quad (36)$$

This implies the scaling form

$$m(t, h) = \left(\frac{h}{\kappa}\right)^{2/3} \mathcal{F}\left(\frac{t}{\kappa^{2/3} h^{1/3}}\right). \quad (37)$$

Let

$$u \equiv \frac{t}{\kappa^{2/3} h^{1/3}}, \quad (38)$$

and define $y(u)$ as the positive real root of

$$y^3 - uy^2 - 1 = 0 \quad \implies \quad \mathcal{F}(u) = y(u)^2. \quad (39)$$

The associated crossover scale is

$$|t| \sim \kappa^{2/3} h^{1/3}, \quad (40)$$

showing that kinked activations lie in a distinct universality class from smooth activations. For literal bias-free ReLU with standard inverted dropout, the microscopic path locks $t = \rho - 1 = -h$, so the $t = 0, h > 0$ direction is a normal-form field direction rather than a directly independent ReLU knob. This path nevertheless satisfies $t/(\kappa^{2/3} h^{1/3}) \rightarrow 0$,

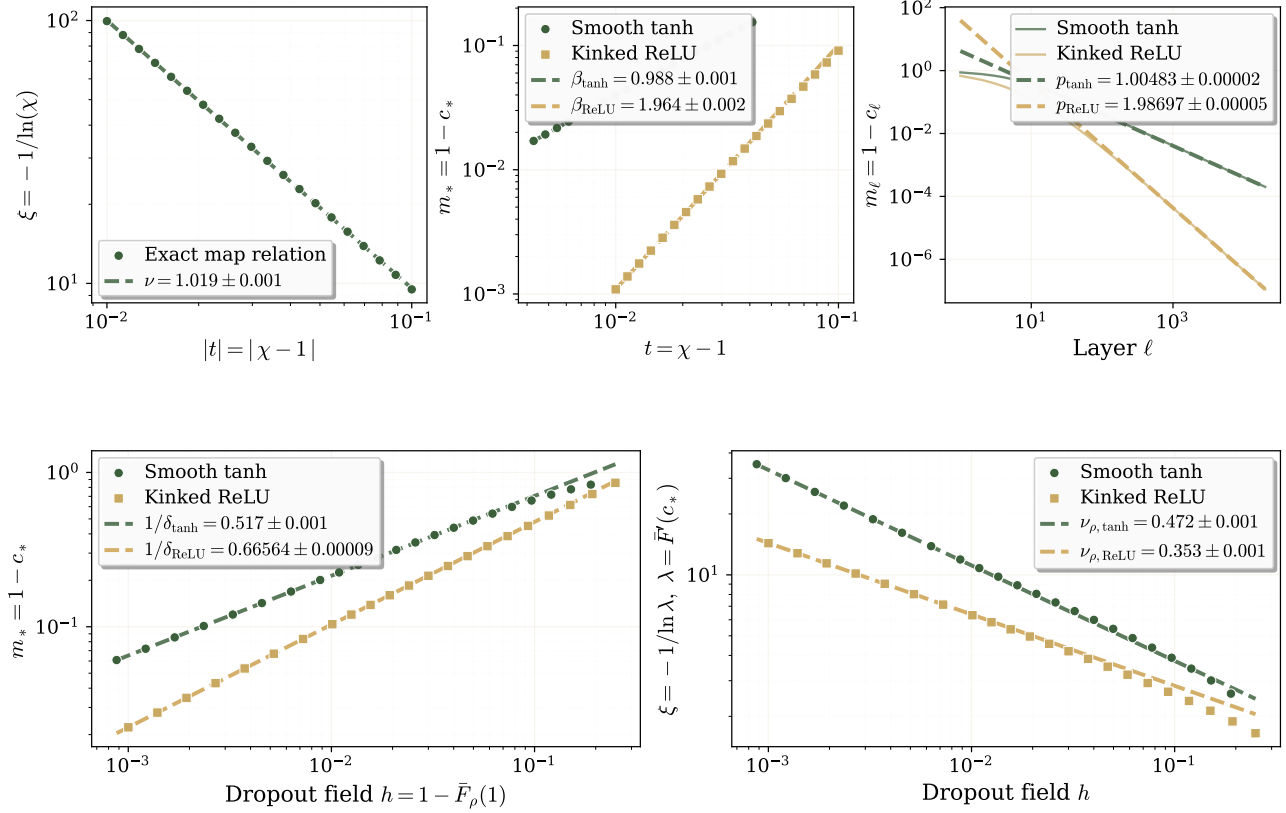


Figure 1. Critical scaling from the mean-field calculation. The top row shows the depth scale $-1/\log \chi$ on the ordered side, fixed-point decorrelation above the edge, and relaxation at criticality. The bottom row varies the dropout field at $\chi = 1$. Dashed lines are log-log fits; quoted errors are regression standard errors on deterministic curves. Finite fitting windows and numerical approximation errors are not included. The tunable ReLU channel is the formal family specified in App. D, rather than an independently tunable finite-variance ReLU initialization.

hence it enters the same field-dominated regime and retains the leading constrained-path law $\xi \sim h^{-1/3}$; App. B gives the short derivation and the nonzero-bias caveat.

Figure 2 shows this collapse for smooth activations. This also clarifies the tanh/ReLU dichotomy observed in ResNet mean-field theory (Yang & Schoenholz, 2017): tanh belongs to the smooth class, where the analytic m^2 term makes weak fields especially important and long correlation lengths are preserved when dropout is kept small; ReLU belongs to the kinked class, where the $m^{3/2}$ branch point broadens the crossover window and makes the dynamics more forgiving to imperfect tuning. In short, smooth channels reward keeping the dropout field very weak, while kinked channels tolerate detuning over a wider range; neither class is uniformly preferable.

3.3. Optimal Dropout Scheduling and Regularization Reach

So far we have taken the keep probability to be constant with depth. Since dropout enters as a relevant field that cuts

off the depth correlation length even at the correlation edge, it is natural to ask how to allocate a fixed dropout budget across layers so as to preserve near-critical propagation as much as possible. Let us introduce a depth-dependent keep probability ρ_ℓ and define the associated dropout field

$$h_\ell \equiv \Delta(\rho_\ell), \quad (41)$$

i.e. the shift of the correlation map at perfect alignment induced by keep probability ρ_ℓ . If variances have settled so that a scalar map applies at each layer, the exact linear response along the trajectory $c_\ell = \bar{F}_{\rho_\ell}(c_{\ell-1})$ is

$$\delta c_L = \left[\prod_{\ell=1}^L \bar{F}'_{\rho_\ell}(c_{\ell-1}) \right] \delta c_0. \quad (42)$$

To obtain a tractable scheduling objective, we approximate each derivative by its local fixed-point value and freeze the near-critical coefficients t and g_ρ at a common reference. This local-stationary approximation neglects variance transients and the lag between the trajectory and changing fixed

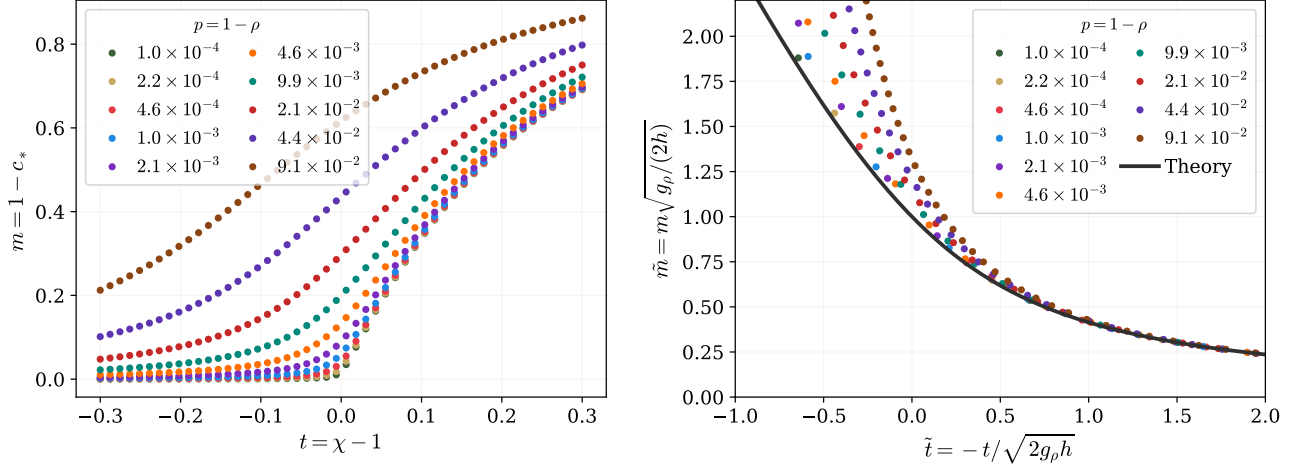


Figure 2. Two-parameter crossover and scaling collapse of the dropout-deformed equation of state for the smooth universality class (tanh). Plots obtained using MFT recursion relations. The curves collapse onto a universal function after rescaling by $\tilde{t}$ and $\tilde{m}$. The kinked counterpart is shown in Fig. 5.

points. In the smooth class it gives (see (86))

$$\lambda(t, h_\ell) \simeq 1 - \sqrt{t^2 + 2g_\rho h_\ell}. \quad (43)$$

For $0 < \lambda_\ell \lesssim 1$, this approximates the propagated perturbation by

$$\delta c_L \sim \prod_{\ell=1}^L \lambda_\ell \delta c_0, \quad \lambda_\ell \equiv \lambda(t, h_\ell), \quad (44)$$

which motivates the following surrogate for the mean decay rate per layer:

$$\xi_{\text{eff}}^{-1} \equiv -\frac{1}{L} \sum_{\ell=1}^L \log \lambda_\ell \simeq \frac{1}{L} \sum_{\ell=1}^L \sqrt{t^2 + 2g_\rho h_\ell}. \quad (45)$$

At the correlation edge $t = 0$, this reduces to

$$\xi_{\text{eff}}^{-1}(t=0) \simeq \sqrt{2g_\rho} \frac{1}{L} \sum_{\ell=1}^L h_\ell^{1/2} \propto \frac{1}{L} \sum_{\ell=1}^L h_\ell^{1/2}. \quad (46)$$

We now fix a total dropout budget and local bounds,

$$\bar{h} \equiv \frac{1}{L} \sum_{\ell=1}^L h_\ell, \quad 0 \leq h_\ell \leq h_{\max}. \quad (47)$$

The budget above is a budget in the dropout field h_ℓ , not directly in the raw dropout probability $p_{\text{drop}, \ell} = 1 - \rho_\ell$. For weak dropout,

$$h_\ell = a p_{\text{drop}, \ell} + \mathcal{O}(p_{\text{drop}, \ell}^2), \quad (48)$$

$$a = \frac{\sigma_w^2}{q^*} \int Dz \phi^2(\sqrt{q^*} z),$$

so a fixed h -budget and a fixed dropout-probability budget agree to leading order. For larger dropout, the nonlinear map $h_\ell = \Delta(\rho_\ell)$ should be used explicitly. Maximizing ξ_{eff} is equivalent to minimizing ξ_{eff}^{-1} , so in the continuum limit ($x = \ell/L$) we obtain the constrained variational problem

$$\min_h \int_0^1 h(x)^{1/2} dx \quad \text{s.t.} \quad \int_0^1 h(x) dx = \bar{h}. \quad (49)$$

Since $h^{1/2}$ is concave, Jensen's inequality gives the maximum at the constant schedule. The chord bound $h^{1/2} \geq h/h_{\max}^{1/2}$ gives the minimum, attained by saturating the box constraint almost everywhere. Equivalently, any schedule that takes values in $\{0, h_{\max}\}$ with active fraction $f = \bar{h}/h_{\max}$ is optimal for the mean-field functional. A front-loaded representative, selected below by regularization reach, is

$$h(x) = \begin{cases} h_{\max}, & 0 \leq x \leq f, \\ 0, & f < x \leq 1, \end{cases} \quad f = \frac{\bar{h}}{h_{\max}}, \quad (50)$$

where $x = 0$ denotes the input side. All permutations of this saturated set give the same ξ_{eff} in the surrogate; we choose an ordering using the regularization-reach heuristic below. For finitely many layers, at most one layer remains unsaturated to meet the exact budget. These conclusions apply to the leading-order objective with weak $h_{\max}$; abrupt schedules need not track their local fixed points, and the full recurrence need not be permutation invariant. For this representative, uniform dropout yields $\xi_{\text{eff}} \sim \bar{h}^{-1/2}$ while the step schedule gives

$$\frac{\xi_{\text{eff}, \text{step}}}{\xi_{\text{eff}, \text{uniform}}} = \sqrt{\frac{h_{\max}}{\bar{h}}} \geq 1. \quad (51)$$

For kinked activations at $t = 0$, one has $\xi^{-1} \propto h^{1/3}$, so the objective becomes $\int_0^1 h(x)^{1/3} dx$, which is also concave and therefore the same conclusion immediately follows.

Since ξ_{eff} in (46) is symmetric in $\{h_\ell\}$, any permutation of a saturated schedule pays the same cost within this approximation. We propose breaking the tie by maximizing downstream exposure to the perturbation (App. A.5). Apply dropout only at layer ℓ . For masked and clean activations $\tilde{y}_k^{(\ell)}$ and y_k at a later layer k , define their angular decorrelation as

$$\eta_k^{(\ell)} \equiv 1 - \frac{\mathbb{E}_p[\langle \tilde{y}_k^{(\ell)}, y_k \rangle]}{\sqrt{\mathbb{E}_p[\|\tilde{y}_k^{(\ell)}\|^2] \|y_k\|^2}}. \quad (52)$$

Thus $\eta_k^{(\ell)} = 0$ means perfect clean-masked alignment. Immediately after the mask, $\eta_\ell^{(\ell)} = 1 - \sqrt{\rho_\ell} = (1 - \rho_\ell)/2 + \mathcal{O}((1 - \rho_\ell)^2)$, while $h_\ell = a(1 - \rho_\ell) + \mathcal{O}((1 - \rho_\ell)^2)$; hence $\eta_\ell^{(\ell)} \propto h_\ell$ to leading order. In an ordered background, linearizing the downstream Gaussian channel gives $\eta_{\ell+r}^{(\ell)} \simeq Ch_\ell e^{-r/\xi_c}$, neglecting variance transients. We use a common background ξ_c when comparing mask locations; at an exactly marginal clean channel, higher-order terms instead govern algebraic relaxation.

The readout perturbation $Ch_\ell e^{-(L-\ell)/\xi_c}$ is larger for later dropout, so readout variance is not the front-loading objective. Our scheduling heuristic uses downstream exposure: how much later computation trains while seeing the mask. A mask inserted early is seen, with decaying strength, by many later blocks; a late mask has less time to decay but fewer blocks left to regularize. We therefore define $\mathcal{D}_\ell$ as the cumulative downstream exposure from placing dropout at layer ℓ , motivating

$$\begin{aligned} \mathcal{D}_\ell &\propto h_\ell \sum_{r=1}^{L-\ell} e^{-r/\xi_c} \\ &\approx h_\ell \xi_c \left(1 - e^{-(L-\ell)/\xi_c}\right) \equiv h_\ell w_\ell. \end{aligned} \quad (53)$$

Under this additive exposure model, $w_\ell = \xi_c(1 - e^{-(L-\ell)/\xi_c})$ decreases with ℓ , so maximizing $\sum_{\ell=1}^L h_\ell w_\ell$ subject to $\sum_{\ell=1}^L h_\ell = L\bar{h}$ and $0 \leq h_\ell \leq h_{\text{max}}$ is a linear program solved by filling the earliest layers first. This is a proposed proxy for regularization; the link from cumulative exposure to generalization remains an empirical question.

We test these schedules with relatively wide MLPs and Vision Transformers (Dosovitskiy et al., 2021) on CIFAR-10/100 (Krizhevsky, 2009) (App. D). The MLP experiments use modest depth and limited augmentation to study dropout placement. The Transformer experiments test whether the same idea transfers to a different architecture; the MLP initialization theory does not establish their training dynamics. The clearest experimental signal is loss: Table 3 shows final cross-entropy reductions of 18–35% in the MLP and

activation-sweep settings, with smaller but consistent reductions in the transformer runs. Keeping the same schedules and averaging each seed’s minimum recorded test loss gives reductions of only 0.9–2.5% across the six comparisons, so the largest differences develop after the loss minimum. This retrospective minimum uses test data to select the epoch; it is separate from the validation-selected checkpoints in the broader benchmark. Accuracy improves as well (e.g., ViT on CIFAR-100: linearly decreasing achieves final accuracy 49.27% vs 48.61% constant), but we treat it mainly as a secondary metric. The appendices check that this is not a single-run effect: the gains persist across $\bar{h}$ -sweeps and large-width sweeps for kinked ReLU MLPs (App. D.4, Fig. 8), across the smooth GELU activation class (App. D.4, Fig. 9), and across transformer schedules and component ablations (Figs. 10 and 12). In these CIFAR experiments, the advantage weakens at high dropout or in narrow networks, consistent with the limitations of the small-field, large-width approximation.

Uncertainty across seeds. Tables 3 and 4 report nominal 95% confidence intervals, pairing runs by their recorded seed or original simulation index. Loss reductions use ratios of seed means, with paired Fieller intervals (Fieller, 1954); accuracy gains use paired Student- t intervals in percentage points. Both use $n - 1$ degrees of freedom and assume independent seed pairs with approximately bivariate normal losses or normal accuracy differences, assumptions that three or five seeds cannot establish reliably. We hold the reported schedules and hyperparameters fixed, without adjusting for their selection or for multiple comparisons, so the intervals describe variation across training seeds on the recorded split. They do not include uncertainty from new data or new splits. In the original experiments, the minimum-loss intervals include zero for the three-seed ReLU sweep and the five-seed ViT ablation; the latter’s accuracy interval also includes zero. The MLP schedule and budget-control rows share their uniform runs and should not be counted as independent replications.

3.4. Front-Loaded Dropout Across Datasets

To test how far the scheduling advantage extends beyond CIFAR, we compare MLPs and Transformers on Speech Commands, Jannis, and Tiny ImageNet, together with a financial MLP on FI-2010. These comparisons span audio, tabular classification, and vision, using the depth-12, width-256 configurations detailed in App. E. Table 4 reports the test loss and accuracy of validation-selected models, alongside the final-epoch endpoints where they were recorded.

For each run, we retain the checkpoint with the lowest validation cross-entropy. We then choose the reported front-loaded profile by averaging these minimum validation losses across the paired seeds, restricting selection to profiles with

Table 3. Original paper experiments. Positive changes favor the named profile over uniform. CE reduction is $100(L_U - L_F)/L_U$ using seed means; accuracy gain is in percentage points. Brackets give nominal paired 95% confidence intervals (Fieller for CE; Student- t for accuracy), conditional on the selected profiles and fixed data split. Profiles minimize mean final test CE among the compared nonuniform schedules and stay fixed across columns. Final CE and accuracy use the last epoch. Minimum CE averages each seed’s lowest recorded test loss; this retrospective endpoint uses test data to choose an epoch, not validation. The first two rows share the same uniform runs. See the main text for the assumptions and selection limits.

Experiment	n	Profile	Final CE reduction (%)	Min. test CE reduction (%)	Final accuracy gain (pp)
CIFAR-10 MLP schedules	25	Step early	+17.86 [17.31, 18.41]	+1.42 [1.19, 1.65]	+0.83 [0.55, 1.11]
CIFAR-10 MLP budget controls	25	Big step	+22.61 [22.06, 23.15]	+2.07 [1.80, 2.33]	+1.08 [0.85, 1.31]
CIFAR-10 ReLU $p=0.1$	3	Big step	+35.43 [33.64, 37.27]	+2.41 [−0.40, 5.21]	+2.04 [0.79, 3.29]
CIFAR-10 GELU $p=0.1$	10	Big step	+29.79 [28.96, 30.61]	+2.45 [1.90, 3.00]	+0.62 [0.16, 1.08]
CIFAR-100 ViT	10	Linear decreasing	+4.15 [3.40, 4.90]	+1.39 [0.56, 2.21]	+0.66 [0.36, 0.96]
CIFAR-10 ViT both-block ablation	5	Step early	+6.32 [3.23, 9.30]	+0.92 [−0.47, 2.30]	+0.52 [−0.11, 1.16]

complete seed coverage; test loss and accuracy enter only the subsequent comparison with uniform dropout. Most cohorts contain five seeds, while the extended Jannis Transformer comparison contains ten. This is a reanalysis of the saved validation histories, with the original hyperparameter searches and data splits held fixed. The selected profile stays the same across the checkpoint and final columns, so a change of endpoint cannot silently change the model being compared.

The clearest improvement occurs for the Speech Commands MLP: early-step dropout reduces checkpoint test loss by 12.71% relative to uniform, with accuracy increasing from 72.23% to 74.06%. Uniform dropout itself reduces loss by 10.82% relative to no dropout, placing this comparison in a regime where regularization already helps. The Transformer gives a smaller, noisier improvement of 4.91% with big-step dropout and an accuracy gain of 1.53 percentage points; both confidence intervals include zero. For the MLP, the loss-reduction interval is [11.01, 14.36]% and the accuracy-gain interval is [1.20, 2.46] percentage points. These audio results use a class-balanced random split of clips from the official training subset, leaving generalization to held-out speakers unresolved.

Jannis provides a tabular comparison with a different learning problem. Validation selects early-step dropout for the MLP, giving a 2.92% reduction in checkpoint test loss and a 1.29-percentage-point accuracy gain. For the standard Transformer cohort, validation selects big step, whose test-loss improvement is only 0.15%. The ten-seed follow-up is effectively a tie: its validation-selected early step increases test loss by 0.27%, despite improving accuracy by 0.39 percentage points. Both Transformer loss intervals include zero, while the MLP interval is [0.49, 5.27]%, leaving the MLP as the clearer signal in this tabular comparison.

On Tiny ImageNet, the selected profiles reduce checkpoint loss by 1.94% and 2.17% for the MLP and ViT at 20,000 training examples, and by 1.26% and 1.01% at 80,000 examples. The larger-data MLP still reaches only 7.95% accuracy, compared with 7.09% for uniform, so its relative gain should be read alongside this weak absolute performance. At the final epoch, the corresponding loss reductions are 8.84% and 1.68%; the difference from the checkpoint comparison makes clear how much of the final-loss gain develops after the validation optimum. For the 80,000-example ViT, both checkpoint and final loss intervals include zero. The FI-2010 MLP gives a 3.57% checkpoint loss reduction with big-step dropout, with interval [2.41, 4.73]%; its 0.67-percentage-point accuracy gain has an interval spanning zero.

These results compare complete training configurations whose learning rates and dropout strengths follow each cohort’s validation protocol; mean dropout can therefore differ between the selected profile and uniform. The fixed-budget CIFAR experiments remain the direct control for allocation. The broader evidence shows a useful but task-dependent advantage, with small effects and losses retained alongside the positive cases. App. E gives the learning curves and all recorded profiles, including the unrestricted nonuniform comparisons in Table 14.

4. Conclusions and Further Scope for Research

By extending mean-field signal propagation to include multiplicative dropout, we established that activation non-analyticity fundamentally dictates universality. This formalism yields a Landau equation of state for the decorrelation order parameter, a universal two-parameter scaling collapse,

Table 4. Frontloaded dropout across tasks. We choose the frontloaded profile with the lowest mean minimum validation CE among complete arms with the same paired seeds as uniform; the profile stays fixed across columns. This reanalyzes saved validation histories, not a fresh blind confirmation. Positive values favor frontloading. CE reduction is $100(L_U - L_F)/L_U$ using seed means; accuracy gain is in percentage points. Brackets give nominal paired 95% confidence intervals (Fieller for CE; Student- t for accuracy); see the main text for assumptions and selection limits. Checkpoint metrics use each run’s minimum-validation-loss checkpoint; final metrics use its last epoch. The extended Jannis Transformer uses weight decay 10^{-7} and fixed mean dropout 0.10; other rows use zero weight decay and allow profile-specific dropout budgets. Learning rate and mean dropout follow each cohort’s validation protocol. Per-epoch test histories were not recorded. Dashes mean missing final test evaluations, not zero change.

Experiment (training N)	n	Profile	Validation-selected test		Final-epoch test	
			CE reduction (%)	Accuracy gain (pp)	CE reduction (%)	Accuracy gain (pp)
Speech Commands (20,020)						
MLP	5	Step early	+12.71 [11.01, 14.36]	+1.83 [1.20, 2.46]	—	—
Speech Commands (20,020)						
Transformer	5	Big step	+4.91 [−9.25, 18.60]	+1.53 [−1.30, 4.36]	—	—
Jannis (3,840)						
MLP	5	Step early	+2.92 [0.49, 5.27]	+1.29 [0.42, 2.16]	—	—
Jannis (3,840)						
Transformer	5	Big step	+0.15 [−1.24, 1.52]	+0.26 [−1.58, 2.11]	—	—
Jannis extended (3,840)						
Transformer	10	Step early	−0.27 [−1.16, 0.61]	+0.39 [−0.31, 1.10]	—	—
Tiny ImageNet (20,000)						
MLP	5	Step early	+1.94 [1.33, 2.55]	+1.40 [1.28, 1.51]	—	—
Tiny ImageNet (20,000)						
Transformer	5	Step early	+2.17 [1.28, 3.05]	+1.63 [0.92, 2.34]	—	—
Tiny ImageNet (80,000)						
MLP	5	Step early	+1.26 [0.93, 1.59]	+0.86 [0.29, 1.44]	+8.84 [8.38, 9.29]	+0.79 [0.42, 1.16]
Tiny ImageNet (80,000)						
Transformer	5	Linear decreasing	+1.01 [−0.34, 2.35]	+0.29 [−0.50, 1.08]	+1.68 [−1.26, 4.54]	+0.11 [−0.88, 1.09]
FI-2010 (40,000)						
MLP	5	Big step	+3.57 [2.41, 4.73]	+0.67 [−0.29, 1.63]	—	—

and distinct critical exponents for smooth versus kinked channels. Mean-field theory is often best at clarifying the mechanisms inside wide networks, and less often at producing directly testable design rules; here it motivates one such rule. Treating dropout as a depth-dependent field gives a local-stationary propagation objective favoring saturated allocation, and an additive regularization-reach heuristic places that allocation near the input. Their use for abrupt schedules and trained networks is tested empirically. These schedules reduce final test loss by 18–35% in MLP settings and by 4–6% in Vision Transformer settings relative to constant dropout at the same budget (Table 3), with secondary accuracy gains and no extra computational cost. Comparing minimum recorded test loss gives smaller reductions of 0.9–2.5% for the same schedules. The broader comparisons select checkpoints and profiles from validation loss and show smaller, task-dependent gains, including losses to uniform dropout in the extended Jannis Transformer cohort. Extending this mean-field analysis to attention mechanisms and incorporating finite-width corrections are natural directions for future work.

5. Limitations

Our analysis is the leading-order, large-width description: finite-width corrections, controlled perturbatively by L/N , are not computed explicitly. They can accumulate with depth; we have not established whether they preserve the leading exponents or how they change a useful scheduling profile. The scheduling objective also freezes local coefficients and replaces trajectory-dependent derivatives by stationary values, while the reach argument assumes additive exposure at a common decay scale. Finite dropout, multiple masks and feature learning can change both approximations.

Architectural scope is also restricted. We treat dropout most explicitly for MLPs; App. A.4 argues that infinite-channel CNNs and residual branches inherit the same smooth/kinked split through the local Gaussian kernel, but we do not derive the full spatial-mode CNN Landau theory, run CNN experiments, or develop a dropout-deformed ResNet theory. Skip connections in particular modify large-depth propagation while preserving the local channel, so promoting residual-branch strength to a control parameter would put the Transformer experiments on firmer footing.

Finally, the theory is an initialization theory of *forward* correlation dynamics. We give the leading backward covariance recursion and note that dropout creates the same diagonal/off-diagonal asymmetry for gradients, but do not develop the full backward critical theory (finite-width gradient susceptibilities, training-time mask correlations, feature-learning effects), nor do we model how observables evolve after feature learning substantially reshapes representations, e.g. in catapult regimes (Zhu et al., 2024). Extending the same approach to training-time regularizers, dropout warm-up, adaptive schedules, L^2 penalties, or broader scaling-law questions (Bahri et al., 2024a) is a natural direction for future work.

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Reproducibility Statement

The repository linked in the introduction contains the original notebooks and saved results, the manuscript source, and the paired seed endpoints and analysis script underlying Tables 3 and 4. App. D reports the original run counts, hyperparameters, dataset splits, hardware, and sweep details. App. E reports the additional benchmark and profile-pilot evidence, including incomplete coverage. The two main gain tables report nominal 95% confidence intervals; existing curve bands and $\pm$ summaries retain their SEM convention. The mean-field fits use deterministic recursion code from the same repository and report fit standard errors separately. The CIFAR-100 archive lacks the configuration needed to verify all reported settings (Table 11).

Impact Statement

The main practical benefit is improved performance at fixed training cost, or potentially reduced training cost when targeting a fixed performance level. We do not anticipate additional ethical risks.

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A. Mean-Field Background and Dropout Recursions

A.1. Notation Guide

The same few letters carry several nearby meanings, so we collect the main conventions here.

Table 5. Notation used across the main text and appendices.

Symbol	Meaning
ρ_ℓ	keep probability in layer ℓ
$p_{\text{drop},\ell} = 1 - \rho_\ell$	raw dropout probability
$p_j^{(\ell)}$	Bernoulli dropout mask variable
$m = 1 - c^*$	decorrelation order parameter at the fixed point
$m_\ell = 1 - c_\ell$	depth-dependent decorrelation used in relaxation laws
$\chi_\perp, \chi_1$	no-dropout angular susceptibility at $c = 1$
χ_ρ	dropout-deformed angular susceptibility $\bar{F}'_\rho(1)$
χ_M	response susceptibility $\partial m / \partial h$
χ, χ_q	generic Hermite/angle susceptibility and variance-channel susceptibility
θ_{rel}	relaxation exponent, $m_\ell \sim \ell^{-\theta_{\text{rel}}}$

A.2. Mean-Field Theory Primer

Even the Standard Model of physics, with only on the order of a few dozen empirical parameters, invites the feeling that some deeper organizing principle is yet to be found. Fermi made the same point more sharply to Dyson by quoting von Neumann: “With four parameters I can fit an elephant, and with five I can make him wiggle his trunk” (Dyson, 2004). Modern machine learning pushes this worry to an extreme: parameters are abundant, so understanding cannot mean tracking each weight individually. A pressing question in modern ML literature is what set of parameters or effective fields can give a useful description of the network’s behavior, and how to derive such a description from the microscopic model.

Mean-field theory gives one such effective description: instead of tracking every neuron and every random weight, one tracks a small number of self-consistent fields. In the signal-propagation version used here, those fields are variances, covariances, and correlations. This is the deep information propagation line of mean-field theory (Poole et al., 2016; Schoenholz et al., 2017), with broader background in statistical-mechanics treatments of deep learning (Bahri et al., 2020; 2024b). The same infinite-width covariance recursion can also be read as the NNGP kernel recursion (Lee et al., 2018); we use the MFT language because our questions concern fixed points, susceptibilities, correlation lengths, and critical exponents rather than the induced function-space prior or training dynamics.

Section 2 gives the main-body version of the signal-propagation story. Here we only fix notation and record the elementary steps used later in the appendices. For a depth- L MLP with activation $\phi : \mathbb{R} \rightarrow \mathbb{R}$, weights and biases are initialized as

$$W_{ij}^l \sim \mathcal{N}\left(0, \frac{\sigma_w^2}{N_{l-1}}\right), \quad b_i^l \sim \mathcal{N}(0, \sigma_b^2), \quad (54)$$

and propagated by

$$z_{i;a}^l = \sum_{j=1}^{N_{l-1}} W_{ij}^l y_{j;a}^{l-1} + b_i^l, \quad y_{i;a}^l = \phi(z_{i;a}^l), \quad (55)$$

where $W^l \in \mathbb{R}^{N_l \times N_{l-1}}$, $y^0 = x$, and a labels the input. We write

$$\int Dz(\cdots) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dz e^{-z^2/2}(\cdots), \quad (56)$$

and use the same convention for multiple Gaussian variables.

The Gaussian closure comes from the fact that each preactivation is a sum of many independently initialized weighted inputs. In the infinite-width limit this central-limit effect becomes exact layer-by-layer, while finite-width corrections are organized perturbatively in the depth-to-width aspect ratio, schematically L/N (Roberts et al., 2022). For $l \geq 2$, the single-input

variance and two-input covariance recursions are the closed equations already quoted in Sec. 2:

$$\begin{aligned}
 q_{aa}^l &= \sigma_w^2 \int Dz \phi^2 \left(\sqrt{q_{aa}^{l-1}} z \right) + \sigma_b^2, \\
 q_{ab}^l &= \sigma_w^2 \int Dz_1 Dz_2 \phi(u_1) \phi(u_2) + \sigma_b^2, \\
 u_1 &= \sqrt{q_{aa}^{l-1}} z_1, \quad c_{ab}^{l-1} = \frac{q_{ab}^{l-1}}{\sqrt{q_{aa}^{l-1} q_{bb}^{l-1}}}, \\
 u_2 &= \sqrt{q_{bb}^{l-1}} \left(c_{ab}^{l-1} z_1 + \sqrt{1 - (c_{ab}^{l-1})^2} z_2 \right).
 \end{aligned} \tag{57}$$

Once the variance relaxes to q^* , these equations induce a one-dimensional correlation map $c_{ab}^l = F(c_{ab}^{l-1})$. Without dropout, perfect alignment is a fixed point, $F(1) = 1$. Its linear stability is controlled by the angular susceptibility

$$\chi_1 = \left. \frac{\partial c_{ab}^l}{\partial c_{ab}^{l-1}} \right|_{c_{ab}=1} = \sigma_w^2 \int Dz [\phi'(\sqrt{q^*} z)]^2, \tag{58}$$

called $\chi_\perp$ in the main text and in (Roberts et al., 2022). The ordered, chaotic, and critical regimes correspond to $\chi_1 < 1$, $\chi_1 > 1$, and $\chi_1 = 1$ respectively; the last case is the edge-of-chaos, where the cross-input correlation length can diverge.

The usual He initialization follows from this criterion in the special case of ReLU. Since $\phi'(z) = \theta(z)$ almost everywhere,

$$\int Dz (\phi'(\sqrt{q^*} z))^2 = \frac{1}{2}, \quad \chi_1 = 1 \implies \sigma_w^2 = 2, \tag{59}$$

which is the variance-2/ N initialization. The same example also shows why variance preservation and correlation criticality are not identical: for ReLU,

$$q_{aa}^l = \frac{\sigma_w^2}{2} q_{aa}^{l-1} + \sigma_b^2, \tag{60}$$

so with nonzero bias the finite- q^* condition requires $\sigma_w^2 < 2$, while exact angular criticality sets $\sigma_w^2 = 2$. In practice one either sets biases to zero or works slightly subcritical to keep the variance scale finite.

Finally, the correlation length used throughout the paper is obtained by linearizing the correlation map around its fixed point. If $r_c = F'(c^*)$, then

$$|c^* - c_{ab}^l| \propto e^{-l/\xi_c}, \quad \xi_c^{-1} = -\log |r_c|. \tag{61}$$

The analogous variance length ξ_q tracks relaxation of q_{aa}^l to q^* , but the edge-of-chaos phenomenon is primarily controlled by ξ_c : it measures how many layers preserve distinctions between different inputs. This is the scale deformed by dropout in the main analysis.

A.3. Derivation of Correlation Map and Derivatives with Dropout

This subsection supplies the steps leading to Eqs. (20) and (21), starting from the two-input correlation recursion that underlies the perfect-alignment shift $\bar{F}_\rho(1) = 1 - \Delta$ in (15).

Fix a layer where the single-input variance has converged to the dropout-shifted fixed point $\bar{q}^*$ defined by (13). For two inputs, let (u_1, u_2) be jointly Gaussian with

$$\mathbb{E}[u_1^2] = \mathbb{E}[u_2^2] = \bar{q}^*, \quad \mathbb{E}[u_1 u_2] = c \bar{q}^*, \quad c \in [-1, 1]. \tag{62}$$

The normalized correlation map is

$$\bar{F}_\rho(c) \equiv \frac{\bar{q}_{ab}^{l+1}}{\bar{q}^*} = \frac{\sigma_w^2 \mathbb{E}[\phi(u_1) \phi(u_2)] + \sigma_b^2}{\bar{q}^*}. \tag{63}$$

At $c = 1$, $u_1 = u_2$ and $\bar{F}_\rho(1) = 1 - \Delta$ by definition, giving (15).

Since $\bar{q}^*$ is independent of c , differentiating (63) yields

$$\bar{F}'_\rho(c) = \frac{\sigma_w^2}{\bar{q}^*} \frac{\partial}{\partial c} \mathbb{E}[\phi(u_1)\phi(u_2)]. \quad (64)$$

Write $u_i = \sqrt{\bar{q}^*}v_i$, where (v_1, v_2) are standard Gaussians with $\mathbb{E}[v_1v_2] = c$, and define

$$G(c) \equiv \mathbb{E}[\phi(\sqrt{\bar{q}^*}v_1)\phi(\sqrt{\bar{q}^*}v_2)]. \quad (65)$$

Price's theorem (Price, 1958) gives

$$\frac{d}{dc}G(c) = \mathbb{E}\left[\frac{\partial^2}{\partial v_1\partial v_2}\left(\phi(\sqrt{\bar{q}^*}v_1)\phi(\sqrt{\bar{q}^*}v_2)\right)\right]. \quad (66)$$

A direct differentiation yields

$$\frac{\partial^2}{\partial v_1\partial v_2}\left(\phi(\sqrt{\bar{q}^*}v_1)\phi(\sqrt{\bar{q}^*}v_2)\right) = \bar{q}^*\phi'(\sqrt{\bar{q}^*}v_1)\phi'(\sqrt{\bar{q}^*}v_2). \quad (67)$$

Therefore,

$$G'(c) = \bar{q}^*\mathbb{E}[\phi'(\sqrt{\bar{q}^*}v_1)\phi'(\sqrt{\bar{q}^*}v_2)]. \quad (68)$$

Taking $c \rightarrow 1$ collapses $v_1 = v_2 = z$ with $z \sim \mathcal{N}(0, 1)$, hence

$$G'(1) = \bar{q}^* \int Dz [\phi'(\sqrt{\bar{q}^*}z)]^2. \quad (69)$$

Substituting into (64) gives

$$\bar{F}'_\rho(1) = \sigma_w^2 \int Dz [\phi'(\sqrt{\bar{q}^*}z)]^2, \quad (70)$$

which is Eq. (20). Differentiating (64) once more:

$$\bar{F}''_\rho(c) = \frac{\sigma_w^2}{\bar{q}^*} \frac{\partial^2}{\partial c^2} \mathbb{E}[\phi(u_1)\phi(u_2)]. \quad (71)$$

Applying Price's theorem again to $f(v_1, v_2) = \phi'(\sqrt{\bar{q}^*}v_1)\phi'(\sqrt{\bar{q}^*}v_2)$ yields

$$\frac{d}{dc}\mathbb{E}[\phi'(\sqrt{\bar{q}^*}v_1)\phi'(\sqrt{\bar{q}^*}v_2)] = \bar{q}^*\mathbb{E}[\phi''(\sqrt{\bar{q}^*}v_1)\phi''(\sqrt{\bar{q}^*}v_2)]. \quad (72)$$

Combining with the previous step gives

$$\frac{\partial^2}{\partial c^2}\mathbb{E}[\phi(u_1)\phi(u_2)] = (\bar{q}^*)^2\mathbb{E}[\phi''(\sqrt{\bar{q}^*}v_1)\phi''(\sqrt{\bar{q}^*}v_2)]. \quad (73)$$

Evaluating at $c = 1$ again collapses $v_1 = v_2 = z$, so

$$\frac{\partial^2}{\partial c^2}\mathbb{E}[\phi(u_1)\phi(u_2)]\Big|_{c=1} = (\bar{q}^*)^2 \int Dz [\phi''(\sqrt{\bar{q}^*}z)]^2. \quad (74)$$

Substituting into (71) yields

$$\bar{F}''_\rho(1) = \sigma_w^2 \bar{q}^* \int Dz [\phi''(\sqrt{\bar{q}^*}z)]^2, \quad (75)$$

which is Eq. (21).

A.4. Extensions to Infinite-Channel CNNs and ResNets

We briefly discuss extensions to architectures not treated in the main analysis. Although our derivation is for MLPs, the universality-class mechanism is tied to the local Gaussian kernel rather than to the MLP architecture itself. Other architectures introduce additional structure that can change the exact exponents, but the qualitative smooth-versus-kinked distinction should remain when the same local channel controls the wide-limit recursion.

For example, the same local channel which appears in MLPs re-emerges in infinite-width CNNs when the number of channels is taken to infinity: preactivations at different spatial positions become jointly Gaussian across channels, and the covariance recursion composes the same bivariate Gaussian activation expectation with a linear operator that averages over convolutional filter offsets (LeCun et al., 1998; Xiao et al., 2018). In schematic form, a spatial covariance $K_{\alpha\beta}^\ell(x, x')$ evolves as $K^{\ell+1} = \sigma_b^2 + \sigma_w^2 \mathcal{A}[V_\phi(K^\ell)]$, where V_ϕ is the scalar Gaussian channel analyzed above and $\mathcal{A}$ is the convolutional averaging operator. Independent dropout shifts the single-input variance entering V_ϕ as in the fully connected case, while the cross-input covariance is still controlled by the same bivariate expectation. Therefore, for an isolated leading spatial mode, convolution changes the relevant eigenvalue and mode shape but not the local smooth-versus-kinked non-analyticity: smooth activations retain the analytic Landau equation, while ReLU-like kinks retain the $m^{3/2}$ branch point. A complete CNN theory would promote m to a spatial covariance-mode amplitude and track boundary conditions, pooling, and degeneracies of $\mathcal{A}$.

ResNets modify depth dynamics in a complementary way: skip connections can change exponential convergence to subexponential or polynomial behavior, and norms may drift rather than relax to a fixed point (Yang & Schoenholz, 2017). Still, the residual branch evaluates the same nonlinear Gaussian channel V_ϕ , so the local analytic distinction is unchanged. The tanh/ReLU asymptotic split observed in (Yang & Schoenholz, 2017) is consistent with our smooth/kinked classes; a dropout-deformed ResNet theory would additionally track residual-branch strength and norm drift.

A.5. Regularization-Reach Derivation

The local-stationary propagation surrogate is invariant under permutations of the depth profile h_ℓ , so it fixes a saturated allocation while leaving its ordering open. We propose choosing that ordering by asking how much downstream computation sees a dropout perturbation before it is absorbed, using a common background decay scale and an additive exposure model.

First apply dropout only at layer ℓ . Write $\tilde{y}_k^{(\ell)}$ for the masked trajectory and y_k for the clean trajectory. We measure the size of the perturbation at layer k by the clean-vs.-masked angular decorrelation

$$\eta_k^{(\ell)} \equiv 1 - \frac{\mathbb{E}_p[\langle \tilde{y}_k^{(\ell)}, y_k \rangle]}{\sqrt{\mathbb{E}_p[\|\tilde{y}_k^{(\ell)}\|^2] \mathbb{E}_p[\|y_k\|^2]}}. \quad (76)$$

Here $\eta_k^{(\ell)} = 0$ means the two trajectories are perfectly aligned at layer k . Immediately after the mask,

$$\eta_\ell^{(\ell)} = 1 - \sqrt{\rho_\ell} = \frac{1 - \rho_\ell}{2} + \mathcal{O}((1 - \rho_\ell)^2). \quad (77)$$

The dropout field used in the main text is the same small parameter up to a constant:

$$h_\ell = a(1 - \rho_\ell) + \mathcal{O}((1 - \rho_\ell)^2), \quad a = \frac{\sigma_w^2}{q^*} \int Dz \phi^2(\sqrt{q^*}z), \quad (78)$$

so $\eta_\ell^{(\ell)} \propto h_\ell$ to leading order. This constant of proportionality does not affect the scheduling optimizer.

Next, we propagate this perturbation downstream. For $k > \ell$, the clean-vs.-masked pair is governed by the same wide-limit Gaussian correlation channel as a two-input pair, up to variance-relaxation corrections. Linearizing around an ordered background fixed point gives

$$\eta_{\ell+r}^{(\ell)} \simeq Ch_\ell e^{-r/\xi_c}, \quad (79)$$

where r is the number of downstream layers and C is independent of h_ℓ to leading order. Here ξ_c is the background correlation length, held fixed across mask locations. At an exactly critical clean channel the linear multiplier is one, and higher-order terms produce algebraic relaxation. If we only looked at the final readout, we would get $Ch_\ell e^{-(L-\ell)/\xi_c}$. This

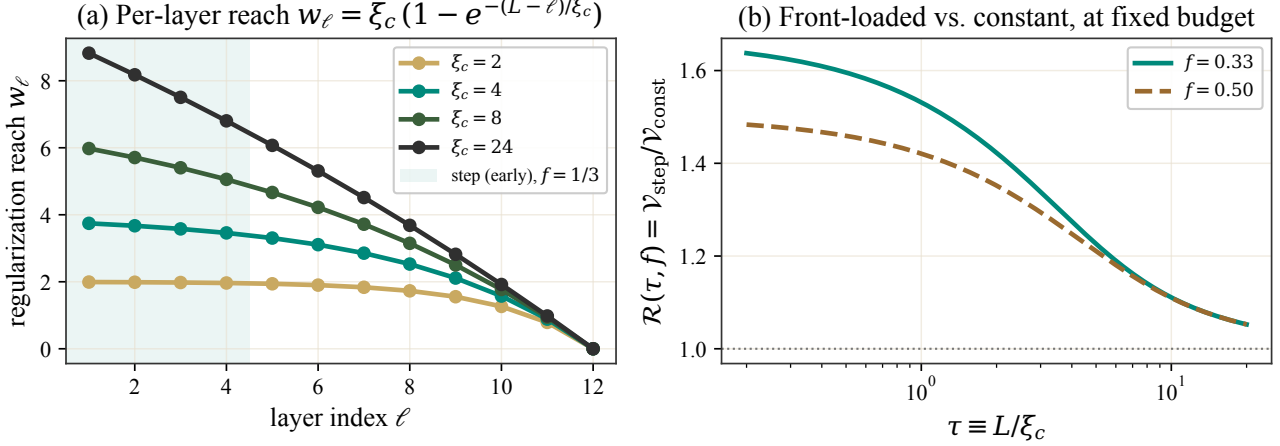


Figure 3. Regularization-reach weights at fixed network depth $L = 12$. (a) Per-layer weight $w_\ell = \xi_c(1 - e^{-(L-\ell)/\xi_c})$ is monotone decreasing in ℓ for any ξ_c ; the linear program over the dropout budget therefore fills the smallest- ℓ layers first (shaded: step-early with $f = 1/3$). For $L \ll \xi_c$ the reach is essentially linear in $L - \ell$; for $L \gg \xi_c$ it saturates at ξ_c . (b) Ratio $\mathcal{R}(\tau, f) = \mathcal{V}_{\text{step}}/\mathcal{V}_{\text{const}}$ of front-loaded to constant schedules at fixed budget, with $\tau \equiv L/\xi_c$. Here $\mathcal{V}$ denotes summed exposure at a common background ξ_c . In this continuum comparison the ratio approaches $2 - f$ as $\tau \rightarrow 0$ and one as $\tau \rightarrow \infty$.

is larger for later dropout, because late perturbations have fewer layers over which to decay. That is why readout variance is not the object that explains front-loading.

For scheduling, we use the amount of downstream computation exposed to the mask as a proxy for regularization. We assume that a later layer's exposure is proportional to the remaining mask-induced decorrelation and add these exposures across layers. This assumption does not establish a corresponding improvement in generalization, and interactions between several masks or changes during training may alter it. We denote this cumulative exposure by $\mathcal{D}_\ell$. It is not additional dropout strength; it is the downstream sum of the remaining perturbations, proportional to $\sum_{r=1}^{L-\ell} \eta_{\ell+r}^{(\ell)}$.

Substituting the exponential decay into this exposure sum gives

$$\mathcal{D}_\ell \propto h_\ell \sum_{r=1}^{L-\ell} e^{-r/\xi_c} \approx h_\ell \int_0^{L-\ell} e^{-s/\xi_c} ds = h_\ell \xi_c \left(1 - e^{-(L-\ell)/\xi_c}\right) \equiv h_\ell w_\ell. \quad (80)$$

The approximation replaces the discrete sum by its continuum integral; using the exact geometric sum changes only constants and leaves the same monotone ordering. The weight $w_\ell = \xi_c(1 - e^{-(L-\ell)/\xi_c})$ decreases with ℓ : earlier masks are visible to more downstream layers before they are absorbed (Fig. 3).

Within the propagation surrogate, every permutation of the same saturated dropout set pays the same cost. We choose the permutation that gives the largest downstream exposure under the preceding assumptions:

$$\max_{\{h_\ell\}} \sum_{\ell=1}^L h_\ell w_\ell \quad \text{s.t.} \quad \sum_{\ell=1}^L h_\ell = L\bar{h}, \quad 0 \leq h_\ell \leq h_{\max}. \quad (81)$$

This is a linear program. Since $w_1 > w_2 > \dots > w_L$, it is solved by filling the largest weights first:

$$h_\ell^* = \begin{cases} h_{\max}, & 1 \leq \ell \leq k, \\ r, & \ell = k+1 \leq L, \\ 0, & \ell > k+1, \end{cases} \quad k = \left\lfloor \frac{L\bar{h}}{h_{\max}} \right\rfloor, \quad r = L\bar{h} - kh_{\max}. \quad (82)$$

The residual r enforces the budget when $L\bar{h}/h_{\max}$ is not an integer. This exposure model selects a front-loaded step. It predicts that sliding a fixed saturated block B toward later layers reduces $\sum_{\ell \in B} w_\ell$; whether performance follows that ordering is an empirical test of the heuristic.

B. Critical Exponents and Non-Analyticity

The Hermite decomposition in App. C.3 diagnoses which expansion is valid here. In the Gaussian channel,

$$F(c) = \frac{\sigma_w^2 \sum_{n \geq 0} a_n^2 c^n + \sigma_b^2}{q^*}. \quad (83)$$

The smooth-class derivative moments make this series differentiable to the order needed at $c = 1$, giving the finite-order Taylor/Landau equation below. This does not require analyticity or convergence of an infinite Taylor series. A kink produces a power-law Hermite tail, so high-degree modes contribute collectively as $c \rightarrow 1$; the Taylor expansion is replaced by the branch-point term $m^{3/2}$, which is the source of the kinked exponents.

B.1. Smooth Networks

The standard static exponents follow directly from (24). Along the critical isotherm $t = 0$,

$$h = \frac{g_\rho}{2} m^2 \quad \Rightarrow \quad m \sim h^{1/2}, \quad (84)$$

so $\delta = 2$. The analogue of the magnetic susceptibility is $\chi_M(t) \equiv \partial m / \partial h|_{h \rightarrow 0}$. On the subcritical side $t < 0$, the analytic branch satisfies $m \simeq -h/t$, hence $\chi_M(t) \sim |t|^{-1}$ and $\gamma = 1$. Along $h = 0$, the physical branch is $m = 0$ for $t \leq 0$ and $m = 2t/g_\rho$ for $t \geq 0$, so $m \sim t$ as $t \rightarrow 0^+$ and $\beta = 1$.

The depth correlation length follows from linearizing the correlation dynamics about the displaced fixed point. To leading order,

$$\lambda \equiv \bar{F}'_\rho(c^*) = \bar{F}'_\rho(1 - m) \simeq \chi_\rho - g_\rho m = 1 + t - g_\rho m, \quad (85)$$

and substituting (25) gives

$$\lambda(t, h) = 1 - \sqrt{t^2 + 2g_\rho h}. \quad (86)$$

For $\lambda \simeq 1$ with $\lambda > 0$, the depth correlation length satisfies

$$\xi_{c,\rho}^{-1} \equiv -\log \lambda \simeq 1 - \lambda = \sqrt{t^2 + 2g_\rho h}. \quad (87)$$

Thus $\xi_{c,\rho}(t, 0) \sim |t|^{-1}$ and $\nu_t = 1$, while $\xi_{c,\rho}(0, h) \sim h^{-1/2}$ and $\nu_\rho = 1/2$. Equivalently, $y_t = 1/\nu_t = 1$ and $y_\rho = 1/\nu_\rho = 2$, so dropout is the more relevant perturbation of the correlation edge.

At the marginal point $t = h = 0$, the same smooth normal form gives the depth relaxation law:

$$m_{\ell+1} - m_\ell \simeq -\frac{g_\rho}{2} m_\ell^2. \quad (88)$$

In a continuum-depth approximation this gives $m(\ell) \sim \ell^{-1}$, hence $\theta_{\text{rel}} = 1$ for the smooth class.

A pseudo-free-energy functional consistent with (24) is

$$f(t, m; h) = -\frac{t}{2} m^2 + \frac{g_\rho}{6} m^3 - h m, \quad (89)$$

since $\partial f / \partial m = 0$ is equivalent to (24). This object should be read in the same spirit as a low-order one-particle-irreducible (1PI) effective action in the sense that it is built so its stationarity condition gives the equation of state, while its on-shell singular part organizes the thermodynamic-style exponents; analytic m -independent backgrounds can be added without changing the physics of the recursion.

At $h = 0$, substituting the physical branch $m = 0$ for $t \leq 0$ and $m = 2t/g_\rho$ for $t \geq 0$ yields

$$f_{\text{on}}(t, 0) = 0 \quad (t \leq 0), \quad f_{\text{on}}(t, 0) = -\frac{2}{3g_\rho^2} t^3 \quad (t \geq 0). \quad (90)$$

Therefore the analogue of the specific heat, $C \equiv -\partial_t^2 f_{\text{on}}$, stays finite and vanishes linearly:

$$C(t) = \frac{4}{g_\rho^2} t \Theta(t) \sim |t|, \quad (91)$$

where $\Theta(t)$ is the Heaviside step function. In standard exponent language $C_{\text{sing}}(t) \sim |t|^{-\alpha}$, this corresponds to

$$\alpha = -1, \quad (92)$$

up to an additive analytic background $f_0(t, h)$ that does not affect the equation of state.

Collecting exponents for the smooth universality class:

$$\beta = 1, \quad \gamma = 1, \quad \delta = 2, \quad \nu_t = 1, \quad \nu_\rho = \frac{1}{2}, \quad \theta_{\text{rel}} = 1, \quad \alpha = -1. \quad (93)$$

For comparison, standard Ising-like Landau mean-field theory (with $m \rightarrow -m$ symmetry and a quartic interaction) has

$$\beta = \frac{1}{2}, \quad \gamma = 1, \quad \delta = 3, \quad \nu = \frac{1}{2}, \quad \alpha = 0. \quad (94)$$

Our dropout-deformed correlation-edge theory shares $\gamma = 1$, but the lack of a $\mathbb{Z}_2$ symmetry (and the resulting cubic interaction) shifts β and δ , produces $\nu_t = 1$ rather than $\nu = 1/2$, and yields a specific-heat analogue that vanishes at the transition ($\alpha = -1$) rather than exhibiting the Ising mean-field behavior ($\alpha = 0$).

B.2. Kinked Networks

Up to now, the Landau-style expansion of $\bar{F}_\rho(c)$ around $c = 1$ relied on finite derivatives through the required order in

$$m \equiv 1 - c. \quad (95)$$

Under the smooth-class Gaussian integrability assumptions, the first nonlinear correction is of order m^2 , controlled by

$$g_\rho \propto \int Dz [\phi''(\sqrt{q^*}z)]^2. \quad (96)$$

For kinked activations, this logic fails: the map remains smooth for $c < 1$, but as it approaches $c \rightarrow 1$, it is governed by a branch point, and the leading nonlinear correction is non-analytic in m . This changes the scaling of the dropout deformation.

We illustrate this explicitly for the ReLU activation $\phi(x) = \max(0, x)$. To keep formulas compact we set $\sigma_b = 0$ throughout this subsection and use the same inverted-dropout convention as before, with masks drawn independently across inputs.

The mean-field correlation map for ReLU admits the standard arc-cosine closed form (Cho & Saul, 2009). Writing $c \in [-1, 1]$ for the preactivation correlation at some depth,

$$F_{\text{ReLU}}(c) = \frac{1}{\pi} \left[\sqrt{1 - c^2} + (\pi - \cos^{-1} c) c \right]. \quad (97)$$

Near perfect alignment, setting $c = 1 - m$ with $0 < m \ll 1$, one obtains the non-analytic expansion

$$F_{\text{ReLU}}(1 - m) = 1 - m + \kappa m^{3/2} + \mathcal{O}(m^2), \quad (98)$$

$$\kappa \equiv \frac{2\sqrt{2}}{3\pi}.$$

Thus, the Taylor expansion valid for smooth activations would have missed the $m^{3/2}$ contribution.

We now turn on dropout and again view dropout as a field deformation at $c = 1$:

$$\bar{F}_\rho(1) = 1 - h, \quad h > 0. \quad (99)$$

In the present $\sigma_b = 0$ setting, this field is exactly

$$h = 1 - \bar{F}_\rho(1) = 1 - \rho, \quad (100)$$

independent of the activation, since at $c = 1$ the only source of mismatch is the independence of the two dropout masks. We write the fixed point as $c^* = 1 - m^*$ with $m^* > 0$ and impose $c^* = \bar{F}_\rho(c^*)$. For the literal bias-free ReLU map,

$\bar{F}_\rho(c) = \rho F_{\text{ReLU}}(c)$, so $\chi_\rho = \rho$ and $t = \rho - 1 = -h$. Thus the $t = 0, h > 0$ direction below is the kinked normal-form field direction, while standard ReLU dropout follows a constrained path through the same scaling function.

Near $m^* \ll 1$, the kinked analogue of the Landau equation of state is obtained by combining the shift $\bar{F}_\rho(1) = 1 - h$, the linear slope $\chi_\rho = 1 + t$, and the non-analytic term (98):

$$h = \kappa m^{3/2} - tm + \mathcal{O}(m^2), \quad t \equiv \chi_\rho - 1. \quad (101)$$

Along the normal-form critical isotherm $t = 0$, this gives

$$m^* \sim \left(\frac{h}{\kappa}\right)^{2/3}, \quad (t = 0), \quad (102)$$

so the smooth exponent $\delta = 2$ is replaced by

$$m \propto h^{1/\delta_{\text{kink}}}, \quad \delta_{\text{kink}} = \frac{3}{2}. \quad (103)$$

And with this we immediately find the smoking gun telling us that kinked and smooth activations fall into distinct mean-field universality classes under the dropout deformation. As before, the depth scale follows from linearizing the correlation dynamics around c^* . Differentiating (98) gives

$$\lambda \equiv \bar{F}'_\rho(c^*) \simeq 1 + t - \frac{3\kappa}{2}\sqrt{m^*}. \quad (104)$$

At $t = 0$ we therefore have

$$1 - \lambda \propto \sqrt{m^*} \propto h^{1/3}. \quad (105)$$

Using $\xi_{c,\rho}^{-1} \simeq 1 - \lambda$ for $\lambda \simeq 1$ yields

$$\xi_{c,\rho}(0, h) \sim h^{-1/3}, \quad \nu_{\rho,\text{kink}} = \frac{1}{3}, \quad (106)$$

to be compared with the smooth result $\xi_{c,\rho}(0, h) \sim h^{-1/2}$. For ordinary bias-free ReLU dropout, substituting the constrained path $t = -h$ into (101) gives

$$h = \kappa m^{3/2} + hm + \mathcal{O}(m^2). \quad (107)$$

Since the leading solution has $m \sim h^{2/3}$, the hm term is subleading. Moreover,

$$1 - \lambda \simeq h + \frac{3\kappa}{2}\sqrt{m} \sim h^{1/3}, \quad (108)$$

so the same $\nu_{\rho,\text{kink}} = 1/3$ survives as a constrained-path exponent.

The remaining kinked exponents follow from the same equation of state. At $h = 0$ and $t > 0$,

$$\kappa m^{3/2} = tm \quad \Rightarrow \quad m = \left(\frac{t}{\kappa}\right)^2, \quad (109)$$

so $\beta_{\text{kink}} = 2$. On the subcritical side $t < 0$, the small-field balance is $h \simeq |t|m$, giving $\chi_M \sim |t|^{-1}$ and therefore $\gamma_{\text{kink}} = 1$. Substituting the zero-field branch into (104) gives $1 - \lambda \propto t$, so $\xi_{c,\rho}(t, 0) \sim t^{-1}$ and $\nu_{t,\text{kink}} = 1$.

Adding nonzero ReLU bias variance does not restore a finite- q^* critical isotherm. With inverted dropout,

$$\bar{q}_{\ell+1} = \frac{\sigma_w^2}{2\rho} \bar{q}_\ell + \sigma_b^2, \quad (110)$$

so finite variance requires $a \equiv \sigma_w^2/(2\rho) < 1$, while angular criticality requires $\chi_\rho = \sigma_w^2/2 = 1$, incompatible with $\rho < 1$. Equivalently, $h = a(1 - \rho)$ and $t = \rho a - 1 = -h - (1 - a)$, so finite variance adds a subcritical detuning that cuts off the asymptotic dropout law unless one takes the singular double scaling $1 - a \ll h^{1/3}$.

It is also useful, in direct analogy with the smooth case, to package (101) into a pseudo-free-energy. A functional whose stationarity condition reproduces the kinked equation of state is

$$f_{\text{kink}}(t, m; h) = -\frac{t}{2}m^2 + \frac{2\kappa}{5}m^{5/2} - hm, \quad (111)$$

since $\partial f_{\text{kink}}/\partial m = 0$ gives $h = \kappa m^{3/2} - tm$ at leading order. In the effective-action language, the $m^{5/2}$ term is the pseudo-free-energy avatar of the branch point in the ReLU correlation map; this is precisely where the kinked universality class departs from the analytic Landau polynomial above.⁶ At zero field the physical branch is $m = 0$ for $t \leq 0$ and $m = (t/\kappa)^2$ for $t \geq 0$, so the on-shell free energy behaves as

$$f_{\text{kink,on}}(t, 0) = 0 \quad (t \leq 0), \quad f_{\text{kink,on}}(t, 0) = -\frac{1}{10\kappa^4}t^5 \quad (t \geq 0). \quad (112)$$

Thus the analogue of the specific heat, $C \equiv -\partial_t^2 f_{\text{kink,on}}$, stays finite and in fact vanishes as

$$C(t) = \frac{2}{\kappa^4}t^3\Theta(t) \sim |t|^3. \quad (113)$$

In the standard exponent convention $C_{\text{sing}}(t) \sim |t|^{-\alpha}$, this corresponds to

$$\alpha_{\text{kink}} = -3. \quad (114)$$

Although we used ReLU to compute the near-alignment expansion in closed form, the resulting fractional power is not a peculiarity of ReLU. Rather, it is a generic feature of the mean-field Gaussian channel whenever the activation has an ordinary kink: ϕ is continuous and piecewise C^2 , with a finite jump in slope and a locally bounded, Gaussian-integrable regular second derivative. In that case the correlated-Gaussian expectation defining the normalized correlation map remains smooth for $c < 1$, but its approach to $c \rightarrow 1$ is controlled by a square-root branch associated with the thin Gaussian tube around the diagonal $u_1 = u_2$. Writing $c = 1 - m$ with $0 < m \ll 1$, the transverse fluctuations scale as $\sqrt{m}$, so the probability that two nearly aligned preactivations fall on opposite sides of the kink scales as $\mathcal{O}(\sqrt{m})$; in that straddling region a finite-slope kink produces activation mismatches of size $\mathcal{O}(\sqrt{m})$, so the leading non-linear correction to the kernel scales as $\mathcal{O}(\sqrt{m}) \times \mathcal{O}(m) = \mathcal{O}(m^{3/2})$. Thus, at the marginal zero-field point and under these regularity assumptions, one obtains

$$F(1 - m) = 1 - m + \kappa_\phi m^{3/2} + \mathcal{O}(m^2), \quad (115)$$

with an activation-dependent coefficient κ_ϕ , while the exponent $3/2$ is fixed by the tube geometry.

Finally, it is useful to recall what happens when dropout is switched off exactly at the marginal point. Setting

$$\rho = 1, \quad t = 0, \quad (116)$$

the non-analytic correction controls the approach to $c = 1$. Writing $m_\ell \equiv 1 - c_\ell$ and using (98),

$$m_{\ell+1} - m_\ell \simeq -\kappa m_\ell^{3/2}. \quad (117)$$

In a continuum-depth approximation $m_{\ell+1} - m_\ell \rightarrow \partial_\ell m$, we obtain

$$\partial_\ell m \simeq -\kappa m^{3/2} \quad \Rightarrow \quad m(\ell) \sim \ell^{-2}, \quad (118)$$

which reproduces the characteristic $1 - c_\ell \sim \ell^{-2}$ relaxation of ReLU activations at the edge-of-chaos (Hayou et al., 2019) and gives $\theta_{\text{rel}} = 2$.

B.3. Non-Analytic Expansion of the ReLU Correlation Map Near $c = 1$

Here we derive the non-analytic expansion

$$F_{\text{ReLU}}(1 - m) = 1 - m + \kappa m^{3/2} + \mathcal{O}(m^2), \quad \kappa = \frac{2\sqrt{2}}{3\pi}, \quad (119)$$

⁶As before, f_{kink} is defined only up to addition of an m -independent analytic background $f_0(t, h)$, which does not affect the equation of state.

starting from the closed form ReLU correlation map (Cho & Saul, 2009)

$$F_{\text{ReLU}}(c) = \frac{1}{\pi} \left[\sqrt{1 - c^2} + (\pi - \cos^{-1} c) c \right], \quad c \in [-1, 1]. \quad (120)$$

Set $c = 1 - m$ with $0 < m \ll 1$.

First expand the square root term,

$$\begin{aligned} \sqrt{1 - c^2} &= \sqrt{1 - (1 - m)^2} = \sqrt{2m - m^2} = \sqrt{2m} \sqrt{1 - \frac{m}{2}} \\ &= \sqrt{2m} \left(1 - \frac{m}{4} + \mathcal{O}(m^2) \right) = \sqrt{2} m^{1/2} - \frac{\sqrt{2}}{4} m^{3/2} + \mathcal{O}(m^{5/2}). \end{aligned} \quad (121)$$

Next expand $\cos^{-1}(1 - m)$. Write $\theta = \cos^{-1}(1 - m)$ and use

$$\cos \theta = 1 - \frac{\theta^2}{2} + \frac{\theta^4}{24} + \mathcal{O}(\theta^6), \quad (122)$$

together with an ansatz $\theta = am^{1/2} + bm^{3/2} + \mathcal{O}(m^{5/2})$. Matching $\cos \theta = 1 - m$ order by order gives $a = \sqrt{2}$ and $b = \sqrt{2}/12$, hence

$$\cos^{-1}(1 - m) = \sqrt{2} m^{1/2} + \frac{\sqrt{2}}{12} m^{3/2} + \mathcal{O}(m^{5/2}). \quad (123)$$

Now expand the product term in (120),

$$\begin{aligned} (\pi - \cos^{-1} c) c &= (\pi - \cos^{-1}(1 - m))(1 - m) \\ &= \left(\pi - \sqrt{2} m^{1/2} - \frac{\sqrt{2}}{12} m^{3/2} + \mathcal{O}(m^{5/2}) \right) (1 - m) \\ &= \pi(1 - m) - \sqrt{2} m^{1/2} + \frac{11\sqrt{2}}{12} m^{3/2} + \mathcal{O}(m^2). \end{aligned} \quad (124)$$

Adding (121) and (124) cancels the $\mathcal{O}(m^{1/2})$ terms and yields

$$\begin{aligned} \sqrt{1 - c^2} + (\pi - \cos^{-1} c) c &= \pi(1 - m) + \left(-\frac{\sqrt{2}}{4} + \frac{11\sqrt{2}}{12} \right) m^{3/2} + \mathcal{O}(m^2) \\ &= \pi(1 - m) + \frac{2\sqrt{2}}{3} m^{3/2} + \mathcal{O}(m^2). \end{aligned} \quad (125)$$

Dividing by π gives (119) with $\kappa = 2\sqrt{2}/(3\pi)$.

The key structural point is the cancellation of the $\mathcal{O}(m^{1/2})$ pieces, leaving a leading non-analytic correction of order $m^{3/2}$, which encodes the kink.

B.4. Origin of the $m^{3/2}$ Term for Kinked Activations

The $m^{3/2}$ term is a near-alignment effect. Write $c = 1 - m$ with $m \ll 1$. Then the two Gaussian preactivations differ only by a small transverse fluctuation. Away from a kink, the activation is locally smooth and the usual Taylor expansion produces only integer powers of m . The non-analytic term comes from the small set of samples where the two nearly identical preactivations fall on opposite sides of the kink. For ReLU, the main text gives the closed-form expansion

$$F(1 - m) = 1 - m + \kappa m^{3/2} + \mathcal{O}(m^2), \quad (126)$$

with $\kappa = \frac{2\sqrt{2}}{3\pi}$ for ReLU. The coefficient is specific to ReLU and to the normalization convention, but the same $3/2$ power appears for a continuous activation with an ordinary finite-slope kink and the piecewise regularity assumed above.

Let (u_1, u_2) be jointly Gaussian with

$$\mathbb{E}[u_1] = \mathbb{E}[u_2] = 0, \quad \mathbb{E}[u_1^2] = \mathbb{E}[u_2^2] = 1, \quad \mathbb{E}[u_1 u_2] = c. \quad (127)$$

The normalized correlation map takes the form

$$F(c) \propto \mathbb{E}[\phi(u_1)\phi(u_2)], \quad (128)$$

with proportionality fixed by the variance normalization used in the main text.

We study the approach to perfect alignment by writing

$$c = 1 - m, \quad 0 < m \ll 1. \quad (129)$$

Introduce the sum and difference coordinates

$$s = \frac{u_1 + u_2}{\sqrt{2}}, \quad d = \frac{u_1 - u_2}{\sqrt{2}}. \quad (130)$$

A direct covariance computation gives

$$\text{Var}(s) = 1 + c = 2 - m, \quad \text{Var}(d) = 1 - c = m, \quad \mathbb{E}[sd] = 0. \quad (131)$$

Hence as $m \rightarrow 0$ one has the scaling

$$s \sim O(1), \quad d \sim O(\sqrt{m}). \quad (132)$$

Thus the average coordinate s remains order one, while the separation d has standard deviation $\sqrt{m}$. Geometrically, the joint Gaussian measure concentrates in a thin tube of transverse thickness $\sqrt{m}$ around the diagonal $u_1 = u_2$. Only this tube can see the kink: the probability of straddling the kink is of order $\sqrt{m}$, and inside that region the activation mismatch produced by the slope jump is also of order $\sqrt{m}$. In the kernel this gives an order- m local correction carried by an order- $\sqrt{m}$ region, hence the leading non-analytic contribution scales as $m^{3/2}$.

This argument extends beyond ReLU. Put the kink at $u = 0$ and let its finite one-sided slopes be $a_{\pm}$. For an activation that is piecewise C^2 with bounded regular second derivative near the kink, write locally

$$\phi(u) = \phi(0) + a_- u + (a_+ - a_-) \text{ReLU}(u) + r(u), \quad r(u) = \mathcal{O}(u^2) \text{ as } u \rightarrow 0, \quad (133)$$

where r has no slope jump. In Price's second derivative, the slope jump contributes a delta function, whose paired Gaussian expectation diverges as $m^{-1/2}$; integrating twice gives the $m^{3/2}$ term. The bounded regular parts contribute at order m^2 , under the stated Gaussian integrability assumptions. Thus the leading fractional power is fixed by the kink, while its amplitude depends on the slope jump, the Gaussian density at the kink and the variance normalization. Mere continuity of r' would not suffice for the stated remainder.

C. Hermite and Variational Perspective

We start from the variational question that first led us to the smooth/kinked split: how does the activation shape affect the mean-field correlation length? We study this through the angular susceptibility, whose stationary points at fixed scale admit a simple spectral description. Since wide-network preactivations are Gaussian, this is naturally a problem in a function space with Gaussian measure. Hermite polynomials are the standard orthonormal basis for such spaces, so expanding the rescaled activation in Hermite modes turns the susceptibility calculation into a spectral problem. This appendix makes that reduction explicit.

Before doing this, we remove a trivial scale freedom: $\phi \mapsto a\phi$ can be absorbed by $\sigma_w^2 \mapsto \sigma_w^2/a^2$ without changing the mean-field fixed point. To focus on shape rather than scale, we fix a variance fixed point q^* and the bias variance σ_b^2 , and determine σ_w^2 from the fixed-point condition. With this convention, the remaining freedom is the shape of a function $f \in L^2(Dz)$ with Gaussian inner product

$$Dz \equiv \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz, \quad \langle f, g \rangle \equiv \int Dz f(z)g(z). \quad (134)$$

C.1. A Scale-Fixed Variational Problem for χ

We define the fixed-point rescaled activation

$$f(z) \equiv \phi(\sqrt{q^*}z). \quad (135)$$

Assuming ϕ is weakly differentiable with $f \in H^1(Dz)$, Price's theorem (Price, 1958) yields the correlation susceptibility used in deep information propagation (Schoenholz et al., 2017)

$$\chi \equiv F'(1) = \frac{\sigma_w^2}{q^*} \int Dz (f'(z))^2. \quad (136)$$

The variance fixed-point equation

$$q^* = \sigma_w^2 \int Dz f(z)^2 + \sigma_b^2 \quad (137)$$

determines

$$\sigma_w^2 = \frac{q^* - \sigma_b^2}{\int Dz f^2}. \quad (138)$$

Substituting gives the scale-fixed factorization

$$\chi = \left(1 - \frac{\sigma_b^2}{q^*}\right) \frac{\int Dz (f')^2}{\int Dz f^2} \equiv \left(1 - \frac{\sigma_b^2}{q^*}\right) \mathcal{Q}[f], \quad \mathcal{Q}[f] \equiv \frac{\|f'\|_2^2}{\|f\|_2^2}. \quad (139)$$

Thus, at fixed (q^*, σ_b^2) , extremizing χ over activations reduces to extremizing the Rayleigh quotient (see for example (Horn & Johnson, 1985)) $\mathcal{Q}[f]$ over the shape of f . Criticality $\chi = 1$ is equivalent to the constraint

$$\mathcal{Q}[f] = \left(1 - \frac{\sigma_b^2}{q^*}\right)^{-1}. \quad (140)$$

In this convention, $\mathcal{Q}[f]$ quantifies the activation's effective sensitivity to typical Gaussian fluctuations: shifting spectral weight to higher modes increases $\|f'\|_2$ relative to $\|f\|_2$ and therefore increases χ unless σ_w^2 is retuned.

C.2. The Ornstein–Uhlenbeck Eigenvalue Problem

Because $\mathcal{Q}[f]$ is scale-invariant, we may impose the normalization $\int Dz f^2 = 1$ and extremize $\int Dz (f')^2$. Introducing a Lagrange multiplier λ , consider

$$\mathcal{S}[f] \equiv \int Dz \left[(f'(z))^2 - \lambda f(z)^2 \right]. \quad (141)$$

Gaussian-weighted integration by parts (with boundary terms vanishing under mild decay assumptions) gives the Euler–Lagrange equation

$$(-\partial_z^2 + z\partial_z) f(z) = \lambda f(z), \quad (142)$$

i.e. the Ornstein–Uhlenbeck eigenproblem on $L^2(Dz)$. Its spectrum is

$$\lambda_n = n, \quad n = 0, 1, 2, \dots, \quad (143)$$

and the eigenfunctions form a complete orthonormal basis. Under the unit-norm constraint, these eigenfunctions are stationary points of $\mathcal{Q}[f]$, with $\mathcal{Q}[f] = \lambda$.

C.3. Hermite Basis and Diagonal Correlation Propagation

A natural orthonormal eigenbasis of (142) is given by the normalized Hermite polynomials; this is the usual Hermite/Ornstein–Uhlenbeck diagonalization of Gaussian channels. Hermite expansions have also been used directly to express neural-network covariance recursions in the infinite-width setting (Arratia et al., 2020).

$$h_n(z) \equiv \frac{1}{\sqrt{2^n n!}} H_n \left(\frac{z}{\sqrt{2}} \right), \quad \langle h_n, h_m \rangle = \delta_{nm}, \quad (144)$$

where H_n are the physicists' Hermite polynomials.⁷ They satisfy

$$h'_n(z) = \sqrt{n}h_{n-1}(z), \quad (-\partial_z^2 + z\partial_z)h_n(z) = nh_n(z), \quad \mathcal{Q}[h_n] = n. \quad (145)$$

Moreover, this basis diagonalizes correlated-Gaussian expectations: if (z_1, z_2) are jointly Gaussian with unit variances and correlation $c \in [-1, 1]$, then

$$\mathbb{E}[h_n(z_1)h_m(z_2)] = c^n \delta_{nm}. \quad (146)$$

Thus each Hermite degree propagates independently through the Gaussian channel with attenuation c^n .

Expanding an arbitrary $f \in L^2(Dz)$ in this basis,

$$f(z) = \sum_{n \geq 0} a_n h_n(z), \quad a_n = \langle f, h_n \rangle, \quad \sum_{n \geq 0} a_n^2 = \int Dz f^2, \quad (147)$$

we obtain

$$\mathbb{E}[f(z_1)f(z_2)] = \sum_{n \geq 0} a_n^2 c^n. \quad (148)$$

Consequently, the correlation map takes the form

$$F(c) = \frac{\sigma_w^2 \sum_{n \geq 0} a_n^2 c^n + \sigma_b^2}{q^*}, \quad q^* = \sigma_w^2 \sum_{n \geq 0} a_n^2 + \sigma_b^2. \quad (149)$$

The decay of $|a_n|$ controls the derivatives of F at $c = 1$: finite weighted sums $\sum_n n(n-1)\cdots(n-k+1)a_n^2$ give the derivatives needed for a Taylor expansion through order k . Smoothness and rapid decay need not give a convergent Taylor series at one; the Landau argument only uses finite-order regularity. The power-law tail produced by an ordinary kink makes the second derivative diverge and gives the fractional-power correction as $c \rightarrow 1$.

Using $h'_n = \sqrt{n}h_{n-1}$, we can compute the susceptibility

$$\chi = F'(1) = \frac{\sigma_w^2}{q^*} \sum_{n \geq 1} n a_n^2. \quad (150)$$

In the zero-bias case $\sigma_b^2 = 0$ this becomes

$$\chi = \frac{\sum_{n \geq 1} n a_n^2}{\sum_{n \geq 0} a_n^2} = \mathcal{Q}[f], \quad (151)$$

so χ is the mean Hermite degree under weights proportional to a_n^2 , and criticality $\chi = 1$ corresponds to unit mean degree.

For completeness, the variance-channel susceptibility $\chi_q \equiv \partial q^{(\ell+1)} / \partial q^{(\ell)}|_{q^*}$ can be written in the same basis as

$$\chi_q = \frac{\sigma_w^2}{q^*} \left(\sum_{n \geq 1} n a_n^2 + \sum_{n \geq 0} \sqrt{(n+1)(n+2)} a_n a_{n+2} \right). \quad (152)$$

Thus χ depends only on the distribution $\{a_n^2\}$ across Hermite degrees, whereas χ_q is additionally sensitive to relative signs/phases within each parity subsector through the $a_n a_{n+2}$ couplings.

Pure modes $f \propto h_n$ are stationary points with $\mathcal{Q}[h_n] = n$. On the mean-zero subspace, $n = 1$ is the lowest eigenmode, but by itself it gives a linear network. Moving spectral weight to higher degrees raises $\mathcal{Q}[f]$ and hence χ at fixed q^* and σ_b^2 . Its effect on correlation depth depends on the stable branch: in the ordered regime $\xi_c = -1/\log \chi$ increases as χ approaches one, while on the chaotic branch the relevant multiplier is $F'(c^*)$. The stationary points of this Rayleigh quotient therefore do not by themselves solve the global correlation-length optimization problem.

⁷Equivalently, $h_n(z) = \text{He}_n(z)/\sqrt{n!}$ in terms of the probabilists' Hermite polynomials He_n .

C.4. Smooth and Kinked Activations

For the smooth examples considered here, the Hermite coefficients decay rapidly and the required derivative moments are finite, allowing the finite-order expansion near $c = 1$. For kinked activations such as ReLU, $\phi'(u) = \theta(u)$ almost everywhere, so

$$\chi = \sigma_w^2 \int Dz \theta(u) = \frac{\sigma_w^2}{2}, \quad \chi = 1 \Rightarrow \sigma_w^2 = 2. \quad (153)$$

Table 6. Orthonormal Hermite coefficients $a_n = \int Dz f(z) h_n(z)$ for two standard activations, shown for the unscaled choices $f(z) = \text{ReLU}(z)$ and $f(z) = \tanh(z)$. For $f(z) = \tanh(sz)$, use $a_n(s) = \int Dz \tanh(sz) h_n(z)$, with $a_{2k}(s) = 0$ by odd parity.

n	$h_n(z)$ parity	a_n for ReLU(z)	a_n for tanh(z)
0	even	$\frac{1}{\sqrt{2\pi}}$	0
1	odd	$\frac{1}{2}$	0.60570551
2	even	$\frac{1}{2\sqrt{\pi}}$	0
3	odd	0	-0.14843719
4	even	$-\frac{1}{\sqrt{48\pi}}$	0
5	odd	0	0.06254752
6	even	$\frac{1}{4\sqrt{10\pi}}$	0
7	odd	0	-0.03144542
8	even	$-\frac{15}{\sqrt{80640\pi}}$	0
9	odd	0	0.01741993
10	even	$\frac{105}{\sqrt{7257600\pi}}$	0
11	odd	0	-0.01029184

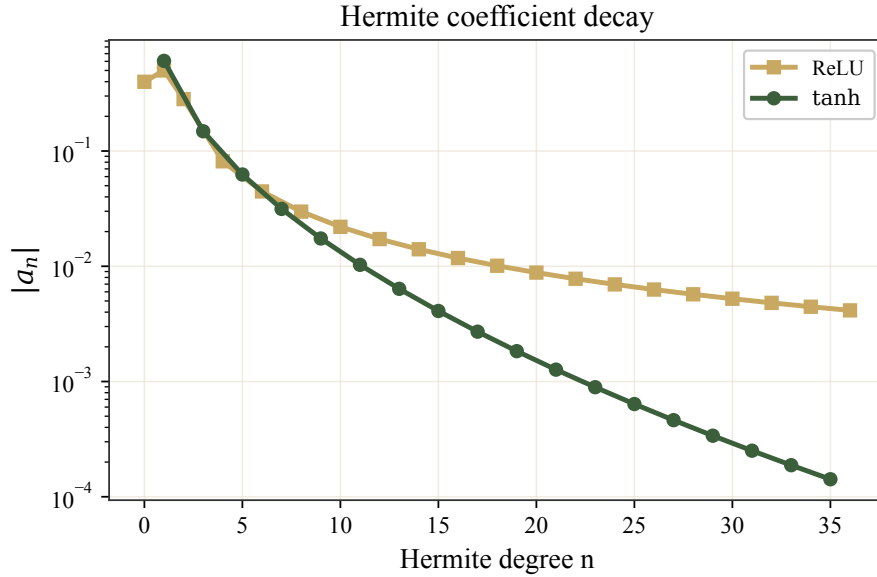


Figure 4. Magnitude of Hermite coefficients $|a_n|$ for ReLU versus tanh. ReLU exhibits a slow (power-law) decay, reflecting multi-scale support across Hermite degrees, while tanh decays rapidly and concentrates most spectral mass in the lowest modes. The ReLU coefficients use their exact expression; tanh uses 220-point Gauss–Hermite quadrature.

C.5. Hermite Decompositions for ReLU and tanh

Throughout, $Z \sim \mathcal{N}(0, 1)$ and

$$Dz \equiv \frac{dz}{\sqrt{2\pi}} e^{-z^2/2}. \quad (154)$$

We expand the fixed point rescaled activation $f(z)$ in an orthonormal Hermite basis of $L^2(Dz)$,

$$f(z) = \sum_{n \geq 0} a_n h_n(z), \quad a_n = \int Dz f(z) h_n(z), \quad \int Dz h_n(z) h_m(z) = \delta_{nm}. \quad (155)$$

With physicists Hermites H_n ,

$$H_n(x) \equiv (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}, \quad h_n(z) \equiv \frac{1}{\sqrt{2^n n!}} H_n\left(\frac{z}{\sqrt{2}}\right). \quad (156)$$

C.5.1. ReLU

Let $f(z) = \text{ReLU}(z) = \max(0, z)$. The coefficients have a closed form.

$$a_0 = \frac{1}{\sqrt{2\pi}}, \quad a_1 = \frac{1}{2}, \quad a_{2k+1} = 0 \text{ for } k \geq 1, \quad (157)$$

and for $k \geq 1$,

$$a_{2k} = \frac{(-1)^{k-1} (2k-3)!!}{\sqrt{2\pi} (2k)!}. \quad (158)$$

We use the convention $(-1)!! = 1$. If instead $f(z) = \text{ReLU}(\sqrt{q^*}z)$, then all coefficients scale as $a_n \mapsto \sqrt{q^*} a_n$.

C.5.2. tanh

Let $f(z) = \tanh(sz)$ with $s > 0$. The coefficients are

$$a_n(s) = \int Dz \tanh(sz) h_n(z). \quad (159)$$

Parity gives $a_{2k}(s) = 0$. There is no simple closed form for general s , but the coefficients are easy to compute numerically by Gaussian quadrature using Mathematica, or other specialized packages.

D. Experimental Details and Additional Figures

This appendix collects the experimental details and additional figures supporting the scheduling predictions. The original notebooks and saved results are available in the [icml](#) repository.

D.1. Mean-Field Fits and Scaling Collapse

Table 7 reports fits to the numerical mean-field curves. The repository includes the raw arrays, fitting windows, and a CPU script that regenerates Fig. 1. Tanh expectations use 60-point Gauss–Hermite quadrature. The ReLU calculation uses the formal channel $F(c) = \chi F_{\text{ReLU}}(c) + 1 - \chi/\rho$, where F_{ReLU} is the normalized arc-cosine map. Thus $t = \chi - 1$ and $h = \chi(1/\rho - 1)$. This family probes the local normal form; its independently tuned parameters need not correspond to nonnegative bias variance in a finite-variance ReLU network.

The same scaling-collapse diagnostic can be carried out for the kinked universality class, using ReLU as the representative activation.

Table 7. Critical exponents from log–log fits in Fig. 1. The $\pm$ values are least-squares fit standard errors, not seed uncertainty or bounds on numerical error. Deviations from the asymptotic predictions reflect the finite fitting windows and numerical approximation; small fit errors do not remove these systematic effects.

Exponent	Numerical fit	Asymptotic prediction
ν	1.0190 ± 0.0014	1
$\beta_{\tanh}$	0.9879 ± 0.0007	1
β_{ReLU}	1.9642 ± 0.0019	2
$\theta_{\text{rel,tanh}}$	1.00483 ± 0.00002	1
$\theta_{\text{rel,ReLU}}$	1.98697 ± 0.00005	2
$1/\delta_{\tanh}$	0.5171 ± 0.0007	$1/2$
$1/\delta_{\text{ReLU}}$	0.6656 ± 0.0001	$2/3$
$\nu_{\rho,\tanh}$	0.4716 ± 0.0014	$1/2$
$\nu_{\rho,\text{ReLU}}$	0.3528 ± 0.0010	$1/3$

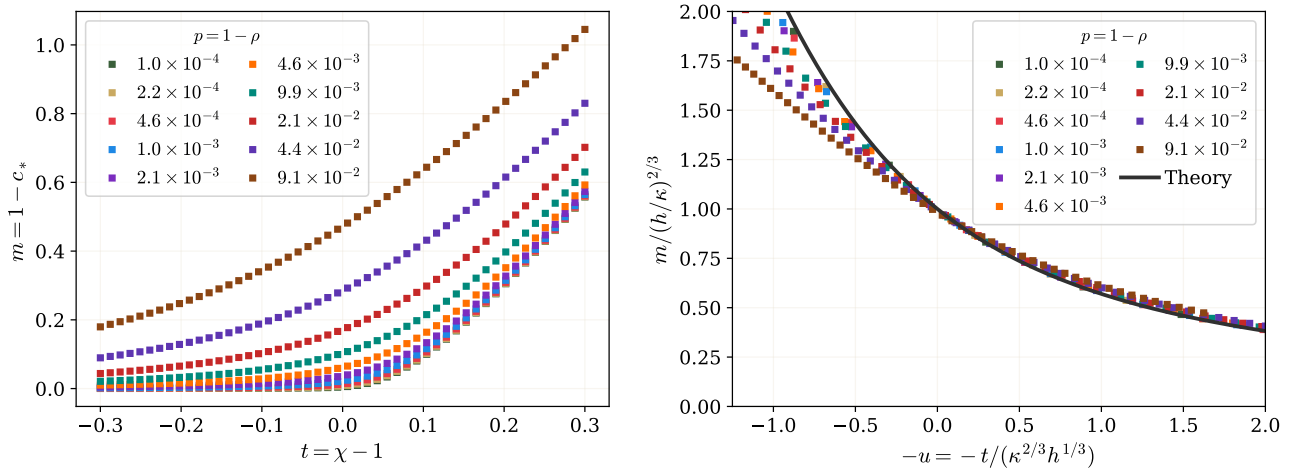


Figure 5. Kinked counterpart to Fig. 2, using the formal two-parameter ReLU channel above. Legends give the fixed probability $p = 1 - \rho$. Rescaling uses $h = \chi(1/\rho - 1)$ at each point, and the horizontal coordinate is $-u$ relative to Eq. (38). This is a normal-form comparison, not a scan of independently tunable finite-variance ReLU initializations.

D.2. Dropout Scheduling Experiments

We write the mean training dropout probability as $\bar{p}$, distinct from the theoretical field h . In the training figures, the original axis labels h and $\bar{h}$ denote raw dropout probabilities. Accuracy tables report means $\pm$ sample SEM; best accuracy is each seed’s maximum recorded test accuracy, then averaged across seeds.

The scheduling experiments below test Sec. 3.3: MLP schedules in Fig. 6 and Tab. 8, matched-budget controls in Fig. 7 and Tab. 9, robustness sweeps in Fig. 8, ViT CIFAR-100 schedules in Fig. 10 and Tab. 12, and transformer ablations in Fig. 12 and Tab. 13.

D.3. MLPs

Finite-width experiments support these spatial scheduling predictions. The main finding is that concentrating dropout in early layers, where a mask has the largest downstream regularization reach, yields the best generalization. The theoretically motivated schedules improve over constant dropout at no extra computational cost.

To test this prediction in a controlled finite-width setting, we design an experiment in the overfitting regime, where information propagation and regularization are both important factors in determining performance. In the effective-theory treatment of (Roberts et al., 2022), deviations from Gaussian mean-field behavior (Neal, 1996; Lee et al., 2018) generate corrections to the correlation recursions that are suppressed by $1/N$ (with N the layer width) and accumulate across depth as L/N . At the same time, deep information propagation suggests that successful learning is controlled by a modest multiple

of the correlation length (Schoenholz et al., 2017).

We train on 5000 CIFAR-10 images with width 256, depth $L = 6$, and batch size 100. We use Gaussian weights with $\sigma_w^2 = 1.98$ and bias variance $\sigma_b^2 = 0.02$, close to the no-dropout He weight scale. This choice does not ensure stationary variance under dropout: the ReLU variance multiplier is $\sigma_w^2/(2\rho)$, which exceeds one at the tested dropout rates. These are finite-depth training experiments, and the tabulated ξ_{eff} values are reference diagnostics, not measured propagation lengths during training.

We compare uniform dropout, increasing and decreasing linear profiles, and step profiles concentrated in the first or last half of the network, all at the same mean dropout probability. No dropout is a separate zero-budget control (Fig. 6). The training code allocates raw dropout probabilities; these are proportional to the theoretical field only to leading order. Front-loaded profiles outperform both uniform and back-loaded profiles here. Early and late steps have the same reference ξ_{eff} but different test performance, so propagation length alone does not explain the ordering. The regularization-reach heuristic motivates the preference for early placement.

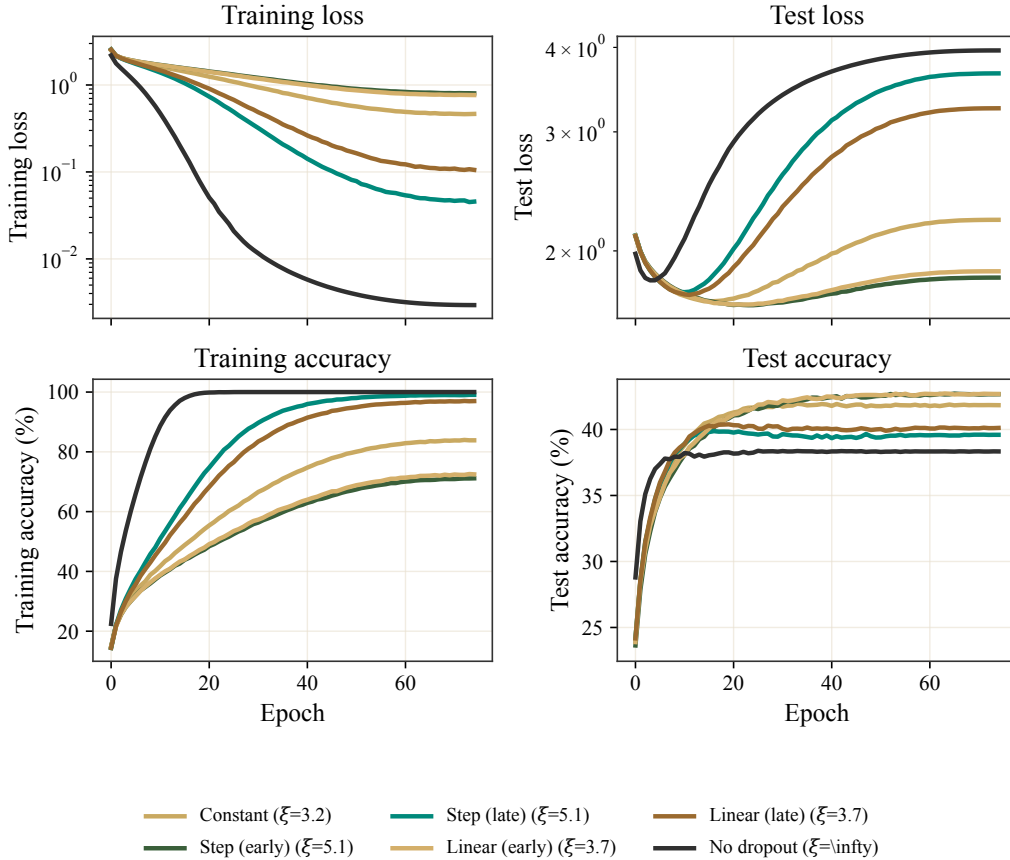


Figure 6. Finite-width MLP training and test curves at fixed mean dropout probability $\bar{p}$ (with $0 \leq p_\ell \leq p_{\max}$). We compare uniform dropout, linear ramps (increasing/decreasing with depth), and step schedules that concentrate dropout in either the first or last half of the network, together with the no-dropout baseline.

The budget-control experiment checks a simpler explanation: early-concentrated dropout may only be winning because it raises the local dropout rate, not because the allocation over depth matters. After all, the step schedule applies dropout at rate $2\bar{p}$ to the first half of the network, so perhaps the same improvement could be achieved by applying this higher rate uniformly throughout.

To disentangle allocation from total dropout strength, we compare constant schedules with probabilities $\bar{p}$, $2\bar{p}$, $3\bar{p}$ against step schedules using $2\bar{p}$ in the first half or $3\bar{p}$ in the first third of the network (Fig. 7, Tab. 9).

Table 8. Performance of different dropout schedules in the overfitting regime for MLPs and CIFAR-10, corresponding to Fig. 6.

Schedule	ξ_{eff}	L/ξ_{eff}	Best Test Acc. (%)	Final Test Acc. (%)
Constant	3.20	1.87	42.54 ± 0.10	41.84 ± 0.10
Step (late)	5.09	1.18	40.56 ± 0.11	39.60 ± 0.13
Step (early)	5.09	1.18	43.13 ± 0.07	42.67 ± 0.09
Linear (increasing)	3.73	1.61	41.01 ± 0.08	40.12 ± 0.12
Linear (decreasing)	3.73	1.61	43.11 ± 0.09	42.69 ± 0.10
No dropout	∞	0	38.77 ± 0.10	38.34 ± 0.12

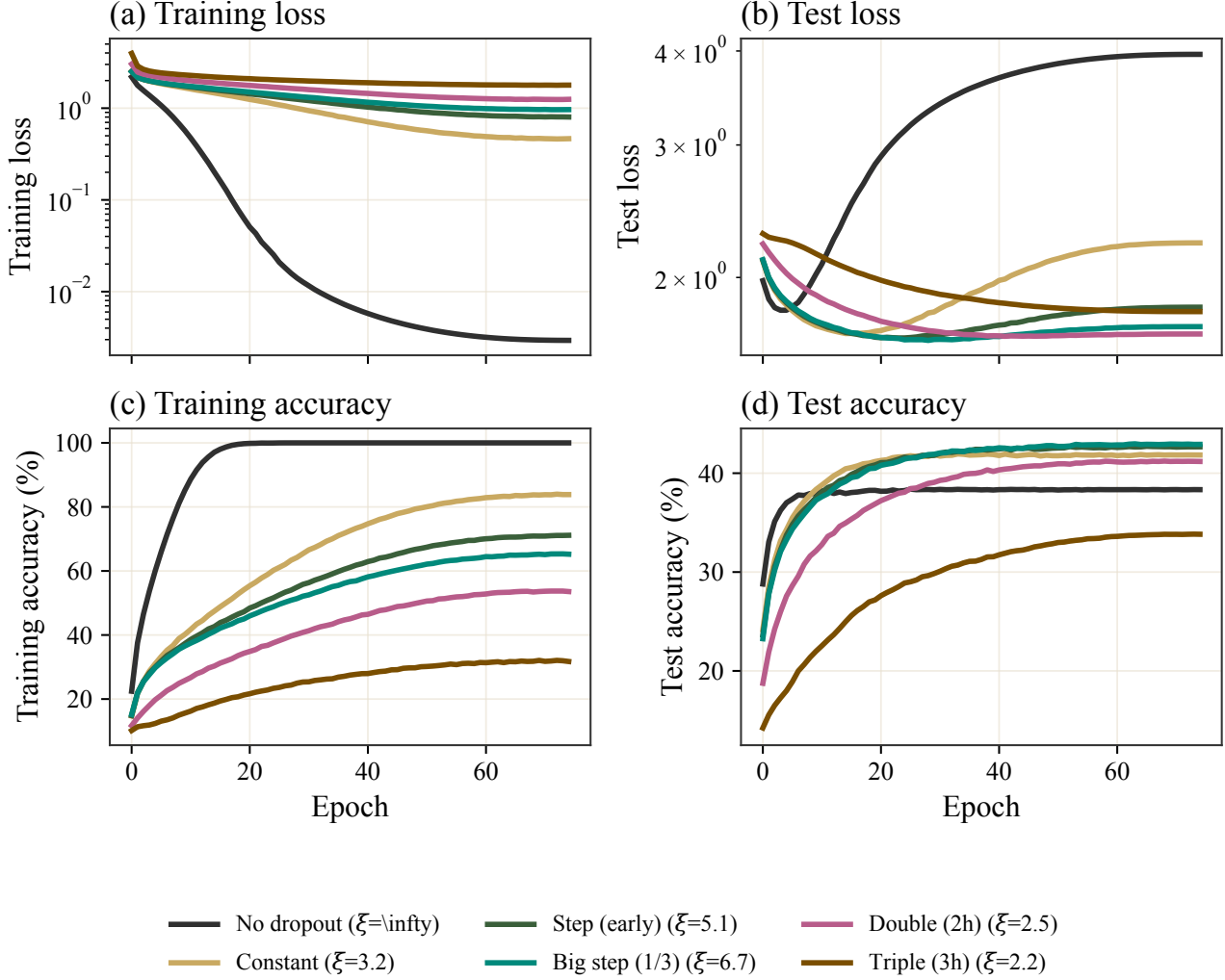


Figure 7. Matched-budget controls comparing front-loaded step schedules against constant dropout probabilities at $\bar{p}$, $2\bar{p}$, and $3\bar{p}$.

If the step schedules succeed merely because they apply locally higher dropout rates, then the uniform schedules with matching dropout should perform at least as well. In contrast, if spatial allocation genuinely matters, the step schedules should outperform their uniform counterparts despite having lower total dropout. The results in Fig. 7 and Table 9 confirm that spatial allocation dominates: Big step ($\xi_{\text{eff}} = 6.7$) achieves the highest accuracy despite having only one-third the total dropout of the schedule with $3\bar{p}$, and front-loaded step ($\xi_{\text{eff}} = 5.1$) outperforms Double despite half the total dropout budget. Moreover, increasing uniform dropout beyond $2\bar{p}$ becomes counterproductive; Triple ($3\bar{p}$) performs worst among all dropout schedules, as excessive regularization throughout the network impairs learning capacity. These controls support a benefit from early allocation beyond simply increasing the uniform dropout rate. They do not establish a monotonic relationship between reference correlation length and generalization: the no-dropout control has infinite reference length and lower accuracy, and early and late steps share a reference length.

Higher-order terms become relevant as dropout increases. The tested probabilities up to 0.3 probe beyond the strictly weak-dropout limit; the theory does not set a validity threshold at this value. The lower-probability comparisons are reported in the budget sweep below.

Table 9. Summary of dropout schedules, correlation lengths, and test performance for MLPs on CIFAR-10, corresponding to Fig. 7.

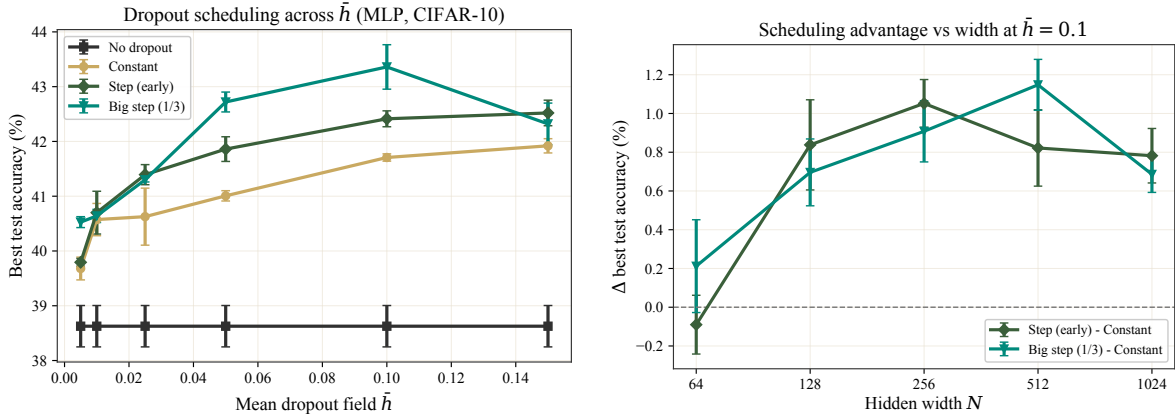
Schedule	ξ_{eff}	L/ξ_{eff}	Best Test Acc. (%)	Final Test Acc. (%)
No dropout	∞	0.00	38.77 ± 0.10	38.34 ± 0.12
Constant	3.20	1.87	42.54 ± 0.10	41.84 ± 0.10
Step (early)	5.09	1.18	43.13 ± 0.07	42.67 ± 0.09
Big step (1/3)	6.67	0.90	43.29 ± 0.08	42.92 ± 0.10
Double ($2\bar{p}$)	2.54	2.36	41.46 ± 0.09	41.19 ± 0.10
Triple ($3\bar{p}$)	2.22	2.70	33.94 ± 0.23	33.82 ± 0.23

Table 10. Experimental and hyperparameter configuration for high overfitting regime. Corresponding performance metrics in Fig. 6.

Parameter	Value / Description
<i>Network Architecture & Initialization</i>	
Network Width	256
Depth Multiplier	2
Weight Variance (σ_w^2)	1.98
Bias Variance (σ_b^2)	0.02
<i>Training Dynamics</i>	
Epochs	75
Batch Size	100
Learning Rate	1×10^{-4}
LR Schedule	Decay to minimum 1×10^{-7}
Mean Dropout Probability ($\bar{p}$)	0.1
Max Dropout Probability (p_{max})	0.2 (Step); 0.3 (Big step)
Step Active Fraction (f)	1/2 (Step); 1/3 (Big step)
<i>General Setup</i>	
Number of Simulations	25
Dataset Usage	CIFAR-10 5,000 train / 5,000 test samples
Hardware	GPU A100

D.4. Robustness Sweeps

To test whether scheduling is tied to one budget, width, or activation, we ran CIFAR-10 MLP sweeps (Figs. 8 and 9). The ReLU sweeps probe mean-field boundaries: at $\bar{p} = 0.15$, Step/Big step use large local dropout probabilities 0.30/0.45; at $N = 64$, finite-width non-Gaussianity weakens the correlation-length prediction. The GELU MLP sweep gives the same best-accuracy ordering at $\bar{p} = 0.1$ —constant 41.96 ± 0.16 , Step 42.24 ± 0.15 , Big step 42.46 ± 0.12 —and Big step falls to 38.97 ± 0.23 at $\bar{p} = 0.15$, matching this perturbative-boundary picture. The final-epoch gains summarized in Table 3 use the last recorded epoch, giving +2.04 pp for the ReLU $\bar{p} = 0.1$ sweep and +0.62 pp for the GELU $\bar{p} = 0.1$ sweep.



(a) Sweep over mean dropout probability at width $N = 256$. (b) Schedule advantage over constant dropout at $\bar{p} = 0.1$.

Figure 8. Robustness sweeps for depth-6 near-critical ReLU MLPs on CIFAR-10. Early schedules improve over constant dropout throughout the large-width regime, while $N = 64$ illustrates the expected finite-width boundary.

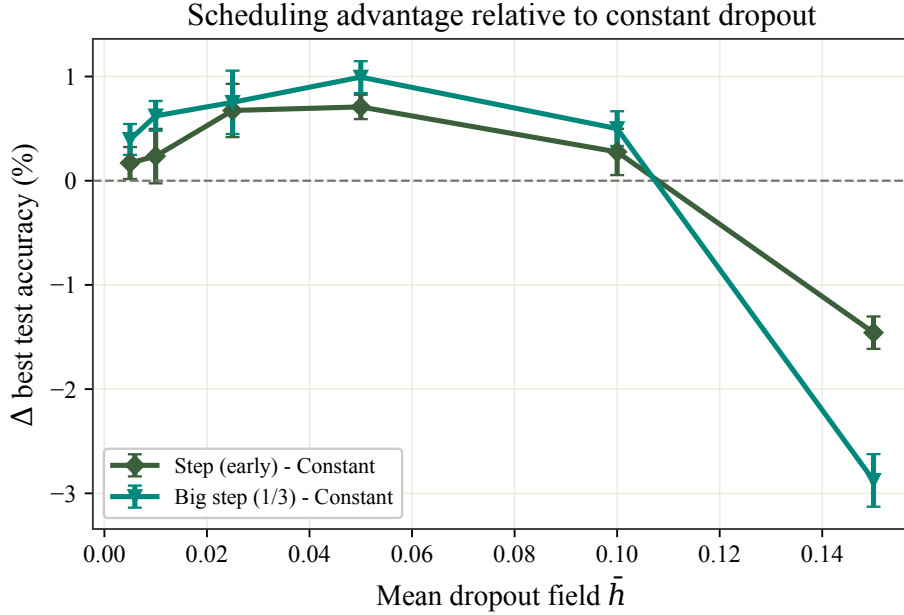


Figure 9. Smooth-activation dropout sweep for depth-6 near-critical GELU MLPs on CIFAR-10 at width $N = 256$. Step-like schedules improve over constant dropout around $\bar{p} = 0.1$, while the largest probability leaves the small-dropout regime where the mean-field perturbation is expected to be predictive.

D.5. Transformers on CIFAR-100

Table 11 and Fig. 10 detail the architectural parameters and corresponding test dynamics for the CIFAR-100 Vision Transformer evaluation (Dosovitskiy et al., 2021; Krizhevsky, 2009); Table 12 reports the schedule comparison.

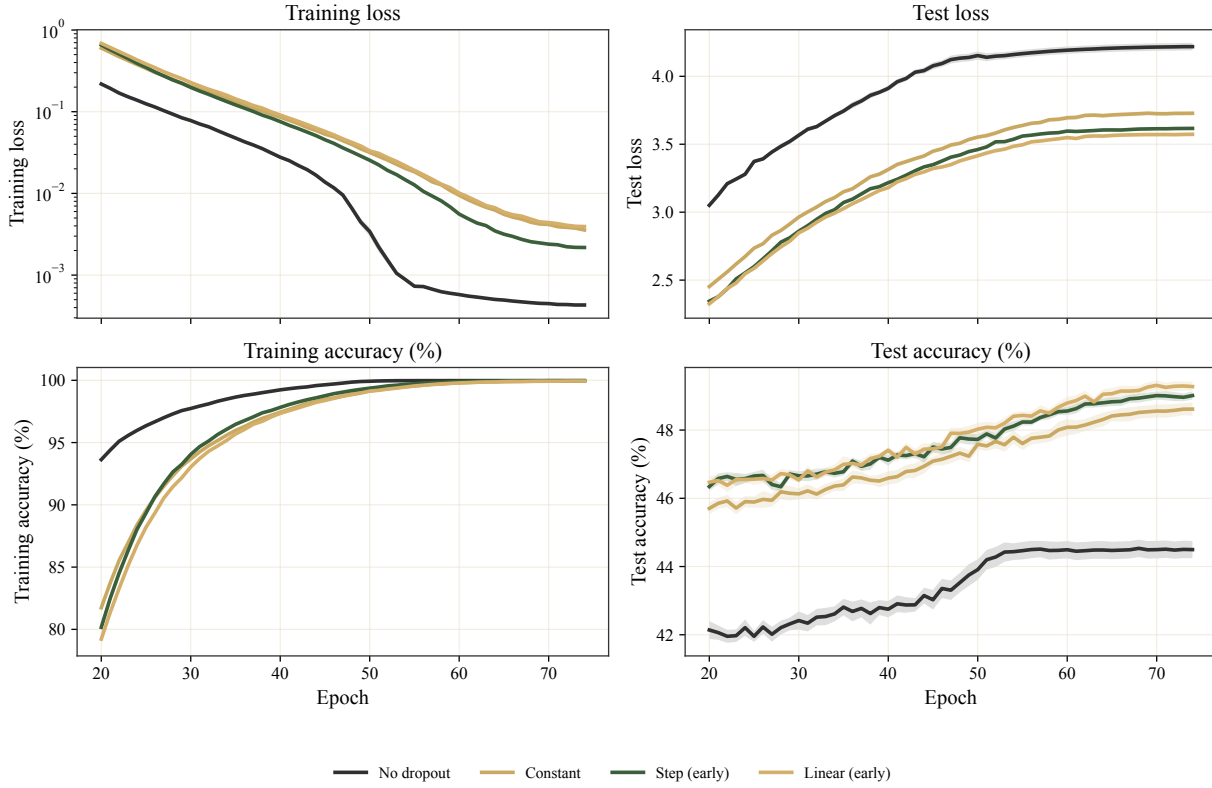


Figure 10. Finite-width Vision Transformer training and test curves at fixed mean dropout probability $\bar{p}$. We compare the no-dropout baseline, uniform dropout, decreasing linear ramps, and early step schedules. The cropped view shows epochs 20–75 for readability; the full curve is also in App. D.

Accuracy curves are shown in Fig. 10, with explicit results in Table 12. Errors in this accuracy table and its accompanying curves are SEM. On CIFAR-100 (and similarly on CIFAR-10), concentrating dropout earlier consistently improved accuracy and reduced overfitting. Interestingly, linear decreasing schedules, which put more dropout in early layers, often performed especially well in transformers; residual connections were introduced in (He et al., 2016), and mean-field analyses show that they ease information propagation (Yang & Schoenholz, 2017). We suspect this makes distributing regularization early preferable empirically, though both early weighted and step-like schedules outperformed the constant baseline.

These experiments test a scheduling idea motivated by MLP initialization theory. Attention, residual connections, and feature learning are outside the derived model, so the observed Transformer gains are empirical evidence for transfer of the schedule, not a verification of its mean-field mechanism. We leave ImageNet-scale evaluation (Deng et al., 2009) to future work.

Table 12. Test accuracy across different dropout schedules for ViT architecture on CIFAR-100, corresponding to Fig. 10.

Schedule	Best Test Acc	Final Test Acc
No dropout	44.61 ± 0.25	44.50 ± 0.25
Constant	48.69 ± 0.18	48.61 ± 0.20
Step (early)	49.10 ± 0.11	49.01 ± 0.12
Linear (decreasing)	49.38 ± 0.14	49.27 ± 0.14

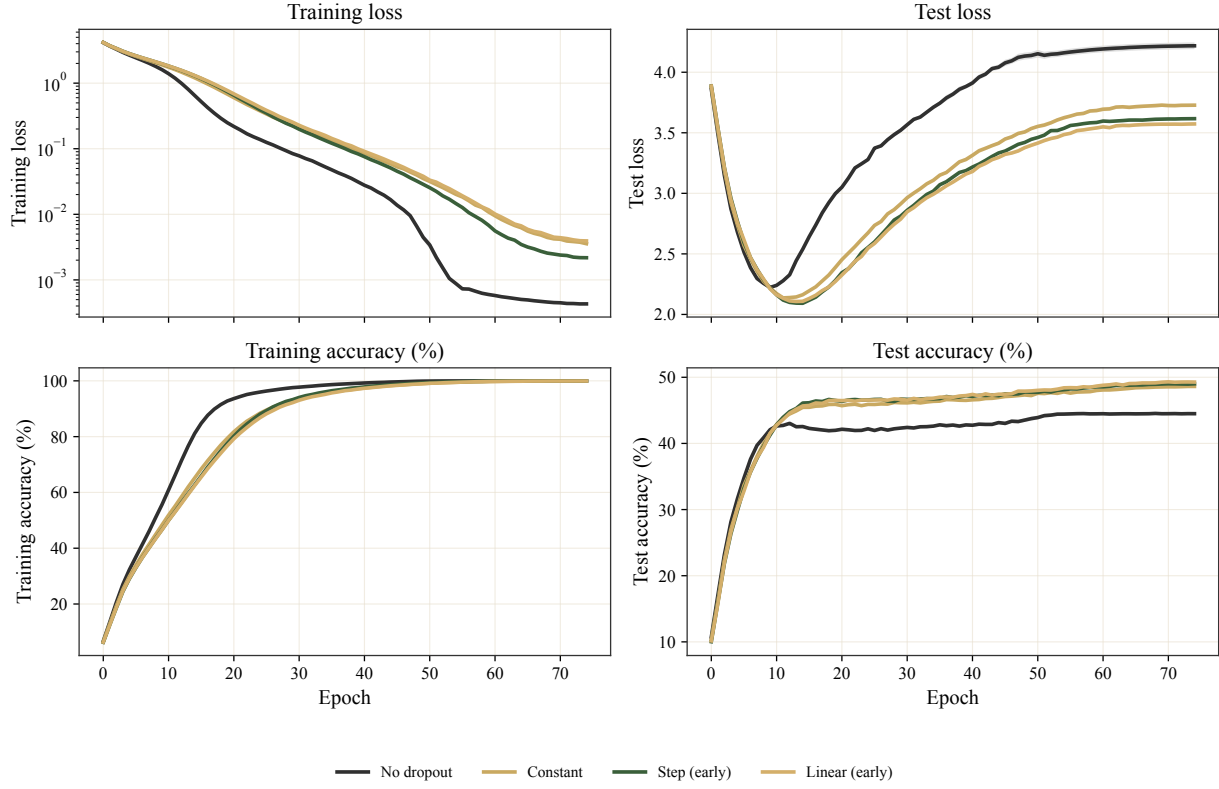


Figure 11. Extended training curves for Figure 10, showing the full 75 epochs.

Table 11. Reported recipe for the CIFAR-100 Vision Transformer. The saved metrics contain ten 75-epoch runs per profile but no run configuration. The archived notebook was set to a one-seed, small-data pilot; the current defaults follow this table. Dataset size and training settings for the historical metrics therefore cannot be independently verified from that archive.

Parameter	Value / Description
<i>ViT Architecture</i>	
Embedding Dimension	128
Depth	10
Number of Heads	16
MLP Ratio	4.0
Patch Size	4
<i>Training Dynamics</i>	
Epochs	75
Batch Size	250
Learning Rate	5×10^{-4}
LR Minimum	5×10^{-6}
Weight Decay	0.00
Mean Dropout Probability ($\bar{p}$)	0.1
Max Dropout Probability ($p_{\max}$)	0.2
Schedules	None, Constant, Step (early), Linear (decreasing)
<i>General Setup</i>	
Number of Simulations	10
Dataset Usage	Full CIFAR-100
Evaluation Frequency	Every 1 Epoch
Hardware	GPU A100

D.6. Transformer Ablations

To isolate the effect of dropout scheduling on distinct transformer components, we performed component-level ablations. We tested transformers with dropout applied exclusively to the attention block, the MLP block, or both, comparing step-like and constant schedules against a no-dropout baseline. We use the full CIFAR-10 dataset, five seeds, and a 12-block ViT with embedding dimension 128, 16 attention heads, MLP ratio 4, and patch size 4. Training runs for 75 epochs on an A100 with batch size 1000, learning rate 5×10^{-4} decaying to 5×10^{-6} , and zero weight decay. Regularized profiles use $\bar{p} = 0.1$ and $p_{\max} = 0.2$; evaluation is recorded every epoch.

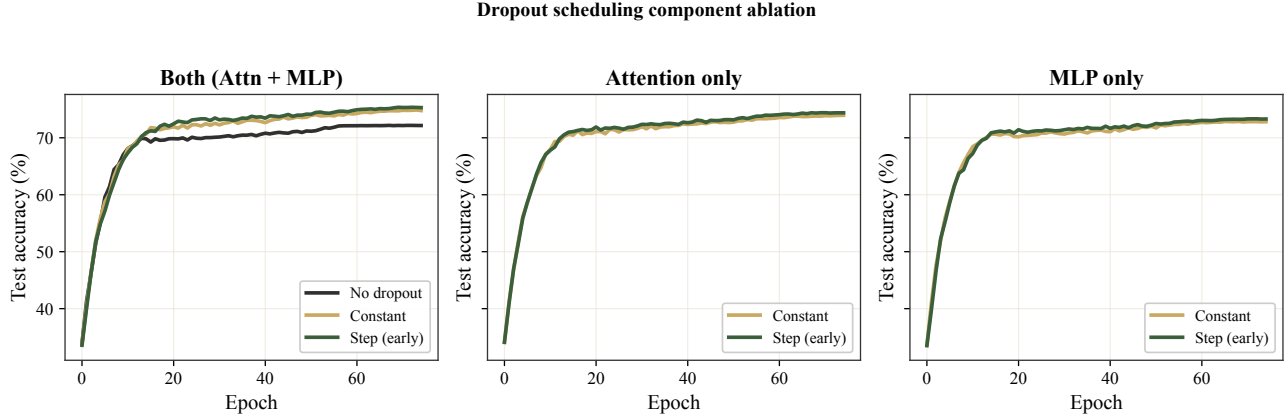


Figure 12. Component ablations applying dropout schedules to the attention block, MLP block, or both. The figure compares no dropout, constant dropout, and step-like dropout.

Table 13. Component-level ablations on CIFAR-10, comparing Constant and Step-like dropout schedules applied exclusively to the MLP block, Attention block, or both. Means $\pm$ sample SEM over five seeds. Best accuracy averages each seed’s maximum recorded test accuracy; final accuracy uses the last epoch. The early-step mean exceeds the uniform mean in all three configurations, without establishing significance for each comparison.

Configuration	Best Test Acc	Final Test Acc
No dropout	72.22 ± 0.16	72.15 ± 0.16
MLP only (Constant)	72.96 ± 0.13	72.82 ± 0.14
MLP only (Step early)	73.38 ± 0.28	73.30 ± 0.30
Attn only (Constant)	74.00 ± 0.10	73.95 ± 0.11
Attn only (Step early)	74.48 ± 0.10	74.38 ± 0.10
Both (Constant)	74.92 ± 0.11	74.78 ± 0.11
Both (Step early)	75.47 ± 0.14	75.30 ± 0.13

E. Extended Experiments Across Datasets

These experiments ask whether spatial allocation survives changes in task, architecture, and training-set size. Each cohort keeps its own split, optimizer settings, and random seeds; we do not pool across weight decay or training duration. Table 14 reports the retained benchmark cohorts and the separate vanilla-MLP study, including comparisons with a missing uniform baseline.

Evaluation and selection. Learning rate and mean dropout follow each cohort’s validation protocol. The budget sweeps tune both separately by profile; the ten-seed Jannis cohort and the two linear follow-ups fix mean dropout at 0.10 and retain their validation-selected learning rates. We select the nonuniform profile with the lowest mean per-seed minimum validation CE among complete arms. Its test loss and accuracy are measured at each seed’s minimum-validation-loss checkpoint; fixed-final-epoch endpoints are listed separately when available. The same selected profile is used throughout a row, including when it loses to uniform. This reanalysis uses existing confirmation runs; it is not a new independent test. Incomplete profiles remain visible but cannot win the main comparison.

Figures. Figures show mean training and validation curves, with SEM bands across confirmation seeds. The right panels show validation metrics; the tables report test endpoints. Training loss uses a logarithmic axis. Training metrics average batches with dropout enabled during optimization, whereas validation evaluates the epoch’s final weights with dropout disabled.

Profiles and architecture. Step early spends the budget in the earliest layers at probability 0.20, with a partially filled last active layer when needed. Big step applies $3\bar{p}$ to the first third of the network, while the linear profiles run between 0 and $2\bar{p}$; big-step and linear profiles can therefore exceed the step profile’s 0.20 cap. In the budget sweeps, mean dropout can differ between profiles, so those comparisons describe full training configurations. Transformer profiles are empirical extensions, with the MLP reference-field diagnostic used only as a proxy.

Coverage. Every candidate in the retained benchmark cohorts has its full seed set. The two linear follow-ups contain a selected linear profile and tuned no-dropout control, but no saved matching uniform baseline. We report their endpoints without borrowing a baseline from a different protocol.

Front-loaded comparisons. Table 4 reports the retained cohorts with a matching uniform baseline. Each row selects among complete front-loaded profiles by mean minimum validation loss, including cases where the selected profile loses to uniform on test. The full comparison below also includes back-loaded profiles and the two linear follow-ups.

Speech Commands evaluation. The audio experiment uses a class-balanced random clip split drawn from the official training subset; speaker identities were not retained in the split construction, so these results do not establish performance on held-out speakers. Final test measurements are unavailable. For the selected profiles in Table 4, the MLP reduces minimum and final validation loss by 13.23% and 13.47%, while the Transformer gives 3.33% and 6.49%. These are validation results, separate from the test columns.

Table 14. Cross-dataset and standalone financial results. Positive values favor the named profile over uniform. Loss reduction is $100(L_U - L_P)/L_U$; accuracy gain is $100(A_P - A_U)/A_U$, with the percentage-point change $A_P - A_U$ in parentheses. All comparisons use seed means and keep the same profile across columns. Final CE uses the last training epoch. Checkpoint CE and accuracy use each run’s minimum-validation-loss checkpoint. Minimum test CE over training is unavailable because test histories were not recorded. A dash means an endpoint or matching uniform baseline is missing. Benchmark profiles minimize mean per-seed minimum validation CE among complete nonuniform arms. The standalone vanilla-MLP aggregates lack validation losses, so those four rows retain their explicitly retrospective choice by mean checkpoint test CE. A dagger marks incomplete candidate coverage. For vanilla MLPs, d and w denote depth and width.

Dataset / setting	Model	n	Nonuniform profile	Final CE reduction (%)	Min. CE reduction (%)	Checkpoint CE reduction (%)	Accuracy gain % (pp)
<i>Multi-dataset benchmark</i>							
Speech Commands	MLP	5	Step early	–	–	+12.71	+2.53 (+1.83)
Speech Commands	Transformer	5	Big step	–	–	+4.91	+1.92 (+1.53)
Tiny ImageNet	MLP	5	Step early	–	–	+1.94	+40.49 (+1.40)
Tiny ImageNet	Transformer	5	Step early	–	–	+2.17	+11.14 (+1.63)
FI-2010	MLP	5	Big step	–	–	+3.57	+1.40 (+0.67)
Jannis	MLP	5	Step early	–	–	+2.92	+2.44 (+1.29)
Jannis	Transformer	5	Big step	–	–	+0.15	+0.44 (+0.26)
<i>Data-regime comparisons</i>							
Tiny ImageNet 80,000 train	MLP	5	Step early	+8.84	–	+1.26	+12.19 (+0.86)
Tiny ImageNet 80,000 train	Transformer	5	Linear decreasing	+1.68	–	+1.01	+0.91 (+0.29)
<i>Architecture and duration follow-ups</i>							
FI-2010 (linear)	Transformer	5	Linear decreasing	–	–	–	–
Jannis (linear)	Transformer	5	Linear decreasing	–	–	–	–
Jannis (100 ep.)	Transformer	10	Linear increasing	–	–	-0.41	-0.10 (-0.06)
<i>Standalone vanilla MLPs</i>							
FI-2010 (d=6, w=256)	MLP	5	Step late	–	–	+1.32	+0.03 (+0.02)
FI-2010 (d=6, w=512)	MLP	5	Linear decreasing	–	–	+0.62	+0.01 (+0.01)
FI-2010 (d=12, w=256)	MLP	5	Linear decreasing	–	–	+0.43	-0.08 (-0.06)
FI-2010 (d=12, w=512)	MLP	5	Step early	–	–	+0.81	+0.02 (+0.02)

E.1. Speech Commands / MLP / Zero weight decay

Depth 12, width 256; 20,020/5,005/10,010 training/validation/test examples; 50 epochs, batch size 128, weight decay 0. The recorded split is held fixed across profiles.

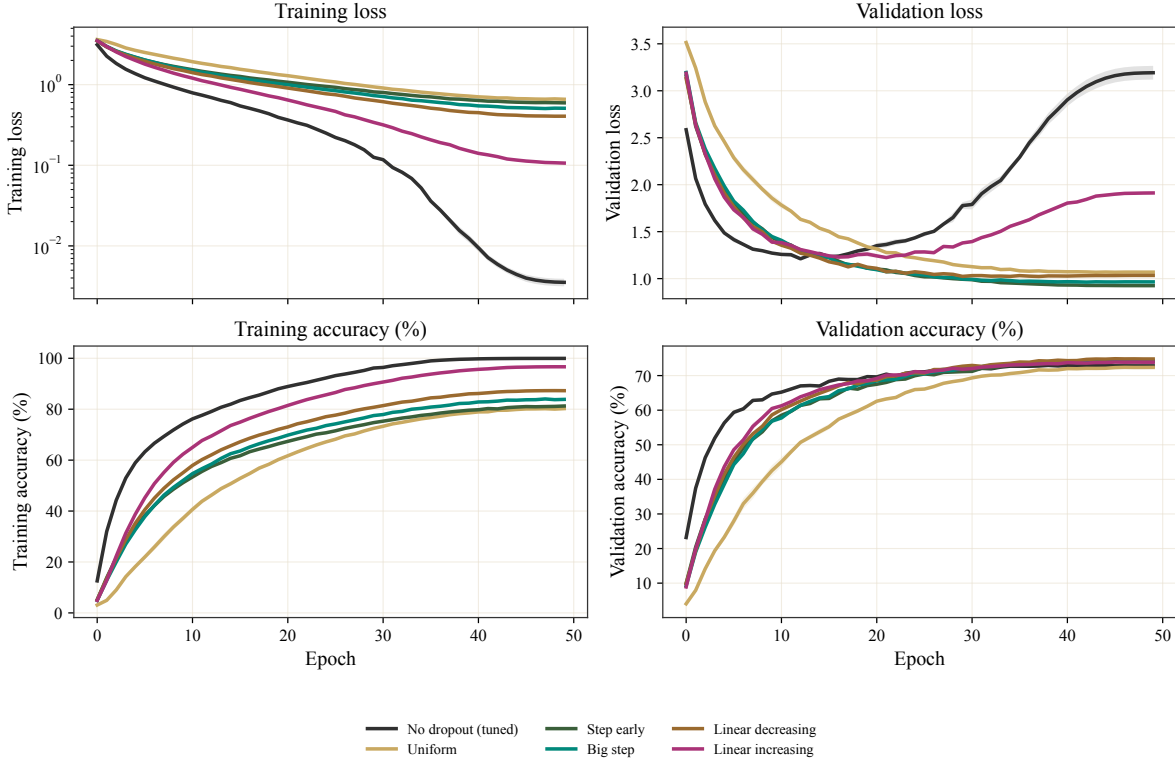


Figure 13. Speech Commands / MLP / Zero weight decay. Full recorded learning curves; bands show SEM across the available seeds for each profile. Sample counts and tuned probabilities are given below.

Profile	n	$\bar{p}$	Learning rate	Test CE	Test acc. (%)	Final test CE
No dropout (tuned)	5	0.00	1e-03	1.203 ± 0.011	67.16 ± 0.35	—
Uniform	5	0.10	1e-03	1.073 ± 0.006	72.23 ± 0.12	—
Step early	5	0.05	1e-03	0.937 ± 0.003	74.06 ± 0.13	—
Big step	5	0.05	1e-03	0.950 ± 0.004	74.07 ± 0.11	—
Linear decreasing	5	0.05	1e-03	1.002 ± 0.005	73.87 ± 0.45	—
Linear increasing	5	0.05	1e-03	1.202 ± 0.009	69.40 ± 0.54	—

For the validation-selected nonuniform profile, the paired test-CE difference (profile minus uniform) is -0.136 ± 0.007 (SEM).

E.2. Speech Commands / Transformer / Zero weight decay

Depth 12, width 256; 20,020/5,005/10,010 training/validation/test examples; 50 epochs, batch size 128, weight decay 0. The recorded split is held fixed across profiles.

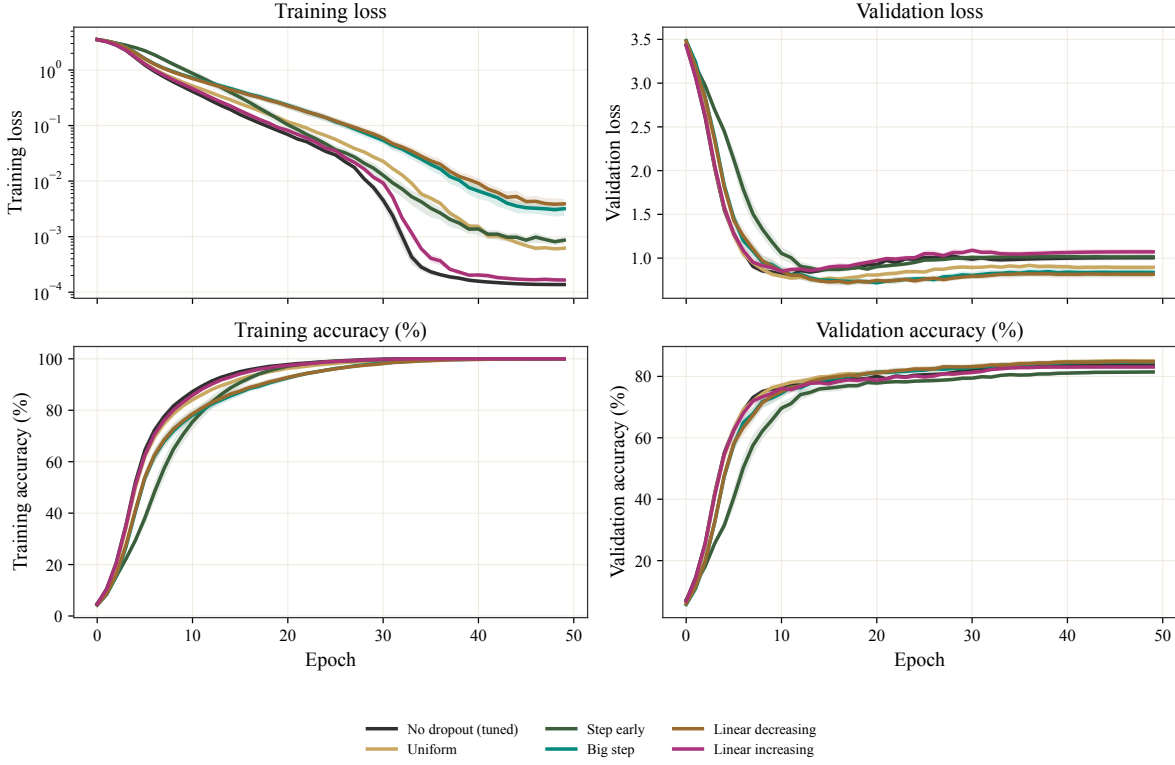


Figure 14. Speech Commands / Transformer / Zero weight decay. Full recorded learning curves; bands show SEM across the available seeds for each profile. Sample counts and tuned probabilities are given below.

Profile	n	$\bar{p}$	Learning rate	Test CE	Test acc. (%)	Final test CE
No dropout (tuned)	5	0.00	3e-04	0.807 ± 0.018	77.65 ± 0.65	—
Uniform	5	0.15	3e-04	0.743 ± 0.017	79.54 ± 0.61	—
Step early	5	0.05	1e-04	0.852 ± 0.027	77.85 ± 0.64	—
Big step	5	0.15	3e-04	0.707 ± 0.036	81.07 ± 0.89	—
Linear decreasing	5	0.20	3e-04	0.711 ± 0.033	81.27 ± 1.05	—
Linear increasing	5	0.10	3e-04	0.832 ± 0.022	77.67 ± 0.67	—

For the validation-selected nonuniform profile, the paired test-CE difference (profile minus uniform) is -0.036 ± 0.037 (SEM).

E.3. Tiny ImageNet / MLP / 80,000 train

Depth 12, width 256; 80,000/10,000/10,000 training/validation/test examples; 75 epochs, batch size 128, weight decay 0. The recorded split is held fixed across profiles.

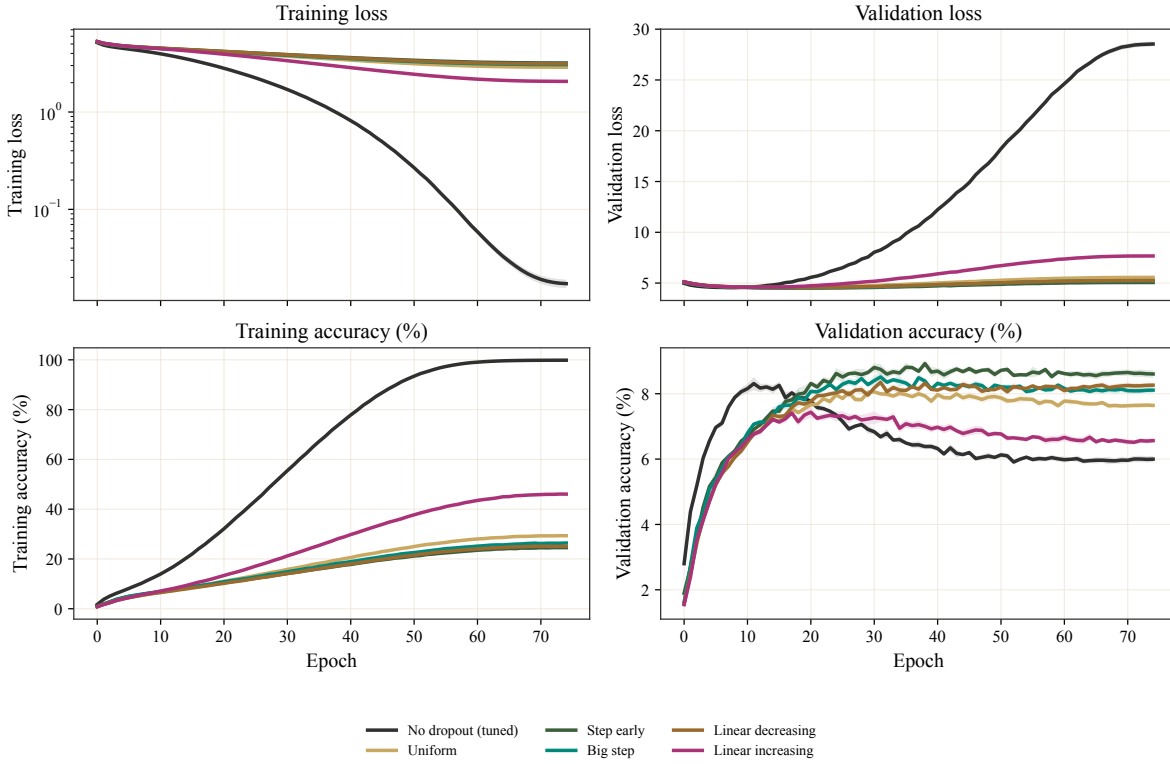


Figure 15. Tiny ImageNet / MLP / 80,000 train. Full recorded learning curves; bands show SEM across the available seeds for each profile. Sample counts and tuned probabilities are given below.

Profile	n	$\bar{p}$	Learning rate	Test CE	Test acc. (%)	Final test CE
No dropout (tuned)	5	0.00	3e-04	4.561 ± 0.004	7.64 ± 0.15	28.876 ± 0.230
Uniform	5	0.05	3e-04	4.575 ± 0.002	7.09 ± 0.14	5.627 ± 0.004
Step early	5	0.05	3e-04	4.518 ± 0.004	7.95 ± 0.10	5.130 ± 0.009
Big step	5	0.05	3e-04	4.541 ± 0.003	7.53 ± 0.12	5.321 ± 0.007
Linear decreasing	5	0.05	3e-04	4.563 ± 0.003	7.19 ± 0.15	5.286 ± 0.013
Linear increasing	5	0.05	3e-04	4.608 ± 0.005	6.75 ± 0.12	7.764 ± 0.023

For the validation-selected nonuniform profile, the paired test-CE difference (profile minus uniform) is -0.058 ± 0.005 (SEM).

E.4. Tiny ImageNet / Transformer / 80,000 train

Depth 12, width 256; 80,000/10,000/10,000 training/validation/test examples; 75 epochs, batch size 128, weight decay 0. The recorded split is held fixed across profiles.

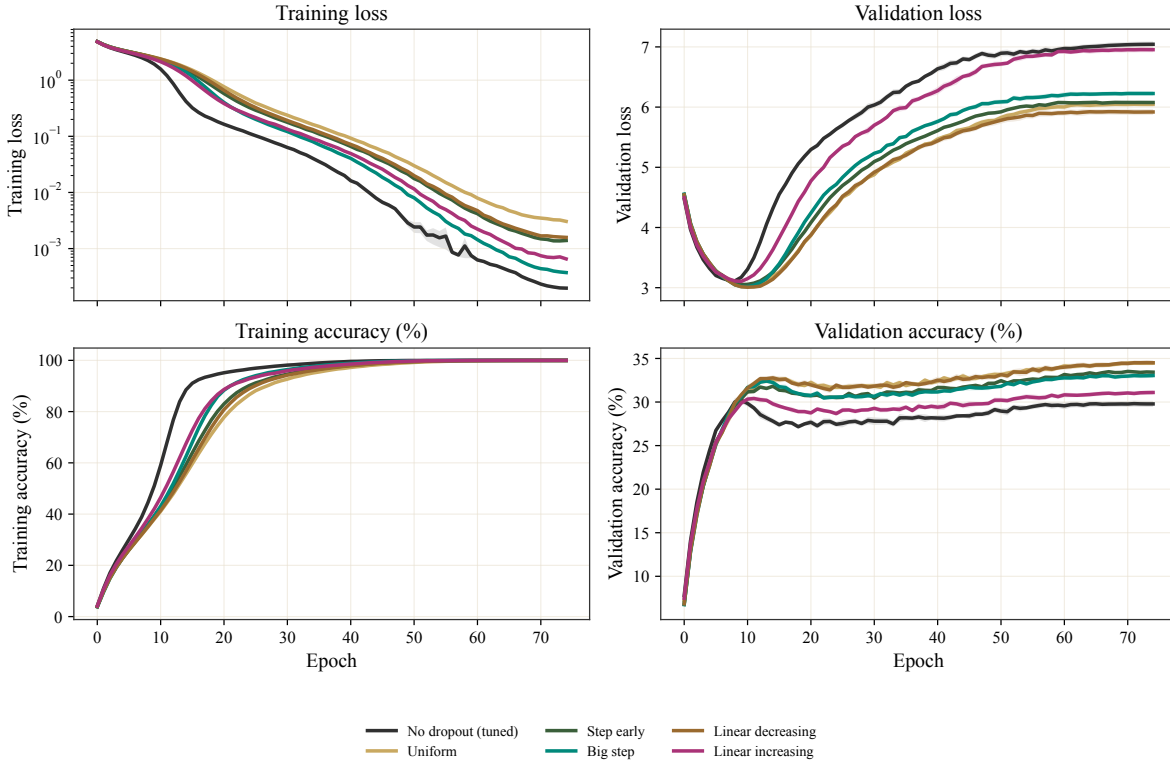


Figure 16. Tiny ImageNet / Transformer / 80,000 train. Full recorded learning curves; bands show SEM across the available seeds for each profile. Sample counts and tuned probabilities are given below.

Profile	n	$\bar{p}$	Learning rate	Test CE	Test acc. (%)	Final test CE
No dropout (tuned)	5	0.00	3e-04	3.125 ± 0.025	29.57 ± 0.44	7.126 ± 0.068
Uniform	5	0.20	3e-04	3.035 ± 0.014	31.49 ± 0.17	6.104 ± 0.043
Step early	5	0.15	3e-04	3.053 ± 0.015	31.22 ± 0.21	6.164 ± 0.025
Big step	5	0.10	3e-04	3.026 ± 0.013	31.35 ± 0.38	6.296 ± 0.034
Linear decreasing	5	0.15	3e-04	3.004 ± 0.016	31.78 ± 0.23	6.002 ± 0.042
Linear increasing	5	0.20	3e-04	3.118 ± 0.017	29.72 ± 0.26	7.026 ± 0.052

For the validation-selected nonuniform profile, the paired test-CE difference (profile minus uniform) is -0.031 ± 0.015 (SEM).

E.5. FI-2010 / Transformer / Linear follow-up

Depth 12, width 256; 40,000/10,000/20,000 training/validation/test examples; 50 epochs, batch size 128, weight decay $1e-07$. The recorded split is held fixed across profiles.

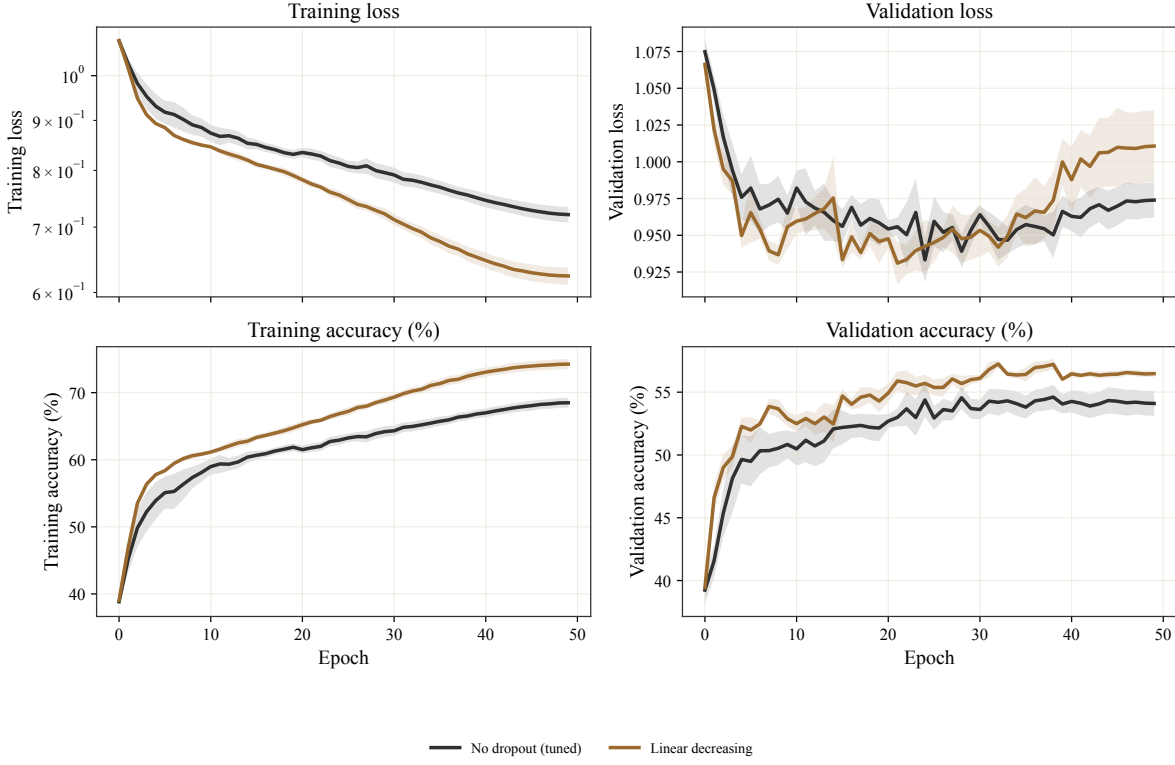


Figure 17. FI-2010 / Transformer / Linear follow-up. Full recorded learning curves; bands show SEM across the available seeds for each profile. Sample counts and tuned probabilities are given below.

Profile	n	$\bar{p}$	Learning rate	Test CE	Test acc. (%)	Final test CE
No dropout (tuned)	5	0.00	$1e-03$	0.955 ± 0.010	55.38 ± 0.39	—
Linear decreasing	5	0.10	$1e-03$	0.930 ± 0.006	57.72 ± 0.83	—

E.6. Jannis / Transformer / Linear follow-up

Depth 12, width 256; 3,840/960/1,920 training/validation/test examples; 50 epochs, batch size 128, weight decay $1e - 07$. The recorded split is held fixed across profiles.

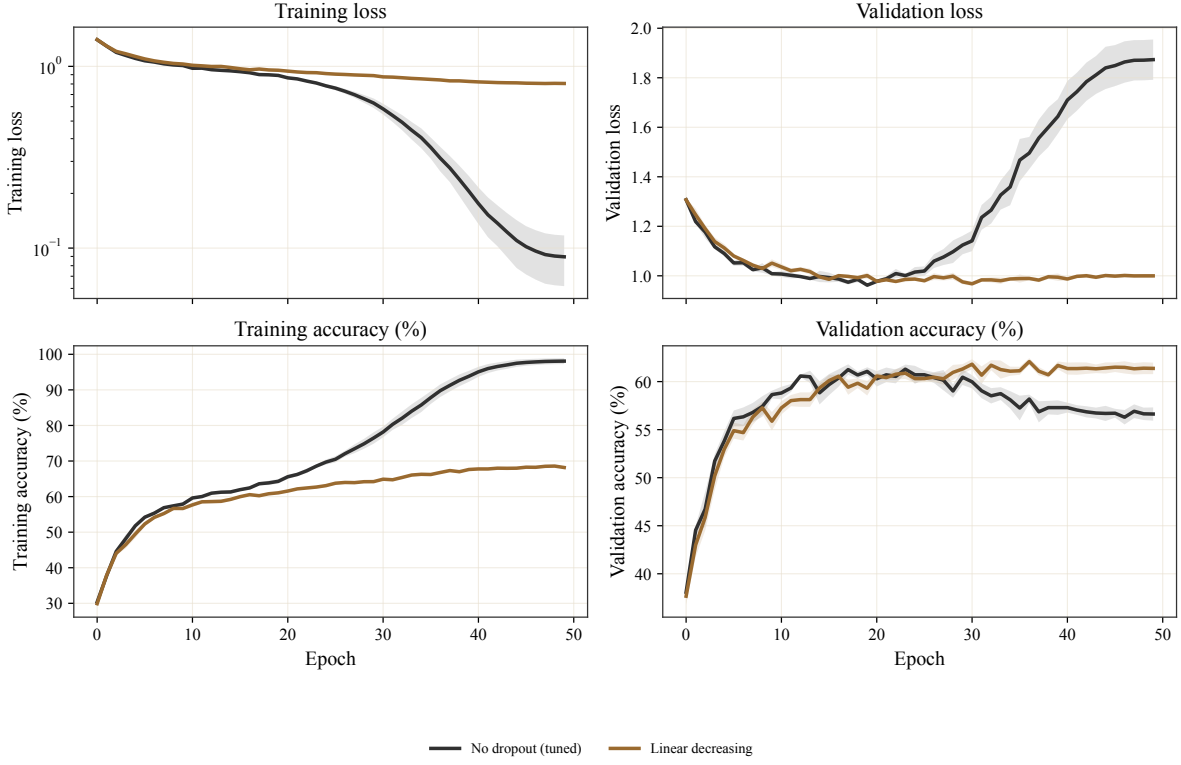


Figure 18. Jannis / Transformer / Linear follow-up. Full recorded learning curves; bands show SEM across the available seeds for each profile. Sample counts and tuned probabilities are given below.

Profile	n	$\bar{p}$	Learning rate	Test CE	Test acc. (%)	Final test CE
No dropout (tuned)	5	0.00	$1e-04$	0.986 ± 0.002	59.07 ± 0.47	—
Linear decreasing	5	0.10	$1e-04$	0.992 ± 0.005	60.27 ± 0.43	—

E.7. Jannis / Transformer / 100 epochs

Depth 12, width 256; 3,840/960/1,920 training/validation/test examples; 100 epochs, batch size 128, weight decay $1e - 07$. The recorded split is held fixed across profiles.

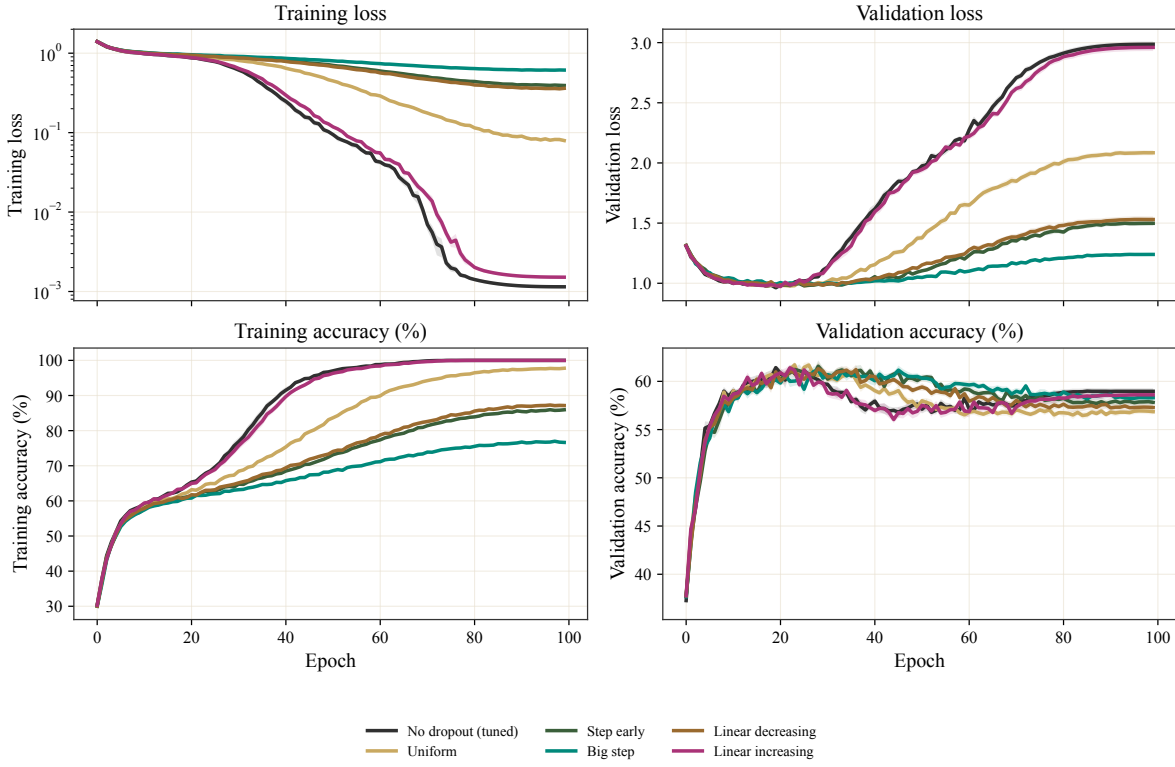


Figure 19. Jannis / Transformer / 100 epochs. Full recorded learning curves; bands show SEM across the available seeds for each profile. Sample counts and tuned probabilities are given below.

Profile	n	$\bar{p}$	Learning rate	Test CE	Test acc. (%)	Final test CE
No dropout (tuned)	10	0.00	$1e-04$	0.985 ± 0.003	59.54 ± 0.16	—
Uniform	10	0.10	$1e-04$	0.988 ± 0.002	59.46 ± 0.23	—
Step early	10	0.10	$1e-04$	0.990 ± 0.004	59.85 ± 0.21	—
Big step	10	0.10	$1e-04$	0.999 ± 0.003	59.36 ± 0.29	—
Linear decreasing	10	0.10	$1e-04$	0.988 ± 0.003	59.75 ± 0.13	—
Linear increasing	10	0.10	$1e-04$	0.992 ± 0.003	59.40 ± 0.25	—

For the validation-selected nonuniform profile, the paired test-CE difference (profile minus uniform) is 0.004 ± 0.003 (SEM).

E.8. Tiny ImageNet / MLP / Zero weight decay

Depth 12, width 256; 20,000/5,000/10,000 training/validation/test examples; 75 epochs, batch size 128, weight decay 0. The recorded split is held fixed across profiles.

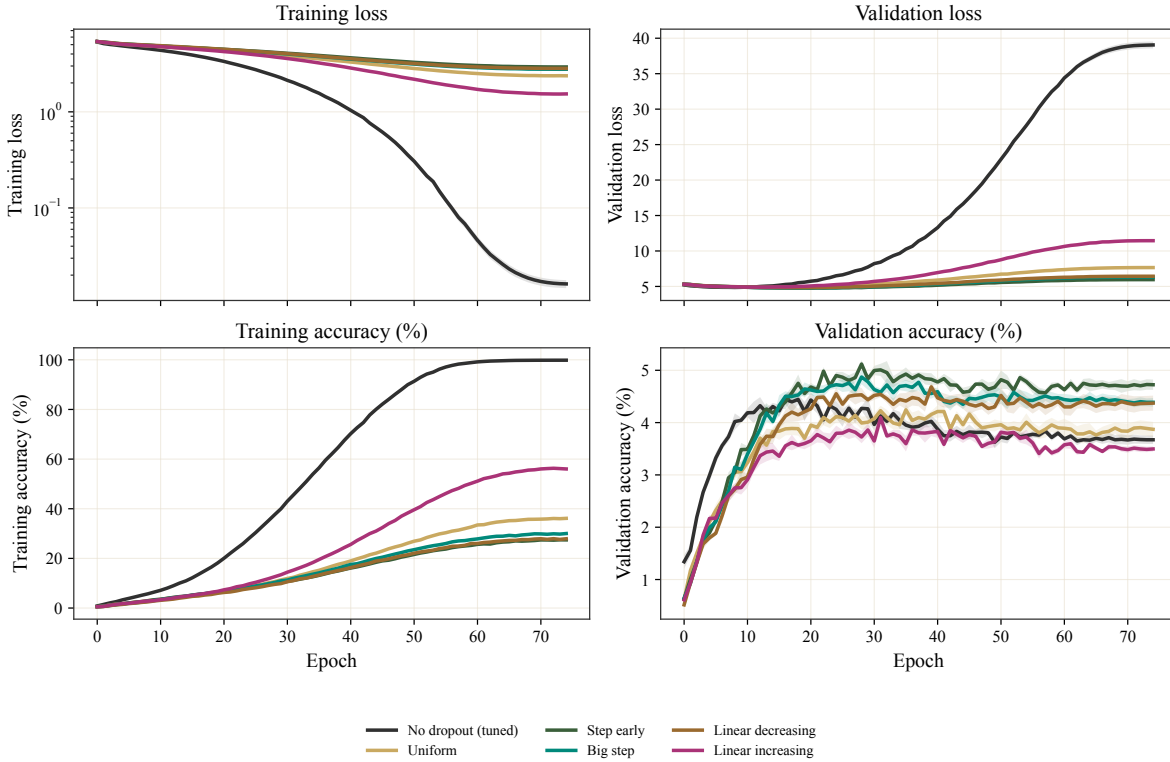


Figure 20. Tiny ImageNet / MLP / Zero weight decay. Full recorded learning curves; bands show SEM across the available seeds for each profile. Sample counts and tuned probabilities are given below.

Profile	n	$\bar{p}$	Learning rate	Test CE	Test acc. (%)	Final test CE
No dropout (tuned)	5	0.00	1e-03	4.856 ± 0.005	3.88 ± 0.05	—
Uniform	5	0.05	1e-03	4.868 ± 0.006	3.45 ± 0.06	—
Step early	5	0.05	3e-04	4.774 ± 0.006	4.84 ± 0.04	—
Big step	5	0.05	3e-04	4.787 ± 0.003	4.61 ± 0.10	—
Linear decreasing	5	0.05	3e-04	4.823 ± 0.004	4.38 ± 0.10	—
Linear increasing	5	0.05	1e-03	4.898 ± 0.007	3.36 ± 0.09	—

For the validation-selected nonuniform profile, the paired test-CE difference (profile minus uniform) is -0.094 ± 0.011 (SEM).

E.9. Tiny ImageNet / Transformer / Zero weight decay

Depth 12, width 256; 20,000/5,000/10,000 training/validation/test examples; 75 epochs, batch size 128, weight decay 0. The recorded split is held fixed across profiles.

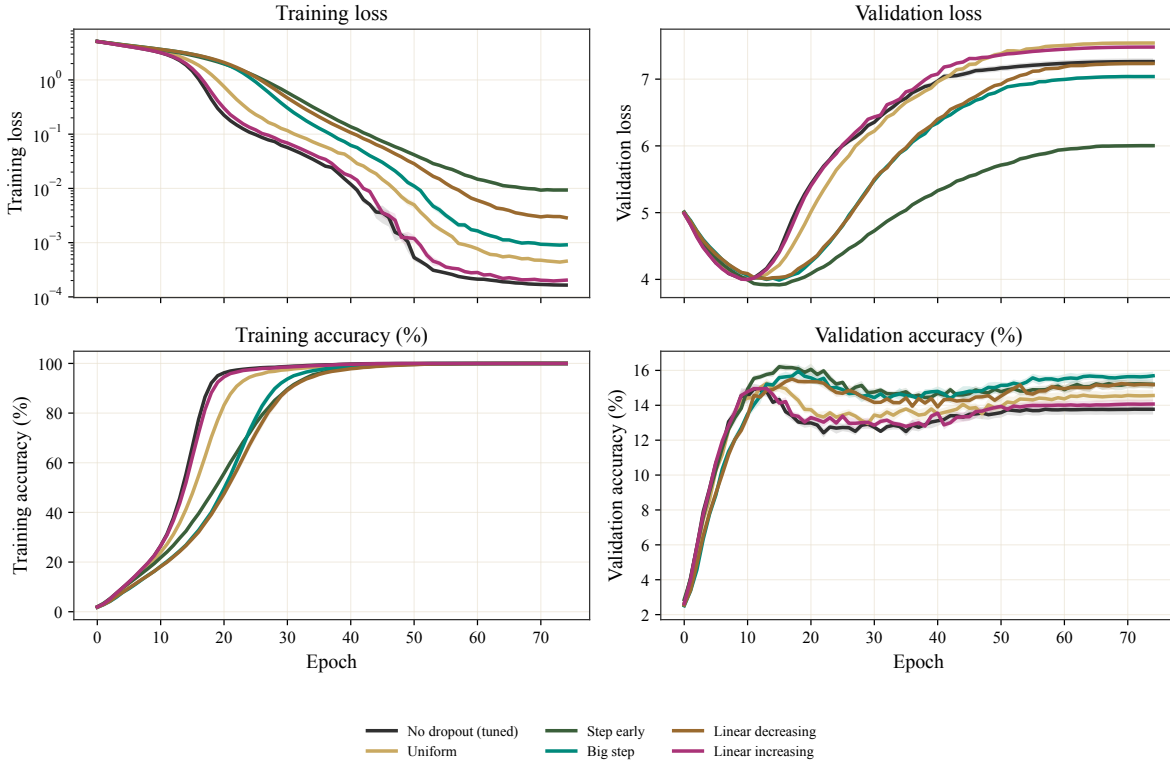


Figure 21. Tiny ImageNet / Transformer / Zero weight decay. Full recorded learning curves; bands show SEM across the available seeds for each profile. Sample counts and tuned probabilities are given below.

Profile	n	$\bar{p}$	Learning rate	Test CE	Test acc. (%)	Final test CE
No dropout (tuned)	5	0.00	3e-04	3.981 ± 0.012	14.92 ± 0.21	—
Uniform	5	0.10	3e-04	3.976 ± 0.007	14.63 ± 0.17	—
Step early	5	0.15	1e-04	3.890 ± 0.008	16.26 ± 0.10	—
Big step	5	0.15	3e-04	3.960 ± 0.012	15.45 ± 0.29	—
Linear decreasing	5	0.20	3e-04	3.980 ± 0.011	14.71 ± 0.22	—
Linear increasing	5	0.05	3e-04	3.981 ± 0.017	14.73 ± 0.30	—

For the validation-selected nonuniform profile, the paired test-CE difference (profile minus uniform) is -0.086 ± 0.013 (SEM).

E.10. FI-2010 / MLP / Zero weight decay

Depth 12, width 256; 40,000/10,000/20,000 training/validation/test examples; 50 epochs, batch size 128, weight decay 0. The recorded split is held fixed across profiles.

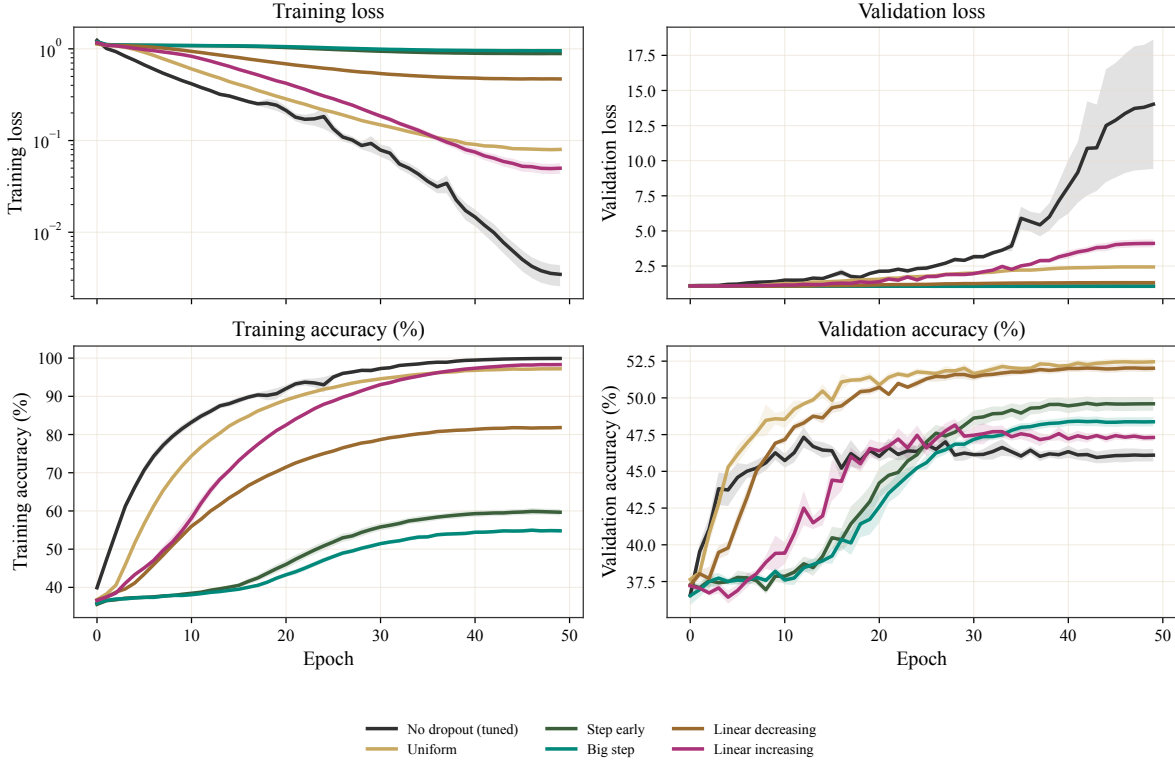


Figure 22. FI-2010 / MLP / Zero weight decay. Full recorded learning curves; bands show SEM across the available seeds for each profile. Sample counts and tuned probabilities are given below.

Profile	n	$\bar{p}$	Learning rate	Test CE	Test acc. (%)	Final test CE
No dropout (tuned)	5	0.00	3e-03	1.118 ± 0.017	42.19 ± 2.20	—
Uniform	5	0.05	1e-04	1.093 ± 0.004	47.70 ± 0.37	—
Step early	5	0.10	3e-05	1.061 ± 0.004	48.19 ± 0.33	—
Big step	5	0.10	3e-05	1.054 ± 0.004	48.37 ± 0.39	—
Linear decreasing	5	0.05	3e-05	1.074 ± 0.008	47.48 ± 0.27	—
Linear increasing	5	0.15	1e-03	1.115 ± 0.011	37.62 ± 0.39	—

For the validation-selected nonuniform profile, the paired test-CE difference (profile minus uniform) is -0.039 ± 0.005 (SEM).

E.11. Jannis / MLP / Zero weight decay

Depth 12, width 256; 3,840/960/1,920 training/validation/test examples; 100 epochs, batch size 128, weight decay 0. The recorded split is held fixed across profiles.

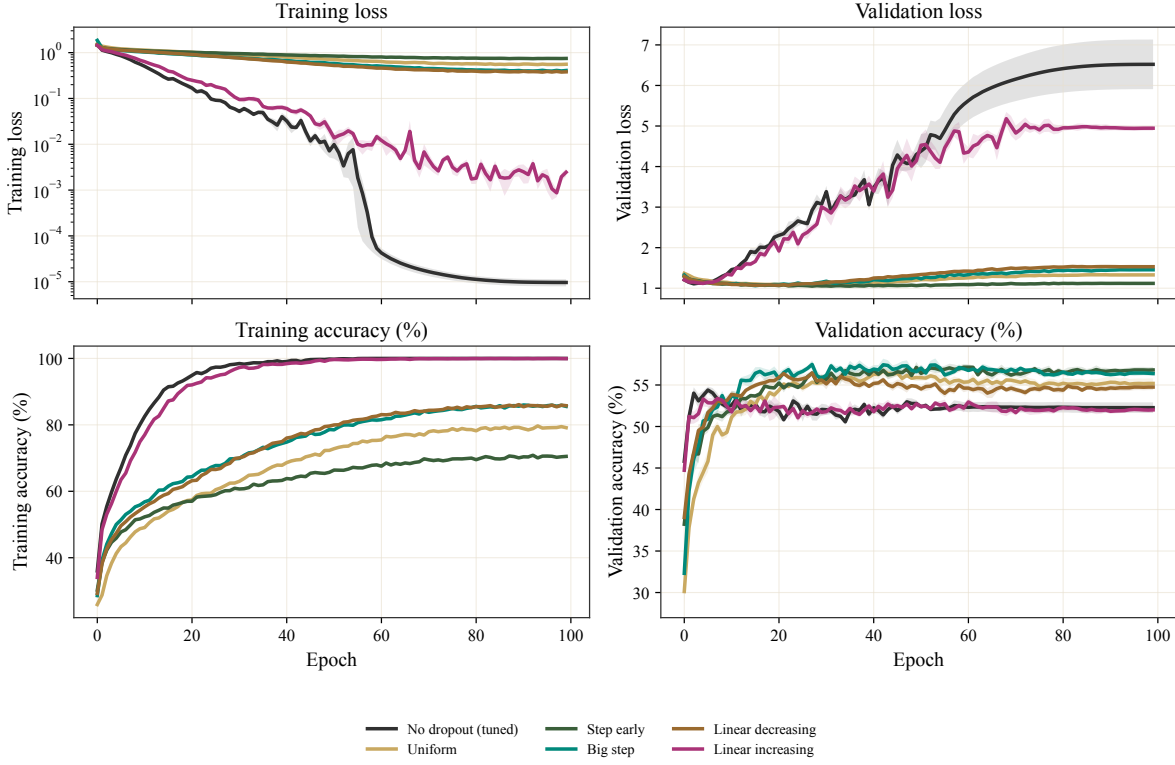


Figure 23. Jannis / MLP / Zero weight decay. Full recorded learning curves; bands show SEM across the available seeds for each profile. Sample counts and tuned probabilities are given below.

Profile	n	$\bar{p}$	Learning rate	Test CE	Test acc. (%)	Final test CE
No dropout (tuned)	5	0.00	3e-03	1.123 ± 0.006	51.56 ± 0.38	—
Uniform	5	0.10	3e-04	1.123 ± 0.007	52.96 ± 0.20	—
Step early	5	0.05	3e-04	1.091 ± 0.004	54.25 ± 0.41	—
Big step	5	0.10	3e-03	1.087 ± 0.003	55.00 ± 0.43	—
Linear decreasing	5	0.05	3e-04	1.110 ± 0.004	53.18 ± 0.52	—
Linear increasing	5	0.05	3e-03	1.140 ± 0.008	51.77 ± 0.71	—

For the validation-selected nonuniform profile, the paired test-CE difference (profile minus uniform) is -0.033 ± 0.010 (SEM).

E.12. Jannis / Transformer / Zero weight decay

Depth 12, width 256; 3,840/960/1,920 training/validation/test examples; 100 epochs, batch size 128, weight decay 0. The recorded split is held fixed across profiles.

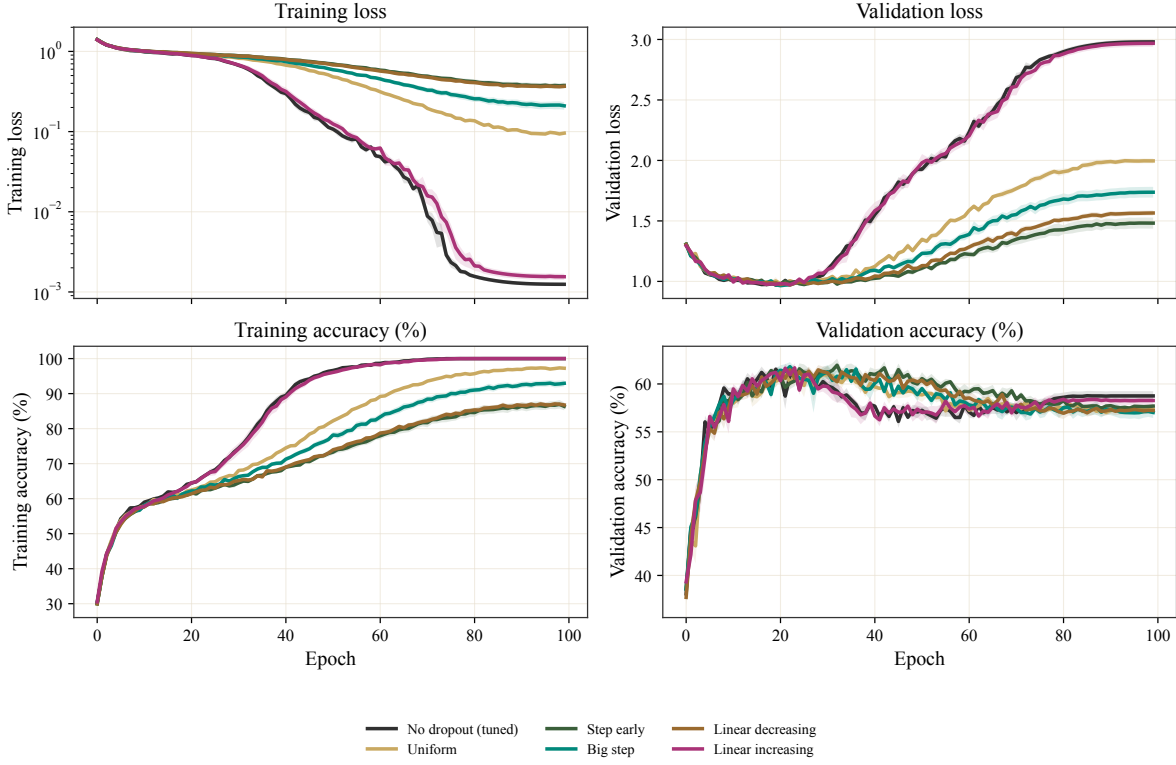


Figure 24. Jannis / Transformer / Zero weight decay. Full recorded learning curves; bands show SEM across the available seeds for each profile. Sample counts and tuned probabilities are given below.

Profile	n	$\bar{p}$	Learning rate	Test CE	Test acc. (%)	Final test CE
No dropout (tuned)	5	0.00	1e-04	0.985 ± 0.002	59.50 ± 0.10	—
Uniform	5	0.10	1e-04	0.993 ± 0.004	58.98 ± 0.19	—
Step early	5	0.05	1e-04	0.988 ± 0.001	59.22 ± 0.21	—
Big step	5	0.05	1e-04	0.992 ± 0.003	59.24 ± 0.57	—
Linear decreasing	5	0.10	1e-04	0.994 ± 0.003	58.90 ± 0.32	—
Linear increasing	5	0.10	1e-04	0.988 ± 0.003	60.28 ± 0.26	—

For the validation-selected nonuniform profile, the paired test-CE difference (profile minus uniform) is -0.002 ± 0.005 (SEM).

E.13. Standalone FI-2010 vanilla MLPs

These plain feedforward ReLU MLPs flatten a 128×40 input window. The study uses day 6 for training, day 7 for validation, and days 8–10 for test. Each architecture is trained for 100 epochs with five paired confirmation seeds (100–104), zero weight decay, and mean dropout 0.10 for nonzero profiles. Weights have variance 1.98/fan_in and biases variance 0.02. The learning rate is chosen using uniform-dropout validation runs and then held fixed across profiles, including no dropout. These are separate from the depth-12 width-256 benchmark cohorts. Test loss, accuracy, and macro-F1 are evaluated once, at the minimum-validation-loss checkpoint. The saved results contain profile means and SEMs, but no validation objectives, per-seed test metrics, or per-epoch test histories. Final-epoch and minimum-over-training test losses are therefore unavailable. The standalone rows in Table 14 retain the retrospective choice of the nonuniform profile with the lowest observed mean checkpoint test CE and use that same profile for accuracy. These rows are separate from the validation-selected benchmark comparisons.

Depth 6, width 256.

Profile	Checkpoint test CE	Test accuracy (%)	Macro-F1 (%)
No dropout	0.8348 ± 0.0071	69.667 ± 0.403	32.96 ± 0.34
Uniform	0.8338 ± 0.0035	70.830 ± 0.030	31.74 ± 0.31
Step early	0.8332 ± 0.0024	70.851 ± 0.027	30.25 ± 0.97
Big step	0.8279 ± 0.0019	70.839 ± 0.010	30.44 ± 1.17
Linear decreasing	0.8312 ± 0.0035	70.851 ± 0.011	28.36 ± 0.54
Linear increasing	0.8293 ± 0.0026	70.865 ± 0.009	31.45 ± 0.56
Step late	0.8227 ± 0.0016	70.855 ± 0.019	31.97 ± 0.45

Depth 6, width 512.

Profile	Checkpoint test CE	Test accuracy (%)	Macro-F1 (%)
No dropout	0.8327 ± 0.0057	69.838 ± 0.392	32.63 ± 1.27
Uniform	0.8318 ± 0.0024	70.854 ± 0.020	31.27 ± 0.70
Step early	0.8341 ± 0.0044	70.851 ± 0.023	31.41 ± 1.05
Big step	0.8330 ± 0.0042	70.853 ± 0.007	30.91 ± 1.27
Linear decreasing	0.8267 ± 0.0071	70.865 ± 0.029	31.15 ± 0.81
Linear increasing	0.8342 ± 0.0037	70.866 ± 0.005	31.17 ± 0.87
Step late	0.8308 ± 0.0033	70.841 ± 0.036	31.58 ± 0.76

Depth 12, width 256.

Profile	Checkpoint test CE	Test accuracy (%)	Macro-F1 (%)
No dropout	0.8260 ± 0.0062	70.884 ± 0.024	29.67 ± 0.70
Uniform	0.8210 ± 0.0029	70.870 ± 0.011	31.93 ± 0.89
Step early	0.8232 ± 0.0032	70.856 ± 0.013	32.16 ± 0.52
Big step	0.8224 ± 0.0017	70.809 ± 0.040	32.43 ± 0.29
Linear decreasing	0.8175 ± 0.0025	70.814 ± 0.035	32.84 ± 0.03
Linear increasing	0.8227 ± 0.0019	70.844 ± 0.017	32.76 ± 0.06
Step late	0.8234 ± 0.0038	70.805 ± 0.020	32.78 ± 0.09

Depth 12, width 512.

Profile	Checkpoint test CE	Test accuracy (%)	Macro-F1 (%)
No dropout	0.8161 ± 0.0034	70.817 ± 0.014	31.53 ± 0.98
Uniform	0.8228 ± 0.0028	70.829 ± 0.016	31.39 ± 0.93
Step early	0.8162 ± 0.0049	70.845 ± 0.030	32.11 ± 0.65
Big step	0.8220 ± 0.0031	70.826 ± 0.019	32.79 ± 0.05
Linear decreasing	0.8201 ± 0.0027	70.849 ± 0.013	32.87 ± 0.02
Linear increasing	0.8175 ± 0.0019	70.853 ± 0.016	32.15 ± 0.67
Step late	0.8238 ± 0.0017	70.835 ± 0.024	31.85 ± 1.05

At depth 12 and width 512, no dropout has slightly lower mean test CE than the best nonuniform profile (0.816111 versus 0.816155). Main-table winners exclude no dropout; this does not establish that nonuniform dropout beats every control.

E.14. CIFAR-10 profile-geometry pilot

The validation-only geometry pilot contains 30 runs: ten profiles at mean probability $\bar{p} = 0.10$, with three seeds per profile, six-layer width-256 ReLU MLPs, 4,000 training examples, 1,000 validation examples, and 75 epochs. It compares profile shape and exact early/late reversals. Test data remain sealed, so this pilot contributes no test-loss winner to the main tables.

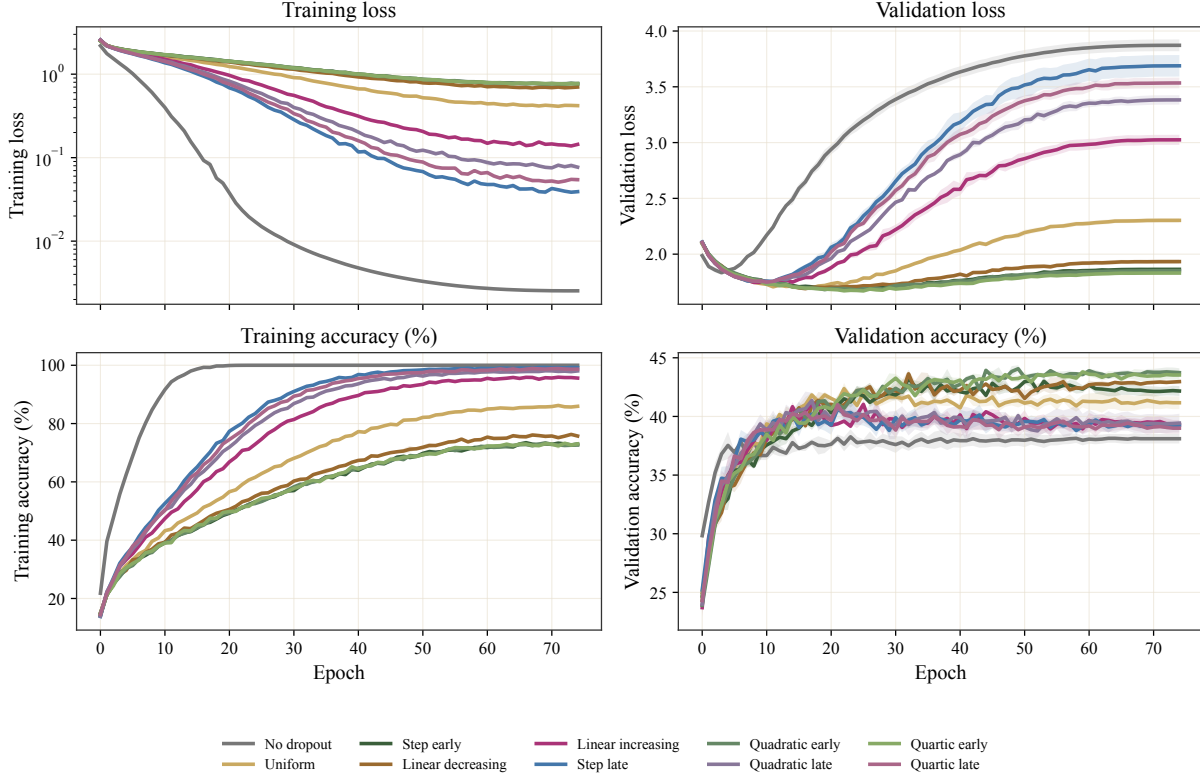


Figure 25. Profile geometry at fixed mean dropout. Training and validation curves use the original paper style and show mean $\pm$ SEM over three seeds.

Profile	Final validation CE	Final validation accuracy (%)
No dropout	3.872 ± 0.053	38.10 ± 0.40
Uniform	2.303 ± 0.019	41.17 ± 0.55
Step early	1.864 ± 0.009	42.17 ± 0.43
Linear decreasing	1.933 ± 0.019	42.97 ± 0.15
Linear increasing	3.025 ± 0.045	39.30 ± 0.40
Step late	3.688 ± 0.097	39.27 ± 0.30
Quadratic early	1.852 ± 0.001	43.73 ± 0.38
Quadratic late	3.382 ± 0.042	39.43 ± 0.80
Quartic early	1.831 ± 0.016	43.53 ± 0.41
Quartic late	3.534 ± 0.048	39.00 ± 0.50

The saved records do not contain a complete width-transfer comparison. The geometry pilot alone does not establish successful μ P transfer.